



PROJECT TITLE

Accumulator Cooling and Packaging - Design Report

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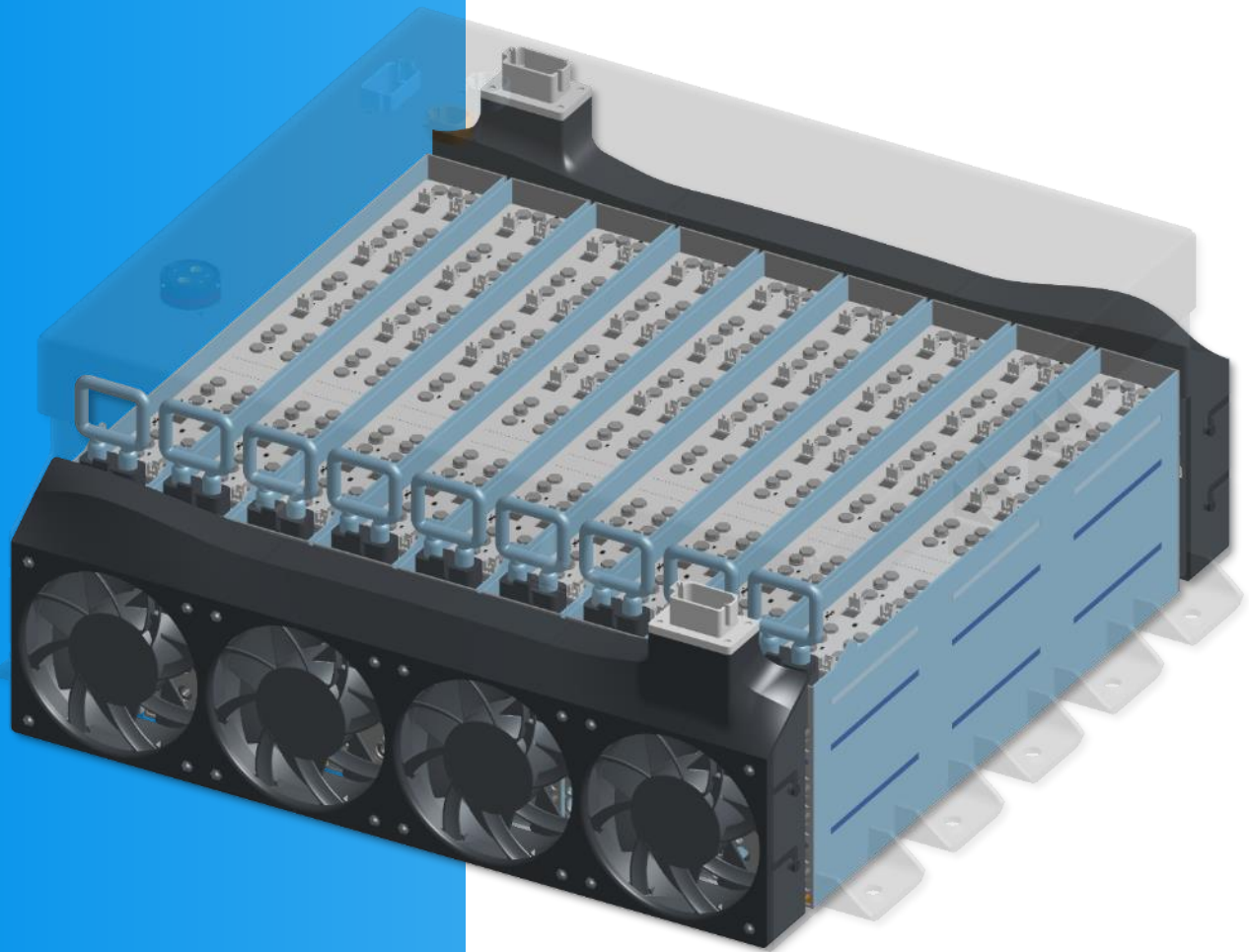
FINAL YEAR PROJECT



Accumulator Cooling & Packaging

DESIGN REPORT

NU23



Michael Dalton
2023



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Introduction

NU Racing is a student team competing in the FSAE Australasia competition. In 2022, NU Racing took their vehicle, EV.Three, to competition in Winton and managed 4th place overall in the electric vehicle division. Being a FSAE car, EV.Three was a single seat electric race car, that featured two electric motors driving the rear wheels capable of producing approximately 60 kW.

This car was extremely competitive, and the 2023 leadership team set out the following goals to develop the new vehicle, NU23:

- Increase serviceability.
- Reduce part count.
- Decrease weight.

With these goals in mind, work begun on one of the cars biggest weaknesses: the accumulator cooling. Strict rules stipulated by FSAE state that accumulators must remain below 60° C. This requirement, and the underperformance of EV.Three's accumulator cooling, directly affected multiple events at the 2022 competition, particularly endurance. Endurance is a 20 km test of a cars reliability and offers the most points out of any event during the competition. EV.Three was so temperature limited that a significant reduction in power was required to remain below 60° C to not overheat. This drastically increased the time taken to finish the event and consequently reduced the points scored.

The accumulator is comprised of several components, the foremost being the battery cells. NU Racing uses a type of battery cell bank called Energus Modules containing 18650 lithium cells, which are housed in separate, identical housings in banks of twelve called "segments". These segments are separated from various high and low Voltage electronics in the accumulator by a structural element called the "top plate".

In EV.Three, NU Racing used an active cooling approach to maintaining the battery cells' temperature. An air-cooled system utilising three centrifugal fans was used to draw fresh air into the accumulator over the cells and exhausting it out the back of the accumulator. While this design was flawed in cooling performance, the overall topology of the accumulator was extremely serviceable and compact. For detailed information and justifications of the accumulator design, Jacob Searle and Lachlan Barrell's reports should be referenced.

This report outlines the improvements made to the cooling, as well as the challenges associated with packaging of the accumulator. This project involved a large volume of design work, shown in Figure 1, as well as the manufacturing of many of these designs, testing, and validation.

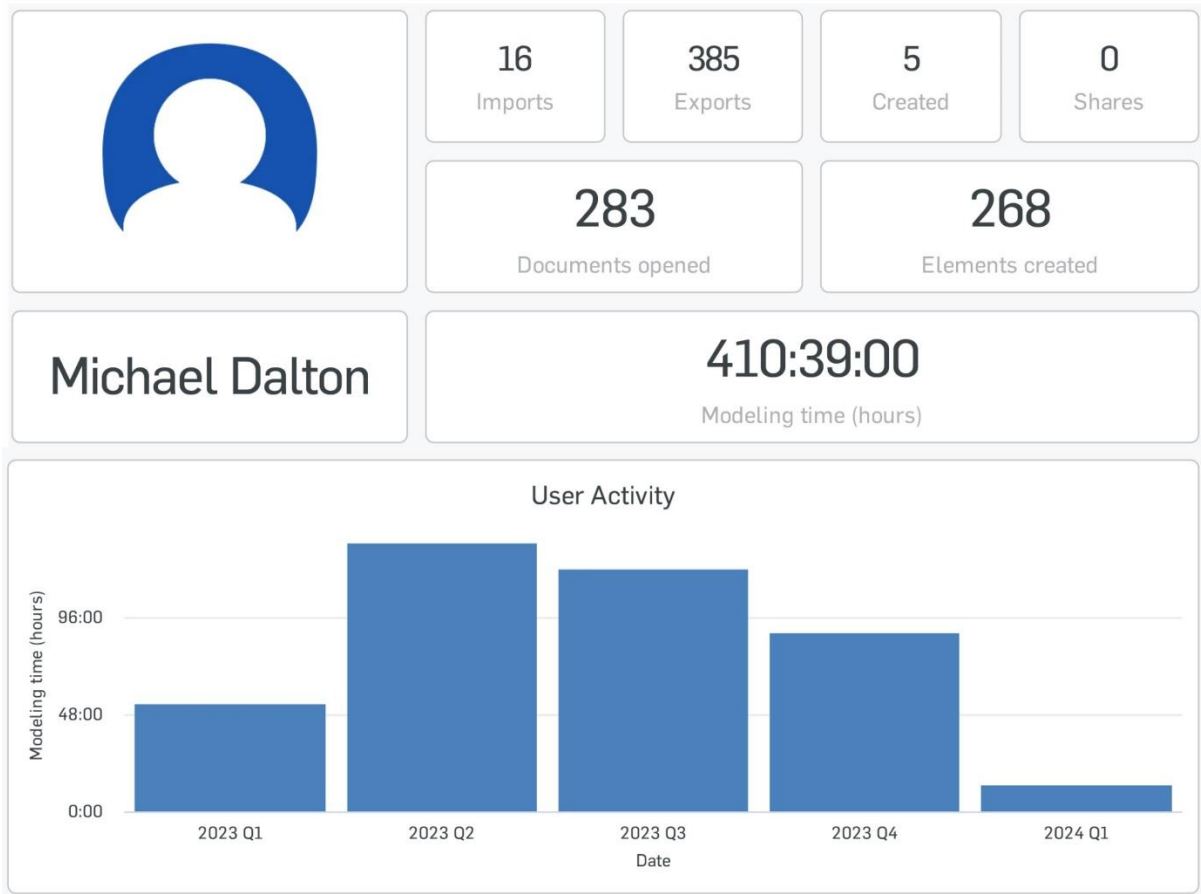


Figure 1 Onshape hours for 2023.

Motivation

To build a foundation for the project, thorough understanding of the accumulator cooling problem was required. A consensus from 2022 NU Racing team members was clear that the accumulator cooling was a significant limit on EV.Three's performance. Motec data from the 2022 competition is a concise summary of the systems shortcomings. The two main events where cooling of the accumulator had the greatest effect was endurance foremost, and autocross.

In autocross, an extremely aggressive vehicle setup was loaded, with maximum power reaching close to 55 kW. This setup allowed for nearly both auto-cross laps to be completed, however on the final corner of the second lap, the accumulator exceeded 60° C resulting in the car shutting down. The car managed to coast to the line, still posting a time, however this fault could have cost the team the event, and valuable competition points, had the car not been able to cross the line. In addition, this fault prevented the second driver from completing their laps, potentially posting an even better time.

Figure 2 shows the maximum temperature in the accumulator during the autocross event. The two distinct peaks are for each of the two laps completed.

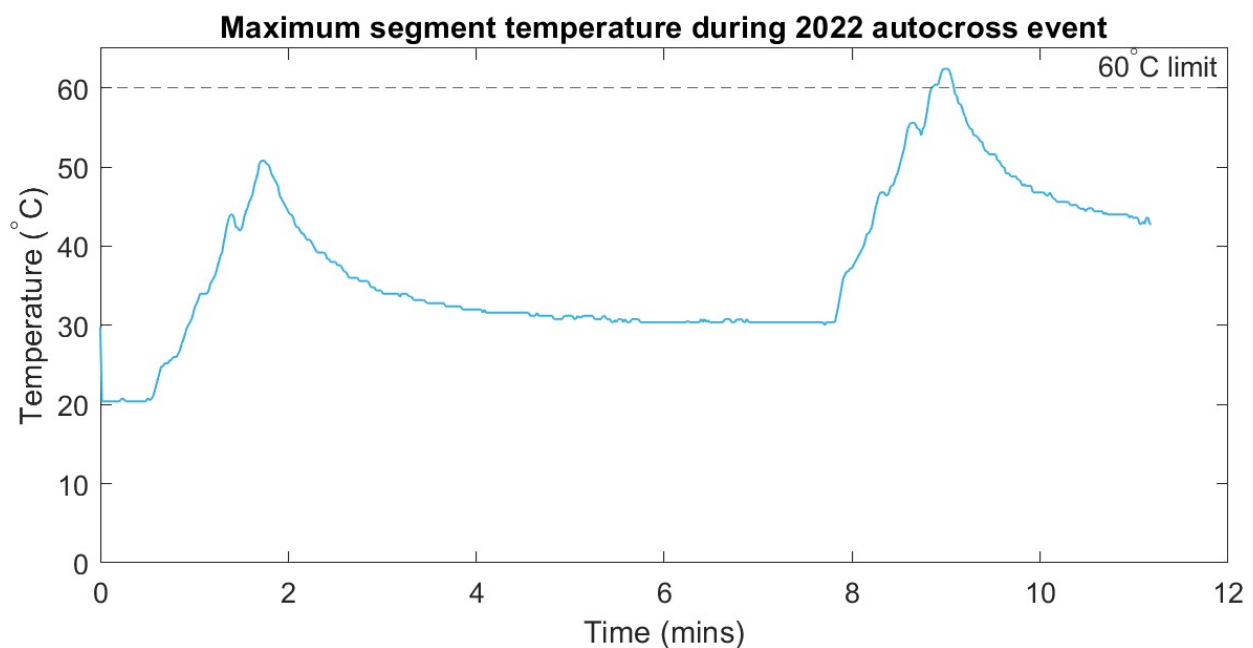


Figure 2 Maximum temperature in the accumulator throughout the autocross event.

While the temperatures did not exceed 60° C in the first lap, in the five minutes between laps, the temperatures did not return to the starting temperature prior to commencing the second lap. Therefore, even though the temperature increase in the second lap was close to that of the first, the accumulator started more than 10° C warmer for the second lap than the first, resulting in a temperature fault.

Given the near identical track layout in autocross and endurance, this indicated that the accumulator would be unable to complete two laps in endurance with the same aggressive power rating, as temperatures would rapidly exceed the 60° C in the second lap without a five-minute cool down.

For endurance, the shortcomings of the accumulators cooling were already known from the previous outing in the autocross event. Therefore, the current draw of the cells was greatly reduced, as shown in Figure 3, to ensure the event was completed without exceeding 60° C. This resulted in the vehicle exceeding 15 kW only a handful of times, and significantly increasing the time taken to complete the event. This forfeited a large number of points, due to other teams completing the event at a much greater pace.

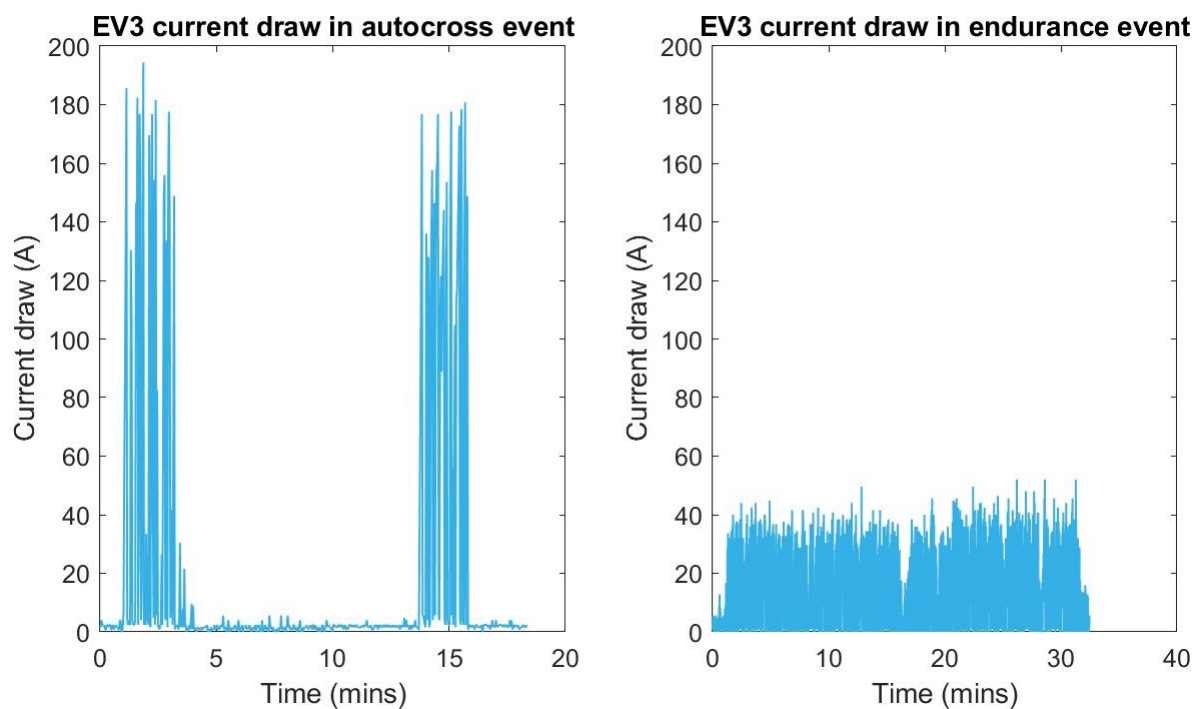


Figure 3 Comparison of current draw for EV.Three in the 2022 autocross and endurance events.

Beyond the cooling performance of the system, there were other minor areas for improvement. As stated previously, improving the serviceability of each system was a goal set out for the 2023 team. While the cooling system for EV.Three was relatively serviceable, the 3D printed manifolds, shown in Figure 4, did represent some difficulty when disassembling and reassembling the accumulator. Given their small form factor, and the nature of the top plate removal, the manifolds were easily damaged during the process of accessing the segments in the lower half of the accumulator. By the end of the service life of EV.Three's accumulator, many cracks and holes populated the manifold, undoubtedly reducing the airflow to the cells, thus reducing the cooling performance.

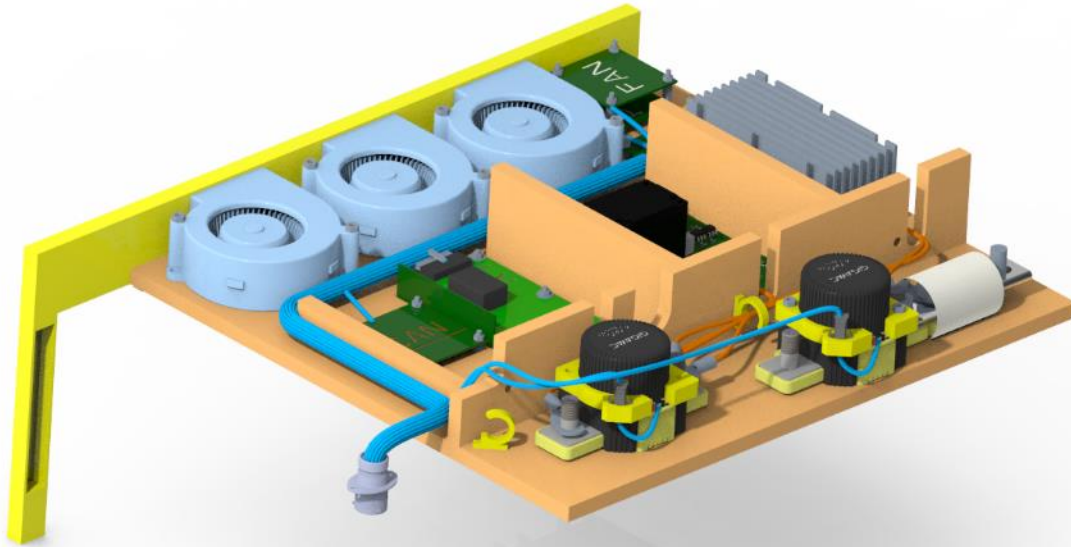


Figure 4 Manifold, shown in yellow, from EV.Three (Barrell, 2020).

Initial System Design

Scoping

To start the year, NU Racing was faced with the possibility of running the same chassis from EV.Three in NU23. The possibility of using the same chassis reduced the accumulator cooling solutions available, with size of the accumulator being essentially fixed due to the nature of its mounting within the chassis. The space for mounting, and its proximity to the firewall shown in Figure 5 illustrates just how little space there was for the accumulator to grow in any dimension.

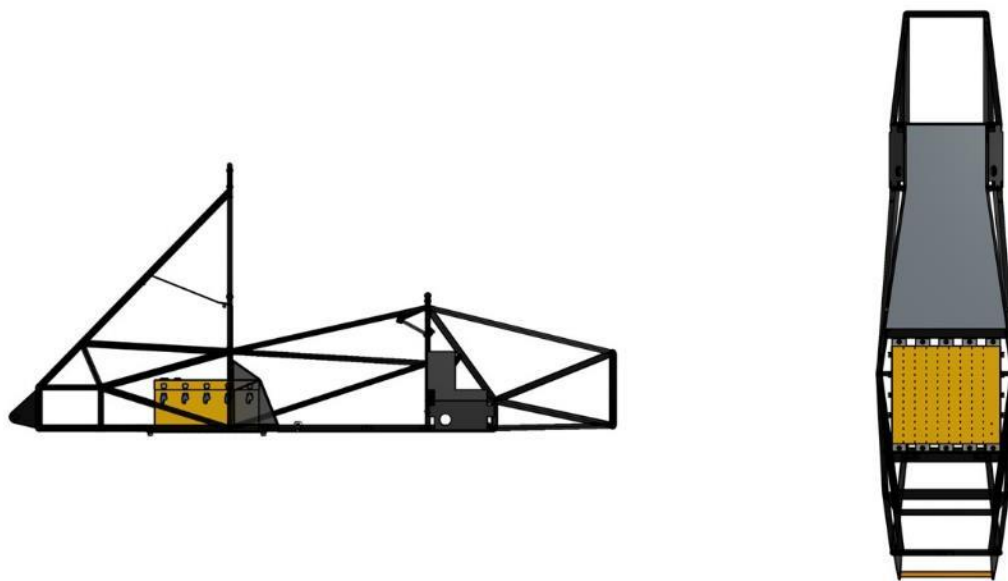


Figure 5 EV.Three's accumulator ,yellow, and chassis from the side, left, and bottom, right.

This meant that prior to committing to the new chassis for NU23, solutions were explored that would still fit within the footprint of the EV.Three accumulator. The first of these solutions was relocating the Orion Battery Management System (BMS) from the lower section of the accumulator to the top plate. When the EV.Three accumulator was designed, it was believed the BMS would require cooling, particularly when balancing cells during charging. Accordingly, the BMS was located in the lower section of the accumulator, with 3D printed manifolds supplying air to it from the cooling fans just like the cells, as seen in Figure 6.

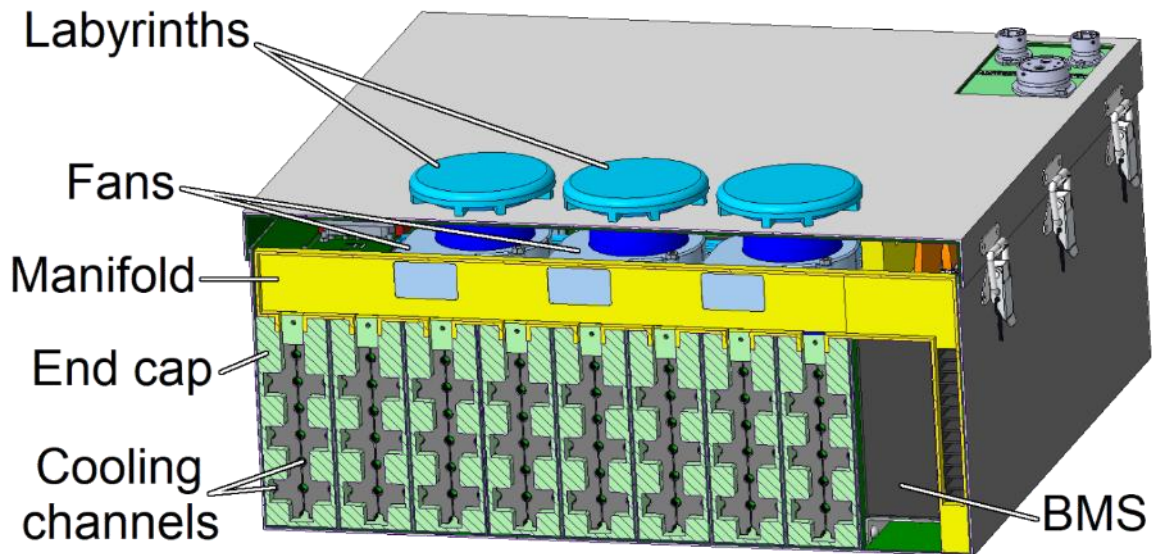


Figure 6 Cooling channels on EV.Three's accumulator (Barrell, 2020)

However, with the benefit of practical testing and charging in 2022, it was deemed that cooling was not necessary, as no significant temperature increase was found in the BMS at any time. With the BMS removed from the lower section, the segments could expand in width from 41.7 mm to 49.6 mm, shown in Figure 7, to consume the space vacated by the BMS.

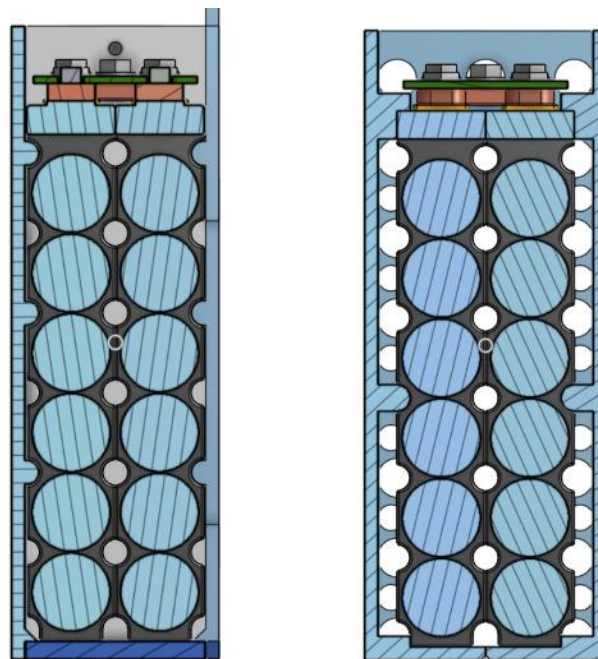


Figure 7 EV.Three segment design, left, and wider design, right, with BMS removed.

It was theorised that this increased width would reduce the restriction on air flow through the system. By expanding the segments, air could now flow along the majority of the side of the segment, rather than just the holes in between individual 18650 cells, like in EV.Three's segments.

Another early investigation was whether the fans used in EV.Three could be upgraded to increase the airflow. The San Ace B97 used was a blower style 12 V DC fan. However, upon looking through available blower fans in similar dimensions and supply Voltage, it became clear that the fans already utilised were the highest performing in their class. In fact, even much larger, more expensive centrifugal fans, to the point of being infeasibly large, could not match the static pressure of the San Ace B97, as seen in Table 1.

Table 1 Comparison of available high performing centrifugal fans.

Name	Pressure (Pa)	Airflow (L/m)	Current (A)	Dimensions (mm)	Price
San Ace B97	1950	1850	6.2	97 x 97 x 33	\$89.75 (RsPro)
Wakefield Thermal DB097	721	1291	1.9	97 x 97 x 33	\$37.79 (Element 14)
Orion Fans ODB17567	268	9571	4.2	175 x 175 x 67	\$155.46 (Element 14)

However, early into these initial investigations, NU Racing leadership were informed that EV.Three's chassis would not be compliant in 2023. This confirmed the chassis overhaul, and resulted in the scope significantly increasing, due to no longer being constrained to the footprint of the EV.Three accumulator. Given the lack of cooling performance in EV.Three, NU23's accumulator cooling was considered as a key variable when designing the new chassis, with the footprint of the new accumulator driving key dimensions in the new chassis.

Early design work completed in collaboration with Jacob Searle, suggested the accumulator footprint should increase. The increases were formulated such that the final spacing within the chassis would be enough for a range of possible cooling solutions. The largest of these possible solutions considered, was multiple axial fans on each side of the accumulator, acting as intake and exhaust fans. To accommodate this arrangement, the accumulator had to be rotated 90° to prevent violating rules. The rules state that any hole in the accumulator container cannot have line of sight to the driver (SAE International, 2023). Without rotating the accumulator 90°, intake holes for axial fans would have line of site to the driver, thus violating these rules.

Just this rotation of the accumulator already changed the dimensions, however they would be further increased with the inclusion of axial fans. As stated, dimensioning was formulated with the largest possible solution in mind. Therefore, the fans considered were 120 mm fans with 38 mm depth. The 120 mm form factor was chosen as it could fit two segments each, with a good increase in segment

width to allow for added airflow. The 38mm depth is the deepest widely available fan in 120 mm size, with the wider the fan generally providing higher static pressure.

These generous estimations were made without committing to this exact cooling set-up. It was the largest of several possible solutions, detailed in the “Flow analysis through pressure testing” section of this report. Time constraints relating to the design and manufacture of the chassis did not allow for a design to be finalised prior to the chassis beginning manufacture. Therefore, it was considered optimal to design around the largest of these configurations, so if a smaller solution was found to be better, it would still fit within the footprint provided.

Axial Exhaust Fan Test

Early in the design phase, it was theorised that the cooling system found in EV.Three suffered from limited airflow through the actual segment. Anecdotal evidence, found by “feeling” the air exiting the cooling holes in the EV.Three accumulator container, had suggested little to no airflow was reaching the end of the segments. A test was designed to try and increase the airflow through the EV.Three segments and measure the temperature difference in the cells.

A two-part fan manifold was designed to achieve this. Due to the nature of the accumulator mounting, very limited clearance was available to add additional parts to the walls of the container. To achieve this, a thin, mounting plate was epoxied to the container wall. Seen on the left in Figure 8, this thin plate was small enough to fit in the gap between chassis and accumulator during the installation of the accumulator.

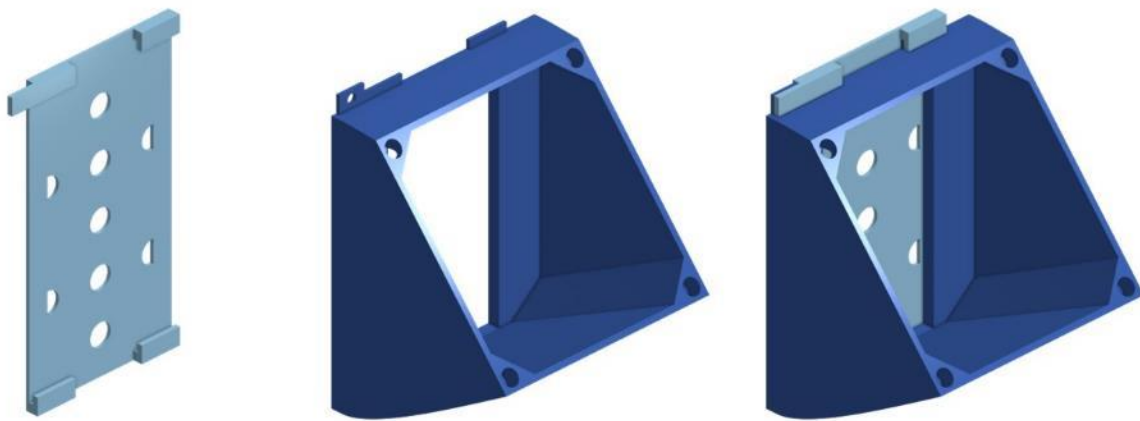


Figure 8 3D printed parts used for exhaust fan testing, left and centre, and their assembled state, right.

The larger part of the manifold, shown in the centre of Figure 8, utilised heat-set threaded inserts to secure a 80 mm axial fan. With the plate epoxied to the accumulator, and the accumulator mounted in place, the manifold could then be slid into the plate, indexing the ribs of the manifold into the slots on the plate as shown on the right of Figure 8.

Shown in Figure 9, the result was just barely able to clear the chain guard. This was ultimately the reason for only one segment being cooled in such a way, as clearance prevented any additional manifolds.

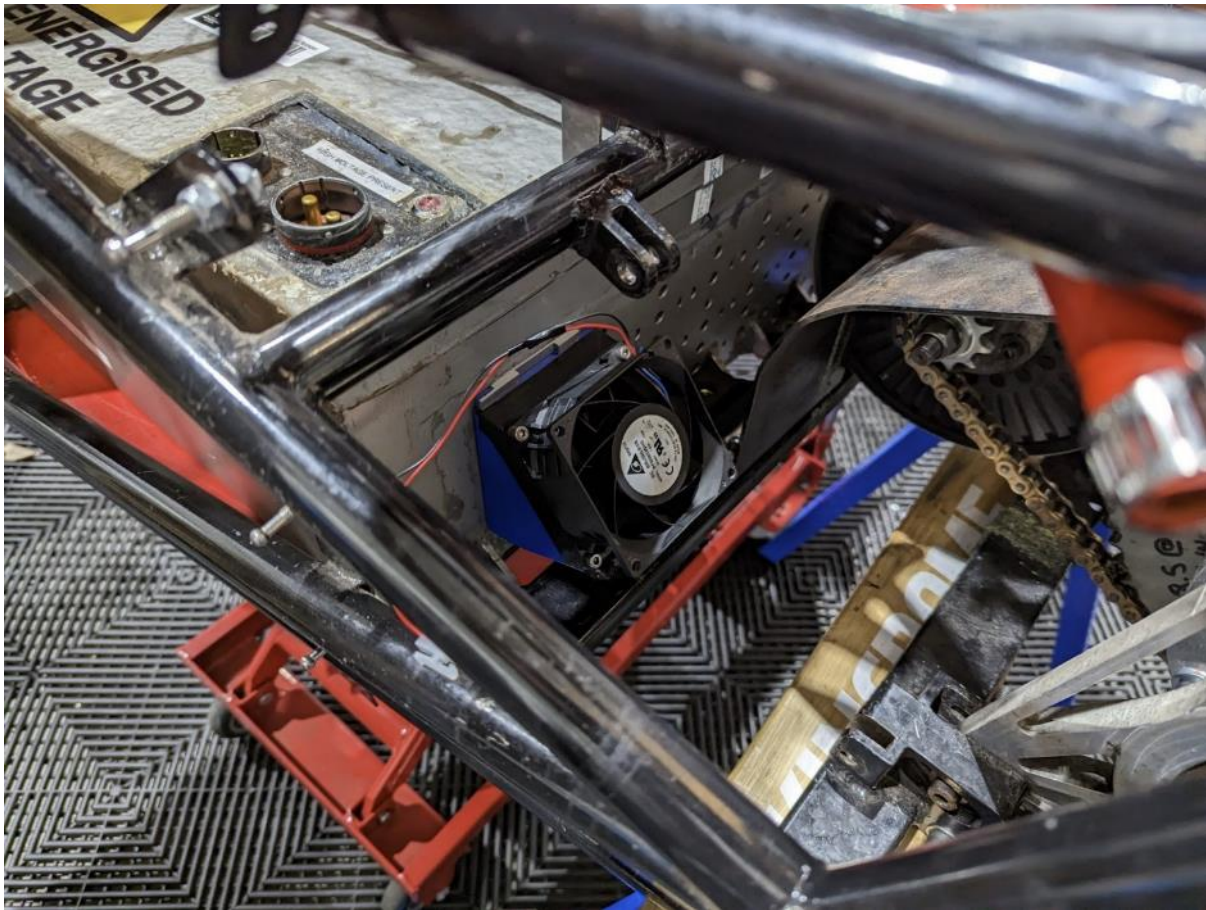


Figure 9 80 mm exhaust fan placed on the EV.Three accumulator.

Collaboration from other team members was required for this project. Nicholas Lyall provided power and ground for the 80 mm fan, while Robert Stanley created an additional Arduino board to measure all Energus modules' temperatures in the segment with the axial fan. EV.Three was only capable of reporting the single highest Energus module in a segment back to Motec. This additional board would provide more insight into the temperature distribution of the cell, and the affect the axial fan was having on the temperatures. This board would store the temperature values onto an SD card, that could be removed and exported in between test runs.

Unfortunately, this board developed undiagnosable issues during track testing. Despite successful bench top testing, approximately only 10 seconds of data would be recorded before the board stopped writing data to the SD card whilst on track. This resulted in essentially unusable data being recorded onto the SD card, as the 10 seconds being recorded would be at the start of the run, without capturing the accumulator under true operating temperatures. However, Motec data was still recorded.

The Motec data seen in Figure 10 shows the maximum temperature in each segment during the testing. The segment with the exhaust fan attached, segment eight, had a significantly lower maximum temperature than all other segments, generally sitting at least five degrees lower.

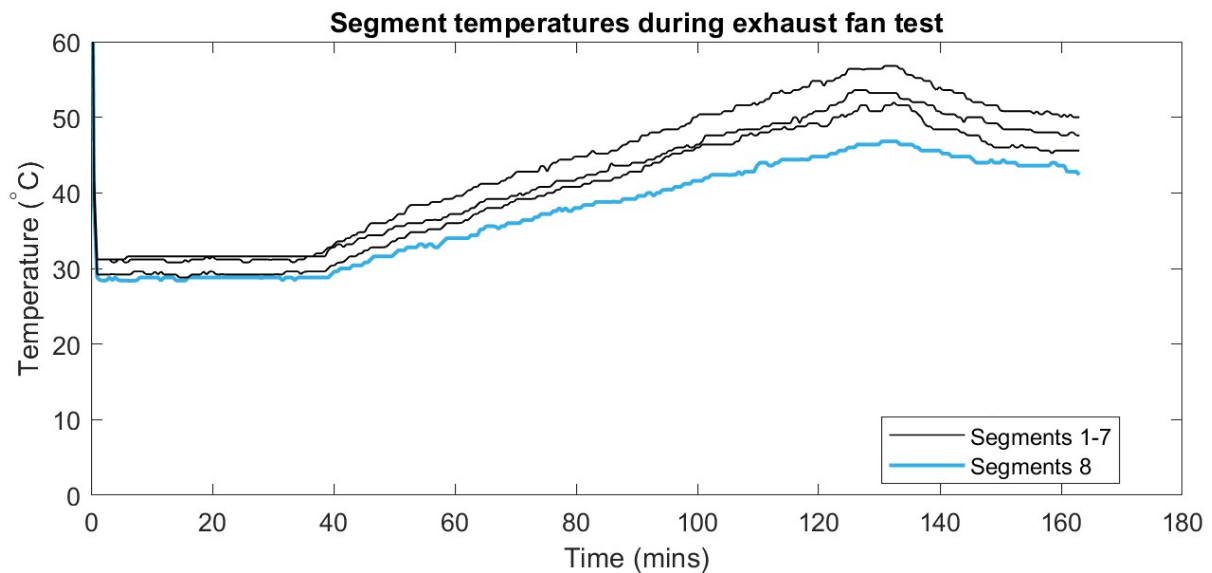


Figure 10 Temperatures from each segment during testing with exhaust fan on segment 8.

Initially, it was thought that this indicated the exhaust fan was improving the cooling on the segment, thus lowering its temperature. However, two key features pointed to other factors affecting the temperature differential. Firstly, on other test days, the same temperature differential could be seen. Segment eight was consistently lower than all other segments previously, even without the added exhaust fan. Secondly, even while the car was at rest with the cells at ambient temperature, as seen at the start of Figure 10, the segment was still measurably lower than all other segments.

This suggests that there is some other variable, causing it to read substantially lower temperatures than all other segments. This could be the placement of the segment, being on the edge of the accumulator, or some other hardware or software related issue. Regardless of what the variable was affecting the temperature in that segment, it appeared to not be the exhaust fan.

Therefore, this testing was considered inconclusive. While the segment appeared to not be any cooler than usual, the potential issue in the temperature reporting could negate any actual results being seen. The test was ultimately abandoned at this stage, as EV.Three was nearing its decommissioning stage in preparation of the transition to the NU23 platform.

Flow Analysis Through Pressure Testing

To aid design selection, objective comparisons between different proposed cooling configurations for the accumulator was sought. An experiment to compare the flow through a mock segment was developed through measuring pressures at various points along the segment with different cooling configurations set up.

The wider segment design mentioned in the “Scoping” section of this report was 3D printed. However, unlike an actual accumulator segment, the design was modified to feature a level of modularity in the intake and exhaust area. This allowed for different manifolds to be attached that could accommodate both the blower used in EV.Three, as well as axial fans, both as intakes and/or exhausts.

The axial fans selected were the Orion 12HBVXE. These fans matched the 120 x 38 mm form factor discussed in the “Scoping” section of this report. These fans were chosen as they demonstrated very strong specifications compared to other fans of the same dimensions. While more reputable brands such as San Ace had marginally better specs, they were not in-stock within Australia, and therefore the Orion fans were chosen as they provided a shorter turn-around for testing. The specifications of the axial Orion fans are compared in Table 2 to the EV.Three blower fans, the San Ace B97. The full data sheets for both fans can be found in the appendices.

Table 2 Comparison between Orion 12HBVXE axial fan vs San Ace B97 Blower fan.

	Orion 12HBVXE	San Ace B97
Air Flow (CFM)	235.5	65.3
Static Pressure (“H₂O)	1.6	7.83
Current Rating (A)	3.8	6.2
Frame	PBT, UL94 V-0 Plastic (Impeller the same rating)	UL94 V-0 Plastic

It can be seen that the axial fans provide much higher air flow rates, at the expense of static pressure. It was theorised this trade-off would be viable, considering the much less restrictive path of air in an axial fan setup, without the need for a complicated intake manifold like in EV.Three’s blower set-up. However, there is a negative relationship between air flow and static pressure. As the pressure decreases, airflow increases. This relationship can be seen for each fan in Figure 11.

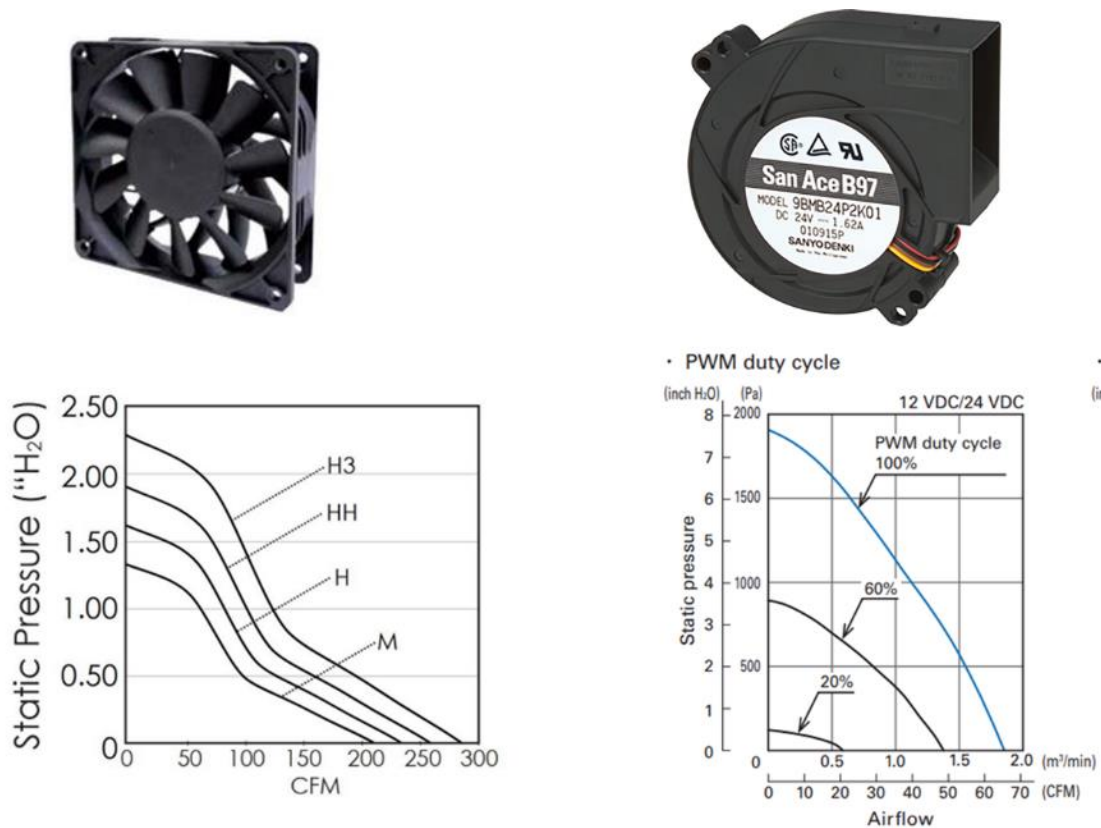


Figure 11 Airflow characteristics of Orion 12HBVXE (left) and San Ace B97 (right)

While there is theoretical method for determining the point on these graphs that the fans will operate, this requires a value for the impedance of the system. This formula is shown in Equation 1. To determine the impedance of the system, expensive equipment and chambers are required that were not accessible. It was therefore determined the best way to identify the best fan was via pressure testing each fan.

Equation 1 The pressure, P as a function of Impedance, Z , and the airflow, Q .

$$P = Z \times Q^2$$

There were five cooling configurations tested in total. The first of which was a replica of the EV. Three cooling configuration. This utilised a single blower fan positioned above the segment as an intake, as shown in Figure 12. This contained similar characteristics to that of EV. Three in terms of a blower fan intake, and the airflow path, with essentially two perpendicular bends before travelling along the segment.

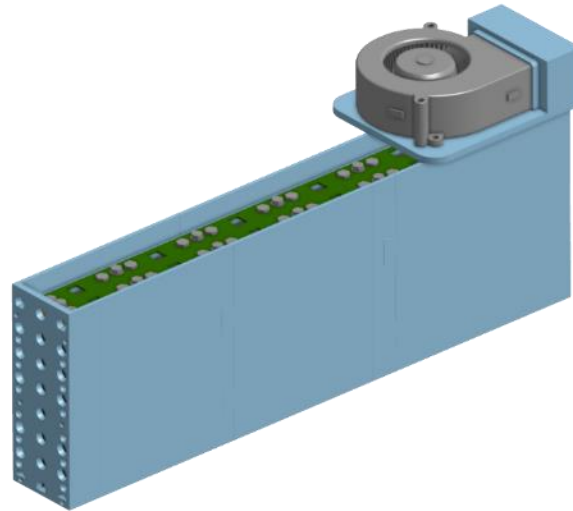


Figure 12 EV.Three Cooling set up with a single blower fan.

However, as previously seen in Figure 6 there were some key differences. The inclusion of the labyrinths at the intake of the EV.Three accumulator essentially added another perpendicular bend to the airflow through the system. It can also be seen that in the testing set up, a single fan serviced one segment, while in the EV.Three accumulator, the three blower fans were distributed via a manifold to eight segments as well as the BMS.

The second cooling configuration tested utilised the same blower intake and manifold, however with the addition of an axial fan acting as an exhaust at the other end. This set up is pictured in Figure 13.

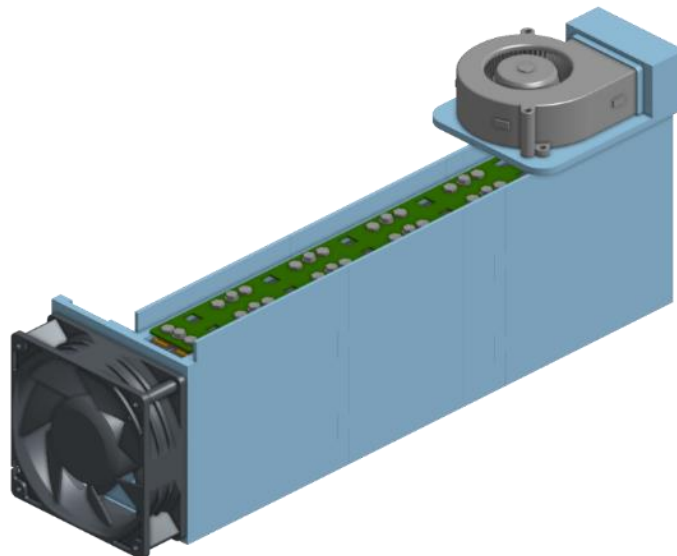


Figure 13 Blower intake with axial exhaust test assembly.

The next two setups utilised a single axial fan at one end of the segment, with no restrictions at the other end of the manifold, shown in Figure 14. This was used for testing the performance of a single axial fan configured as an intake, or as an exhaust. In order to test both, the fan was simply flipped to

change the direction of the airflow. Given that the pressure test points were symmetric along the vertical axis, there would be no effect on the pressure readings.

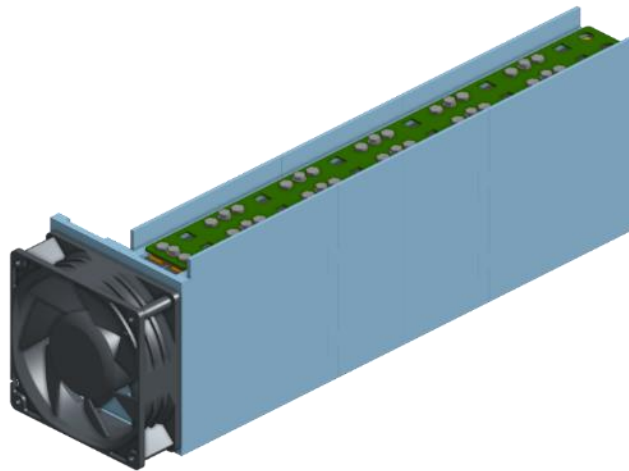


Figure 14 Single axial fan set up for intake and exhaust testing.

The final set up tested was a dual axial set up. This utilised two mounting plates at either end of the segment, with an axial fan on each forcing air the same direction through the segment as an intake and exhaust set up. This is shown in Figure 15.

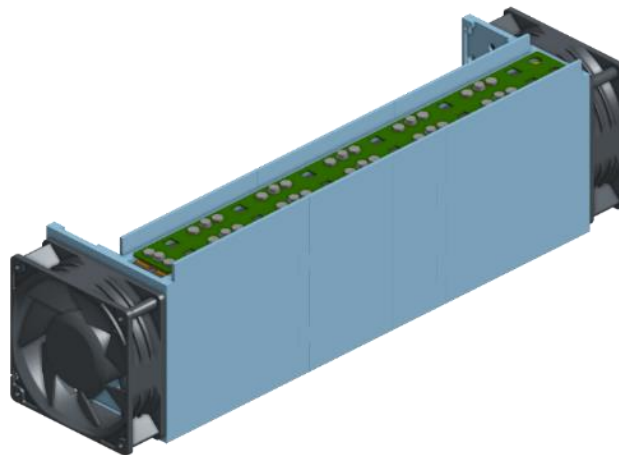


Figure 15 Axial intake and exhaust test set up.

Various assumptions were made to simplify the physical models used for pressure testing. Firstly, rather than using an entire accumulator container, the different cooling configurations would be tested on a single segment.

Additionally, rather than using the real Energus modules in the testing, 3D printed PLA replicas were placed in the mock segment. The design of the system required small tubes to be placed at various points along the segment to measure the pressures at each point. Rather than drilling holes into EV.Three's segment housings to insert these tubes, mock housings were also 3D printed out of PLA

that matched the FR4 housings. In both cases, the surface effects on the air flow by using plastic rather than the true housings and Energus modules were assumed to be negligible. Additionally, as this variable remained consistent across all tested configurations, its affects were further negated.

A barometer was used to probe each tube one by one with the fans operating at steady state. These pressures were recorded, with multiple rounds of recording done to ensure the results were reliable. The complete set up is shown in Figure 16. Not Pictured is the benchtop power supplies used to power the fans. They were configured to deliver 12.5 V and ensure they were voltage limited, to allow each fan draw as much current as required. This attempted to provide consistent results, despite the two different types of fan being tested pulling significantly different amounts of current.

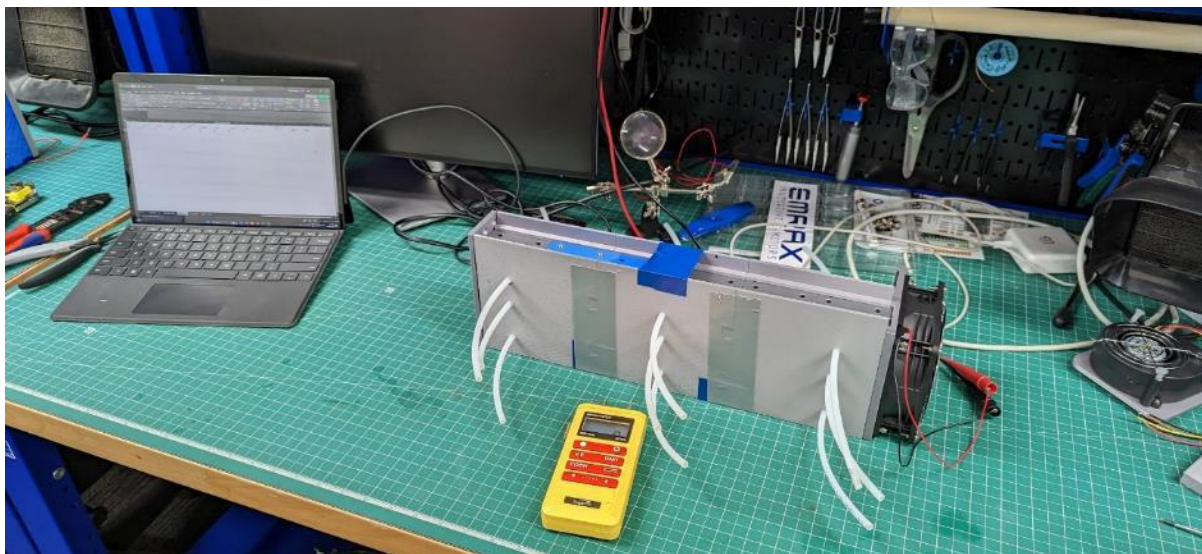


Figure 16 Test segment with pressure measuring.

Each configuration was tested multiple times for the pressures at each point. The tests were completed over multiple days, with each point's values averaged in order to increase the reliability of the results.

It is worth noting the naming convention used for each pressure testing point during the discussion of the results. Figure 17 details how each pressure point is named and discussed throughout the remainder of this section.

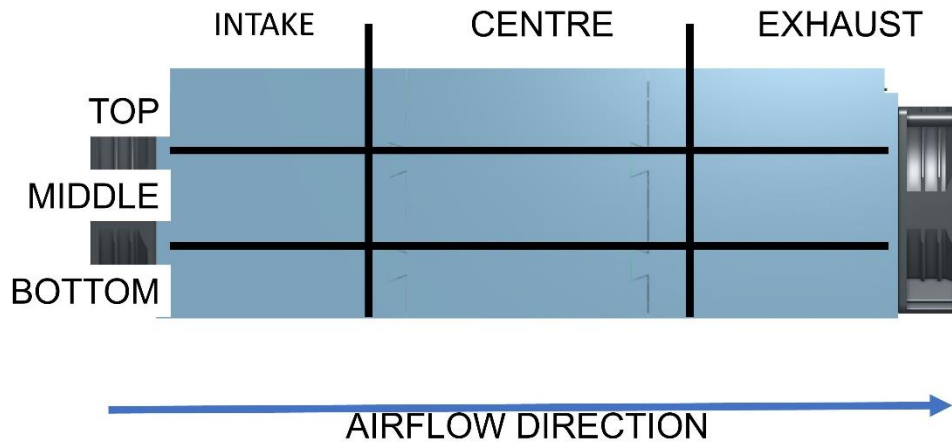


Figure 17 Naming convention for pressure test points in pressure testing.

With testing completed, the results are shown in Figure 18.

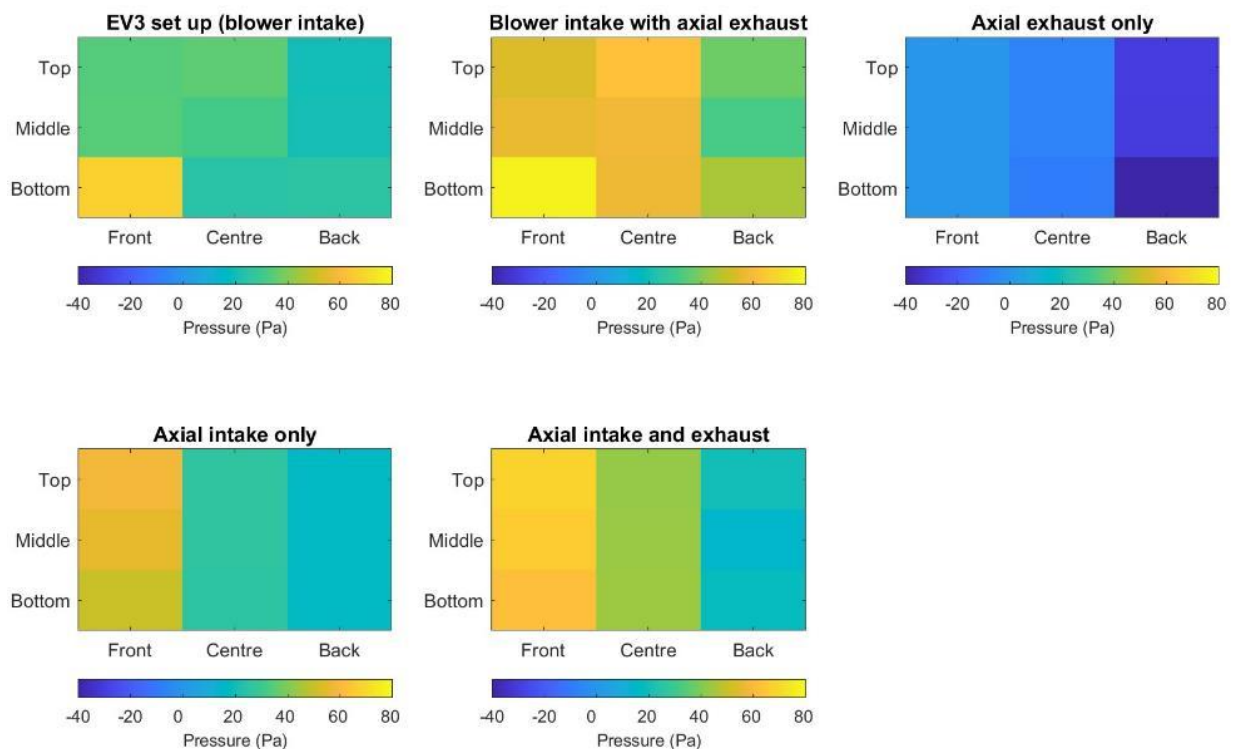


Figure 18 Pressure testing results.

As can be seen in Figure 18, the pressure distribution of the EV.Three cooling configuration shows some major fluctuations between the top middle and bottom. These fluctuations are apparent at the front/intake end of the segment and become somewhat uniform by the exhaust end. This could indicate that the air entering the top of the segment goes straight to the bottom, as the pressure is much higher at the bottom of the segment in comparison to the top when looking at the intake end. When looking at the readings in the centre, it can be seen almost the converse occurs, where the top of the segment is the highest-pressure area, and the bottom of the segment becomes the lowest

pressure. This results in the intake being lower pressure than the middle of the segment when looking at the top measurements. This indicates highly undesirable, turbulent flow in the front half of the segment.

The reason that this flow is considered turbulent, or categorised as non-ideal, is through inspection of the Bernoulli Equation, shown in Equation 2. Assuming incompressible flow, the density of the air becomes a constant, and given there is no elevation change, those associated terms become zero. Therefore, the only variable affecting pressure becomes the velocity of the air.

Equation 2 Bernoulli Equation where P is the pressure, ρ is the density, v is the velocity, h is the elevation and g is acceleration due to gravity. Locations 1 and 2 denoted by subscript.

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

This indicates a large amount of turbulence within the segment in the EV.Three cooling arrangement. While the pressure actually increasing along the top may indicate air is actually flowing from exhaust to intake, this is a large assumption considering the relatively low resolution of the testing. To make this assumption would be implying that the air is only travelling in three, separate streams: top, middle and bottom which would be highly improbable. This assumption could be proven false with the testing re-attempted with a higher number of test points. However, this would likely provide no real benefit, and that even just the testing completed exhibits large amounts of turbulence and would be enough to categorise this as far from ideal flow.

Additionally, if instead of considering point 1 and point 2 in the Bernoulli EquationEquation 2 as intake and exhaust, but rather top and middle points, the flow appears even worse. Doing this calculation finds that there is air moving up and down within the segment, as well as the desired intake to exhaust. By inspection of Equation 2, and given that the elevation changes between top and middle is 18.4 mm, an estimate of the velocity change in the vertical axis can be calculated as 4.3 m/s. This further implies a high level of turbulence in this cooling configuration.

Even with the addition of an axial exhaust fan, also shown in Figure 18, to the EV.Three cooling set up, many of the undesirable traits remain. With the bottom pressure readings being an outlier, the overall pressure gradient does become slightly steeper, with greater pressure drop from intake to exhaust indicating higher velocity flow through the segment. The bottom of the segment does have a slightly lower pressure drop; however it is much more linear, potentially indicating less turbulence in this area.

However, much the same as the EV.Three cooling configuration, the same increase in pressure from intake to the centre of the segment remains for the top of segment, and occurs in the middle of the segment. As previously stated, this is not desirable as it may indicate flow in the opposite direction to

that which is desired. As this pressure increase is now seen in both the top and middle, this makes this scenario more likely. This is unexpected, given that the addition of an exhaust fan was hoped to improve the linearity and steepen the gradient of the pressure distribution from intake to exhaust.

Moving to the configurations using just axial fans, much more linearity in the results shown in Figure 18 can be seen in comparison. Starting with only an axial intake fan, a consistent distribution across the top, middle and bottom of the segment can be seen. This suggests that the axial fan, as expected, is moving the air in only one axis along the segment, unlike the blower fans. The gradient for the most part is linear, however it is clear that the rate of pressure-drop decreases towards the back of the segment, suggesting that the impedance of the system is limiting the airflow created by the fan.

Additionally, it can also be seen that maximum pressure seen in the segment is in fact lower than that seen in both blower fan configurations. A single axial fan achieves a maximum pressure of 58.3 Pa with EV.Three's set up reaching 67 Pa at the bottom and front, and the blower with axial exhaust reaching 76.8 Pa at the bottom front. While a higher pressure would be considered advantageous, if the average of the front three points, being top, middle, and bottom, for each of the configurations was taken, the single axial fan is actually not too far adrift from the blower fan set ups. It is found that EV.Three was 45.2 Pa, a blower and axial exhaust was 61.9 Pa and a single axial fan set up was 54.6 Pa. As stated, the goal is for a linear pressure drop from front to back, in which case a single axial fan does appear advantageous over a single blower despite a lower maximum pressure.

The other single axial fan set up tested was as an exhaust, also shown in Figure 18. There are immediate similarities to the single intake axial set up if the graph were inverted. Again, the rate of pressure drop is much greater the closer the proximity to the fan, once again suggesting that the impedance of the system may be slightly greater than the fan's capabilities. Also, the pressure is very uniform between the top, middle and bottom, just like the axial fan intake set up.

However, the pressure ratings at the intake side of the fan suggest that this may provide little to no airflow over the cells at the intake end of the segment. Rather surprisingly, despite consistent testing, it was found there was 0 Pa gauge pressure at the top, middle and bottom at the intake end. When using the Bernoulli equation, if the Points 1 and 2, denoted in Figure 19 were considered, it would suggest that there would be no velocity in the air between these points, as all other terms would become zero.

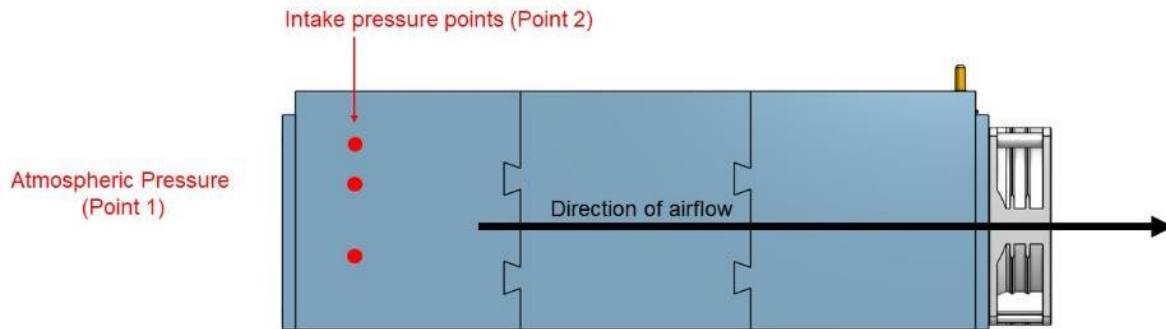


Figure 19 Single axial exhaust fan.

This implies that the air seen travelling through the segment close to the fan must be entering through the top of the segment, between the cells and the segment wall, rather than through the intake as hoped. This is somewhat affirmed when inspecting the pressures at the centre of the segment, as the pressure is slightly higher at the top of the segment, suggest the air is moving from top to bottom ever so slightly. This could be further tested and proved by properly sealing the top of the segment with tape, and once again increasing the resolution of the test points. However, it was deemed this would not be beneficial to the goal of finding the best cooling configuration.

The axial intake and exhaust configuration displayed the best flow characteristics of all the configurations tested. The pressure testing results for this configuration are again found in Figure 18. Of all the configurations tested, the twin axial fan set up gave the most linear pressure gradient from intake to exhaust. This indicates the velocity of the air remained relatively constant throughout the segment, as taking any two points horizontally and inserting their pressures into the Bernoulli equation would yield very similar velocities.

While it didn't reach as high a maximum pressure at 68 Pa as the blower fan with axial exhaust, it did demonstrate the greatest pressure drop from intake pressure points to the exhaust points at an approximate 53 Pa drop. The next highest pressure drop, when considering the absolute maximum pressure relative to the absolute lowest measured in a configuration was in the blower intake only, achieving a 46 Pa pressure drop. This indicates the highest average air velocity across the segment would be found in this configuration.

Similar to the single axial fan, the uniformity between the top, middle and bottom pressure points at each length along the segment is extremely close. The greatest variance between two vertically aligned points is at the intake, between the top and bottom. The difference of 6.7 Pa indicates there is a vertical component to the velocity of the air flow of approximately 3.4 m/s. While this is not substantially lower than the vertical component found at the centre of the blower intake set up, when considered in relation to the horizontal component, the advantages of the twin axial set up become

clear. For the sake of consistency, the vertical component of the velocity in both cooling configurations is found at the centre point of the segment in Table 3.

Table 3 Comparison of vertical and horizontal velocity in EV.Three and NU23's cooling configurations.

Configuration	Average pressure at intake (Pa)	Average pressure at exhaust (Pa)	Average velocity through segment (m/s)	Greatest pressure delta in vertical component (Pa)	Velocity in vertical axis (m/s)	Velocity gradient (rise/run)
EV.Three						
(Single blower intake)	45.2	22.3	6.2	10.7	4.3	0.7
NU23 (axial intake and exhaust)	65.0	17.8	8.9	1.0	1.4	0.2

Table 3 highlights just how much more effective the twin axial setup is at transporting air from the intake of the segment to the exhaust. The air travels horizontally versus vertically at 3.5 times the rate in the twin axial cooling configuration compared to EV.Three's.

This is important not just from an efficiency standpoint. As the fresh air enters the accumulator and begins passing over cells, it begins to heat up. The shortest distance for air to travel to the cells at the exhaust end of the segment is in a straight, horizontal line from intake to exhaust. Vertical components to the velocity causing turbulence only make this path longer. Having to travel further over hot cells would heat the air up more before it reaches the cells at the back of the segment. Inspection of Equation 3 shows that, with the safe assumption that the air remains cooler than the cells, the greater the temperature of the air, the less heat will be removed from the cells.

Equation 3 Heat convection equation, where q is heat flux, h is transfer coefficient, and ΔT is the temperature difference between the surface and the air.

$$q = h\Delta T$$

These desirable traits are what ultimately led to the decision to use two axial fans acting as intake and exhausts to cool the accumulator for NU23. The decision to design the new accumulator, and consequently chassis around this configuration was also aided by the footprint this system required.

Given this configuration demonstrated the largest area required of all the systems tested, this system would provide the most flexibility to change configurations if required, even with the chassis finalised. Due to the critical time constraints in finalising the chassis dimensions, this was considered extremely desirable to ensure the most flexibility in design of the accumulator moving forward.

Whilst this testing was used to justify the cooling configuration, there were improvements that could be made with more time to the testing procedure. The obvious one would be to increase the number of test points to improve the resolution of the testing. This could remove some ambiguity discussed regarding the possible large vertical components of the air velocity seen in the blower cooling configurations.

Another improvement that would be extremely beneficial would be having test points as close as possible to the discharge side of the intake fan. By doing so, the operating pressure of the fan could potentially be measured providing insights into the restriction seen by the cooling system. By getting a more representative operating pressure, the operating point of the fan on a pressure versus airflow graph could be more accurately pinpointed. This would give better insights into how restricted a fan being tested is, and provide some idea of the airflow.

Furthermore, designing a method whereby air velocity could be measured could validate the calculated velocities using Bernoulli's Equation. A simple exhaust pipe of sorts with a known cross-section area, with an anemometer placed within it could not only validate these velocities, but also give an indication of the volumetric and mass flow rate of the cooling configuration. Both values are crucial for determining the configuration's capacity to remove heat from the system, and therefore would undoubtedly further improve the validity of the testing.

Segment Discharge to Test Cooling Configurations.

Prior to finalising the chassis design, a discharge circuit was designed capable of discharging one segment. By discharging a segment, temperature could be generated in the cells equivalent to that of running the car for an event. Temperatures would be monitored across the cells, therefore being able to directly see the differences in the cooling configurations, and thus validate the expected cooling performance from the pressure testing.

In order to design this bench test, a desired current draw first had to be decided. In order to simplify the bench test, a constant current draw was chosen rather than variable current. Given that endurance was the most temperature limiting event at competition in 2022, it was theorised that using the average current draw across the endurance run would provide a suitable value for discharging the segment at a continuous rate. This value was calculated by importing Motec data from the 2022 endurance event into MatLab to find a value of 14 A average draw across the event. This data is visualised in Figure 20.

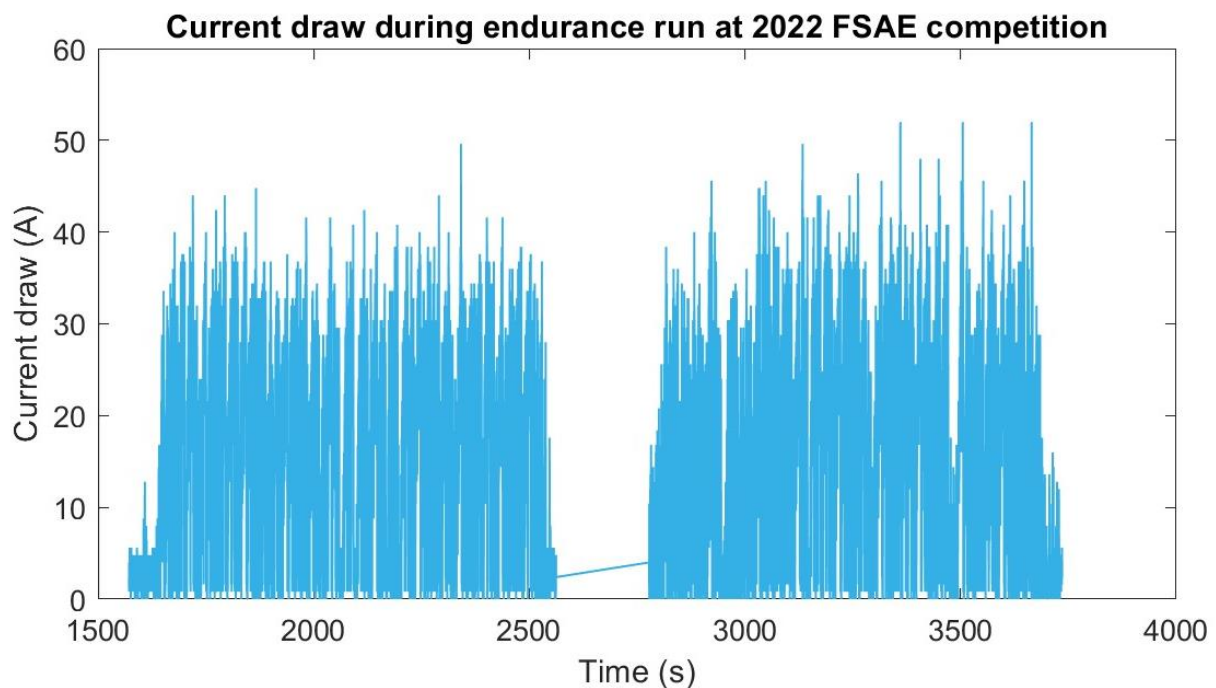


Figure 20 Current draw during FSAE 2022 Competition Endurance for EV.Three.

Using the value of 14 A discovered from MatLab, a suitable discharge bench had to be designed. Given that only one segment was to be discharged, a nominal voltage of 50 V was used in order to calculate the required resistance to draw the desired current. This was done simply using Equation 4.

Equation 4 Resistance required to draw 14 A from 50 V segment.

$$R = \frac{V}{I} = \frac{50 \text{ V}}{14 \text{ A}} = 3.57 \Omega$$

Additionally, considerations had to be made for the power dissipation. At this voltage and current draw, the power to be dissipated was calculated in Equation 5.

Equation 5 Power to be dissipated in segment discharge at 14 A.

$$P = V \times I = 50\text{ V} \times 14\text{ A} = 700\text{ W}$$

With these requirements, a suitable system to safely discharge the cells could be designed. Options considered included heating elements in water, an array of spotlights and even large workshop dischargers. However, simple power resistors were found to meet the required specifications and were considered a safer option. The TE Connectivity HSSC25010RJ was used, capable of dissipating 250 W, a maximum of 2.2 kV and a 10 Ω resistance (Premier Farnell Limited, 2023). Placing three of these in parallel created a total resistance of 3.3 Ω , extremely close to the target of 3.57 Ω . Additionally, three of these resistors provided enough headroom for the power dissipation required.

The circuit was then constructed, with two AIR's connected to each terminal of the segment in order to safely open and close the circuit. Three additional cooling fans were used to lower the temperature of the power resistors, and two multi-meters capable of measuring temperature used to monitor the power resistors and the cells. Two additional multi-meters were used to measure the voltage of the segment and the current. A jumper cable developed by Robert Stanley was used to take temperature values being reported by the cells and feed them into a live serial on Arduino. This complete set up can be seen in Figure 21.



Figure 21 Bench testing a segment for cooling performance.

The first discharge was completed with no fans connected to the segment. The goal of this was to test the discharge setup in the simplest form, but also set a baseline temperature for the current being drawn. However, once the circuit was closed, and discharging began, it was found even with an absence of active cooling on the cells, the rise in temperature over ambient was negligible, despite the circuit being connected for close to 20 minutes.

To try and counteract this, an additional resistor was added in parallel, identical to the other three. This lowered the total resistance from 3.3Ω to 2.5Ω , and consequently raised the current draw from 15.2 A to 20 A. However, similar to the first test, even without any active cooling on the cells, the temperature only increased by 2°C after a 20 minute period.

This led to the conclusion that the average current draw was an inaccurate model to replicate the temperature generation within the cells during actual driving. This likely points to the peak current draws, as were seen in Figure 20, causing the most heat to be generated, where currents reached upwards of 40 A.

Due to time constraints, the discharge testing had to be abandoned at this point, given that this discovery induced significant complexity to the problem. With no more of the 10Ω resistors available to lower the resistance further, it was deemed there was no punctual, safe method to increase the current draw on the segment.

Even with more time, it is debatable whether a constant current draw method would be an appropriate form of testing cooling performance. If the test were to be re-attempted, a method where current draw could be modulated to more resemble the current draw EV. Three placed on the cells, , would be an ideal method to test the cooling performance of different set-ups. This poses many challenges, however such a test apparatus would prove invaluable for the team in multiple future designs for the accumulator cooling.

Segment Design

Eight Segment Accumulator

The function of a segment housing was to contain battery cells and separate them physically and electrically from other segments and systems within the accumulator. This function was stipulated by the FSAE rules, with various requirements relating to maximum energy capacity of the cells each housing contained and the electrically insulating properties of the housing. Due to their requirement to be an electrical insulator within the accumulator, this entailed additional rules, relating the temperature ratings and being self-extinguishing and non-drip (UL94-V0). Furthermore, the accumulator was required to be able to withstand substantial loading, 40 g lateral and 20 g vertical. These requirements, along with the goals of serviceability, reduced weight, and low part count made for a difficult design problem.

NU Racing's design goals at the start of year meant no change in capacity of the accumulator in terms of Voltage, maximum current or energy storage capacity. Considering the restrictions in the rules regarding energy limits per segment, the overall serviceability of the accumulator and its familiar topology in EV.Three, the decision was made to keep many aspects of EV.Three's segment design.

EV.Three utilised a 5-part design for the segment housing, with all 8 housings being identical. Shown in Figure 22, all bar one of the 6 sides of their rectangular prism shape were housed. The base, both longer walls, and one smaller end wall were made from FR4 Glass Epoxy, while the other end wall was made from 3D printed flame-retardant ABS filament.



Figure 22 EV.Three Segment, showing the ABS intake, left, and FR4 housing end, right (Turner, 2022).

This design provided a reliable and strong housing, that presented no major failures throughout the 2022 team's campaign, and through the start of 2023. Given this proven platform, the decision was made to adapt the design for the goals of the 2023 team.

Foremost of the changes to be made, was the adaption to the new cooling configuration. Discussed in the "Flow Analysis Through Pressure Testing" section of this report, the move to a dual axial fan cooling arrangement from the single blower intake fan of EV.Three meant the 3D printed end piece in EV.Three was made redundant. This was a simple modification to make, as it meant the two smaller end walls could be identical in design. It also meant they too could made from FR4, as they were both now a simple, essentially 2D shape.

Another big change to the design from EV.Three was the increase in segment width. While for the FR4 housing itself, this only affected the dimensions of the base, big changes were required for the 3D printed "ridges" along the segment that index on the Energus modules, seen in Figure 23.

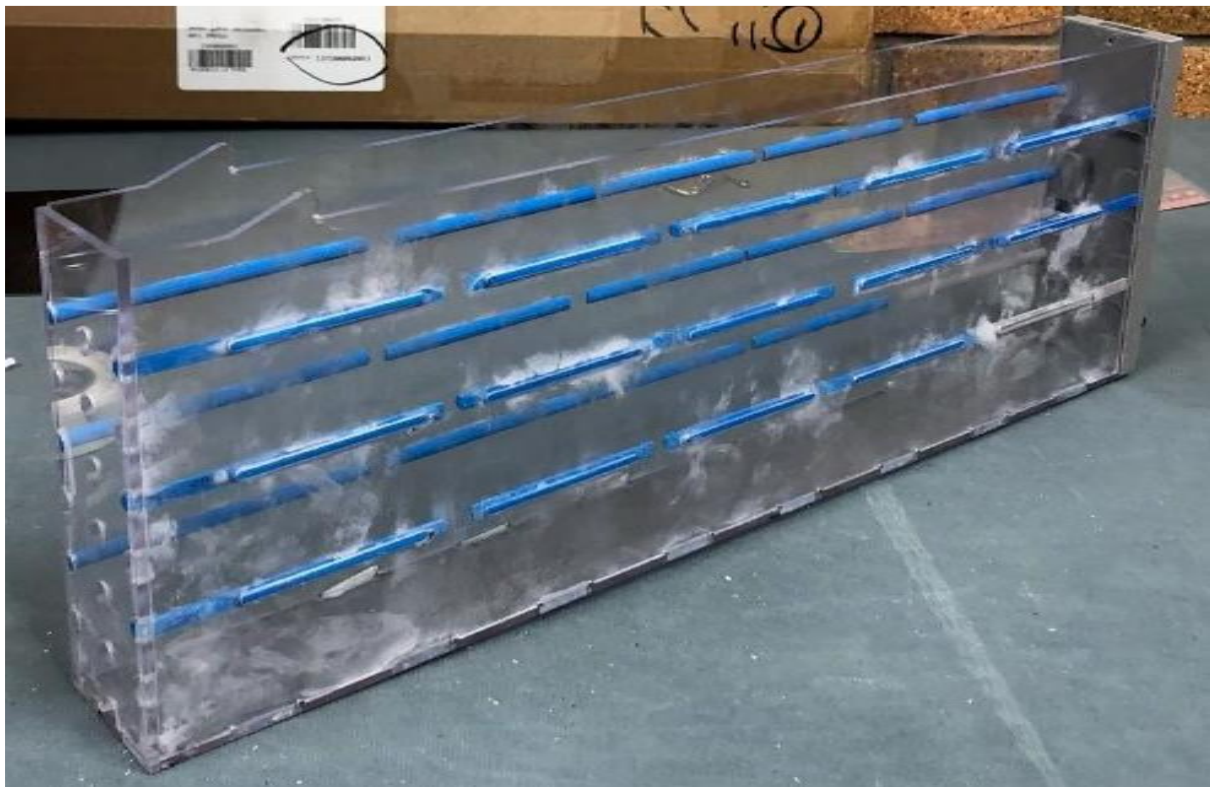


Figure 23 Prototype EV.Three housing showing the "ridges" used (Turner, 2022).

In the case of EV.Three, the ridges acted to secure the cells within the segment. However, due to the Energus modules also being flush against the segment wall, they provided little to know support in the lateral direction for the cells. Additionally, the Energus modules rested on the floor of the segment with the top of the modules compressed by the top plate via 3D printed positive locking devices. This meant the ridges provided little support in the vertical axis as well.

In the case of NU23's segments, while the vertical loading case was to be kept the same, the lateral case was not. Due to the increase in width of the segments, the sides of the Energus modules would not contact the walls of the segment housings. This meant that the entirety of the lateral loading would be placed on the ridges. In the interest of maximising the volume of air in the segment to promote airflow, reducing the number of ridges was considered, provided that the loading would not be excessive.

Preliminary calculations were made to determine the area needed to support the load required by the FSAE rules. This was due to wanting to not populate all the semi-cylinder cut outs on the Energus modules, to try and further improve the airflow. The equation used can be seen in Equation 6.

Equation 6 Stress equation for housing ridges, where force is the mass multiplied by 40, as per FSAE rules.

$$\sigma = \frac{F}{A}, F = \text{Segment mass} \times 40, A = \text{Area supported by ridges}$$

The yield strength of the ABS filament to be used was listed as 45 MPa, found on its data sheet in Appendix 6 Appendix 6: Flame Retardant ABS Filament. However, the yield strength of 3D print is variable based on print orientation, layer height, infill percentage and type and general irregularities due to nozzle calibration and temperature fluctuations during printing.

Major assumptions and simplifications were made for these calculations. The cross-sectional area that was used for the stress calculations was generated by the diameter of the semi-cylinder slots in the Energus modules shown in Figure 24, multiplied by the length of the segment.

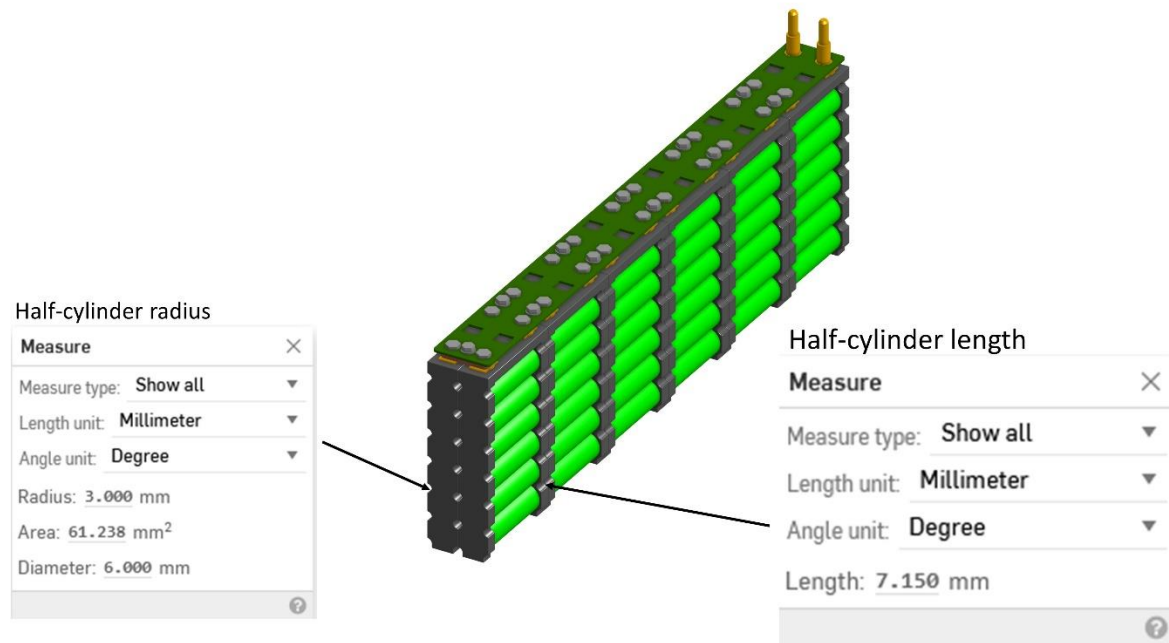


Figure 24 Critical dimensions for calculating the area of the ridges.

With these two measurements, the maximum cross-sectional area per ridge was calculated, shown in Equation 7, finding the area of each ridge to be 514.8 mm².

Equation 7 Area per ridge in accumulator segment.

$$A = \text{Diameter} \times \text{Length} = 6 \text{ mm} \times 7.15 \text{ mm} \times 12 = 514.8 \text{ mm}^2$$

Calculations were then made using Equation 6 and Equation 7 to find the stress in the ridges, dependent on the number of ridges used, with a weight of 2.95 kg per segment. The results of these calculations are shown in Figure 25. As can be seen, the calculated normal stress, even with just one ridge equates to only 2.24 MPa. This is significantly lower than the 45 MPa yield strength claimed by the filament manufacturer.

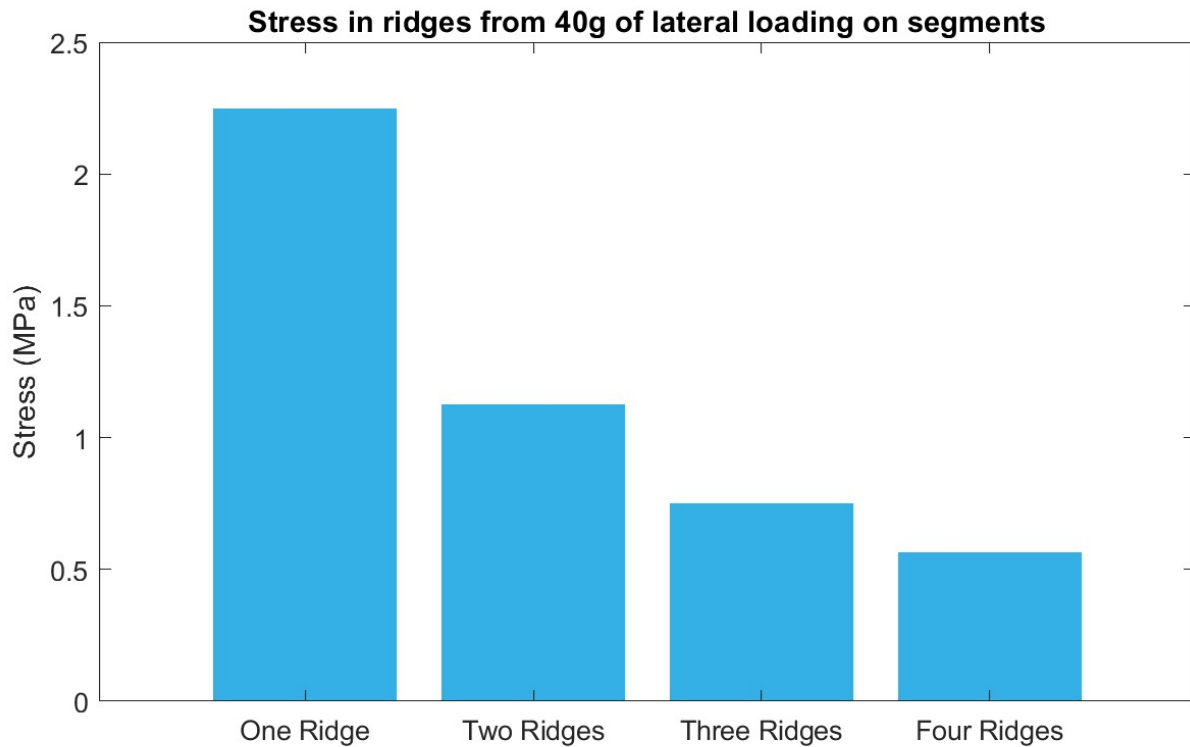


Figure 25 Stress in ridges from 40g of lateral loading on segments.

This was clearly a purely normal force being applied in a lateral loading case, manifested as an acceleration or deceleration front to back of the car. This did not account for any bending that may occur if there were misalignment in the ridges, or if there were in fact vertical movement of the cells. If this loading occurred, increased stress would be placed on the segment ridges. However, due to the extremely low magnitude of the calculated stress, this risk was considered minimal.

It is worth noting that the lateral load would transfer from the 3D printed ridges into the FR4 housing where the two index. However, due to the much greater yield strength of FR4 given by Appendix 5: FR4 Datasheet, the 3D print would fail well prior to the FR4 housing, as the area where the ridge indexed on the FR4 was greater than the diameter of the ridges.

Consequently, it was considered a design incorporating these ridges would likely be sufficient to withstand the loading requirements stipulated by FSAE and that development should proceed. It was however, considered that these calculations may not satisfy the judges requirement for the team to prove the strength of the segments. Therefore, a loading test was anticipated to be completed by the team, however this occurred on the final design of the segments and attached in Appendix 7.

During the initial design phase, there were discussions about how to improve the removal of the segments. EV.Three utilised tape that lapped around both walls and the base, extending above the segment to create tabs that could be gripped to remove the segments. This tape can be seen earlier

in Figure 22. While for the most part this system worked, issues with the tape tearing did lead to more challenging removal of the segments. This was a problem that could not ultimately be solved in what was considered a reliable and serviceable way. A variety of factors, most notably the extremely low volume of space that the solution could take as the segments were essentially entirely filled with cells, and the top plate resting directly on top of them, led to the decision to abandon this development. It was therefore decided that the same method as EV.Three would be utilised, and additional clearances would be added between the segment housing and internal walls.

The endplates of the segments in the eight-segment accumulator were designed to be manufactured via 3D printing. Due to the necessity of these pieces to conform to the insulation rules in the accumulator, they would have to be manufactured from the same flame-retardant ABS as the ridges. While exact thickness of these endplates was undecided, it was planned that test fitting with different thicknesses would be completed with PETG samples, before finalising the ABS pieces. However, due to the eventual overhaul of the accumulator to add the ninth segment, these pieces were never required. The finalised design can be seen in Figure 26.

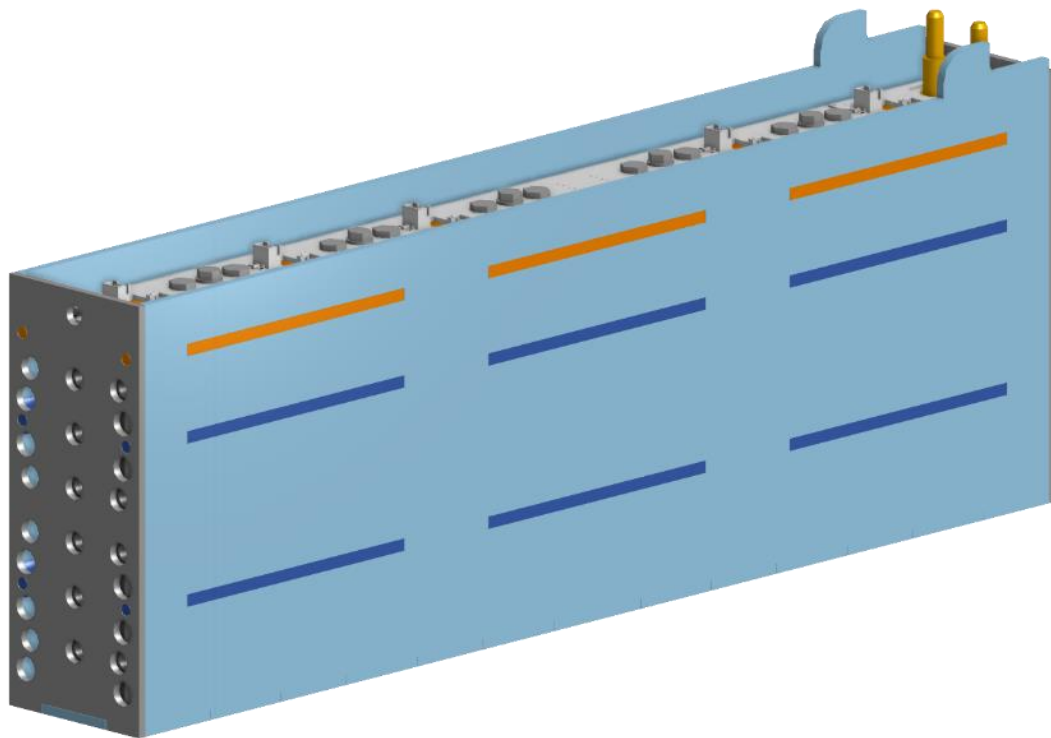


Figure 26 Finalised eight-segment accumulator segment design.

Nine Segment Design

Shortly after successful commissioning of NU23 with the eight-segment accumulator, discussions by the leadership team began regarding potentially increasing the capacity of the accumulator by adding a ninth segment. The addition of a ninth segment would increase the maximum voltage of the accumulator from approximately 400 V to 450 V.

While a more thorough discussion of the justification and benefits in terms of vehicle performance can be found in Josh Dawson's report, there were multiple ramifications for the accumulator and its cooling. Foremost, given the chassis was finalised and would not be remade or modified, the new nine-segment accumulator had to fit in the same footprint as the eight-segment accumulator. This required each segment to grow slightly narrower, as shown in Figure 27.

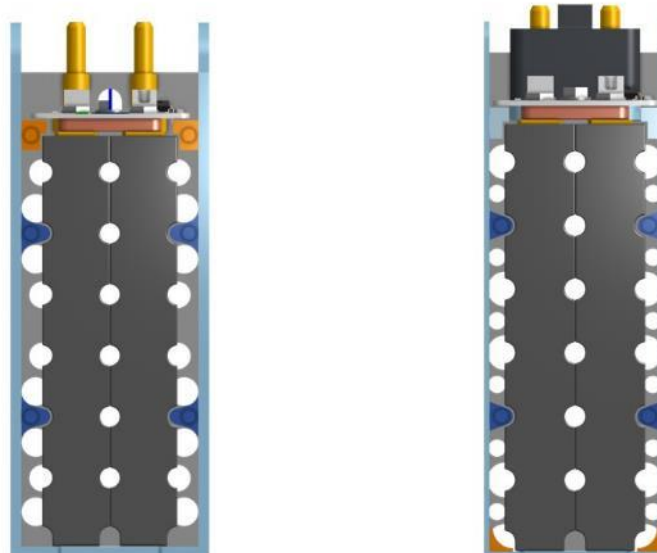


Figure 27 Eight segment accumulator segment, left. Nine segment accumulator segment, right.

It was feared that this reduction in width would be detrimental to airflow. Early in the discovery phase when considering the increase to nine segments, a test was proposed to ensure the cooling performance found in the eight-segment accumulator wasn't purely a function of the increased segment width. In order to try determine this, NU23 was driven in a reduced format endurance run with only half of its fans connected and operating, such that only four of the segments were being cooled by the fans. While a detailed discussion of this testing can be found in the "Commissioning and Validation" section of this report, it was found that the segments that were being cooled by fans were measurably cooler than those without the fans on. This indicated that the improved cooling performance was not only a result of the wider segments, and that the fans were a significant factor.

Other considerations for this overhaul of the accumulator was the incurred costs, time and weight gain. One modification made to combat all these factors was transitioning from 3 mm to 1.6 mm FR4. 1.6 mm thickness was chosen due to its popularity in PCB manufacture, as it is generally considered the default thickness. Due to how common it is, the price compared to 3 mm FR4 was closer to a fifth of the price.

Additionally, this reduction in thickness resulted in nearly half the amount of FR4 per segment, essentially halving the weight of the housings per segment. This was considered one of the biggest benefits of transitioning to the thinner material, as the weight penalty of an additional segment was greatly reduced. Thanks to the work of Jacob Searle, it was found that weight of the nine-segment accumulator only increased to 76.45 kg from 74.1 kg of the eight-segment accumulator.

This change did also improve the previous issue of reduced segment width, as an additional 1.4 mm of gap between the segment wall and the Energus modules was gained by reducing the wall thickness. This resulted in the gap between the Energus modules and the segment wall in the eight-segment accumulator being 6.4 mm compared to 4.4 mm in the nine-segment accumulator.

However, the transition to 1.6 mm did incur some downsides. To minimise the development time, many aspects of the eight segment housings were carried over to the nine-segment design. However, the reduced surface area at the butt join between the long walls and the base was a challenge moving to 1.6 mm FR4. While there were no issues with the 3 mm segment housings and their structural integrity, there were concerns the same wouldn't be the case for the nine-segment housings.

To increase the surface area for the epoxy to bond to, 3D printed corner rounds were designed. Shown previously in Figure 27 in orange, these were to be placed in the corner of the segments to increase the surface area, and as a result the bond strength between the housing base and wall.

These corner pieces were designed around 3D printing manufacturing methods, like the ridges used to locate the Energus modules. These ridges needed minor overhaul to suit the reduction in segment width but remained otherwise unchanged from the eight segment accumulator ridges.

The finalised design of the nine-segment accumulator did also include end pieces. While in the eight-segment accumulator, the fit was found to be slightly too tight with a 3 mm thick end piece and a 1 mm FR3 lining on the wall, it was theorised this tight fit could be improved by changing the dimensions of the lining and the end pieces. The FR4 lining of the accumulator is discussed in greater detail in the "Additional Work" section of this report.

Due to the decision to have the housings and linings manufactured off site by a PCB manufacturing company, and the reduced price of 1.6 mm FR4, this thickness was selected for both the segment end

pieces and the lining of the accumulator. This would create a greater clearance in theory, therefore reducing the tight clearance that was found in the eight-segment accumulator. While this final design was ultimately modified to remove the end pieces all together, detailed in “Commissioning and Validation”, the final design is seen in Figure 28, with a drawing included in the “Drawings” section of this report.

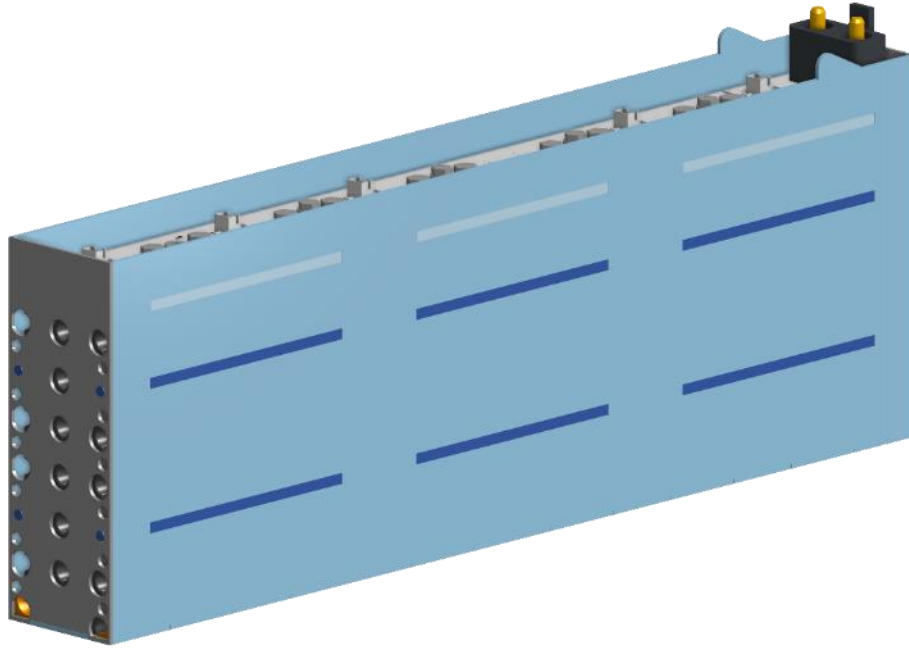


Figure 28 Final design of the nine-segment accumulator segment housing.

Fan Manifold and Accumulator Container Design

Eight Segment Accumulator

While the configuration of the cooling is discussed in the “Initial System Design” section of this report, the method of attachment and location of the fans required attention. As discussed, the cooling configuration was finalised early in the design phase to enable the accumulator container to be designed. This was critical to ensure the chassis was sent for manufacturing in time, as the accumulator’s footprint affected the chassis dimensions.

A fundamental decision was whether the fans would be positioned inside the external walls of the accumulator, or external to the walls. The major benefit of having the fans on the inside of the accumulator was a more tightly packed system, keeping the 12 V cabling required to power the fans within the accumulator. It also made efforts to keep the system waterproof easier, as all electrical being housed within the walls, meaning only the volume within the accumulator had to be considered.

However, there were two main downsides to having the fans internal. The foremost was the likely additional of another internal wall each side to maintain the structural integrity of the accumulator. In order for the segments to be secured without the ends being pressed against the fans, another wall would have to be placed, essentially creating another two additional compartments within the accumulator as shown in Figure 29.

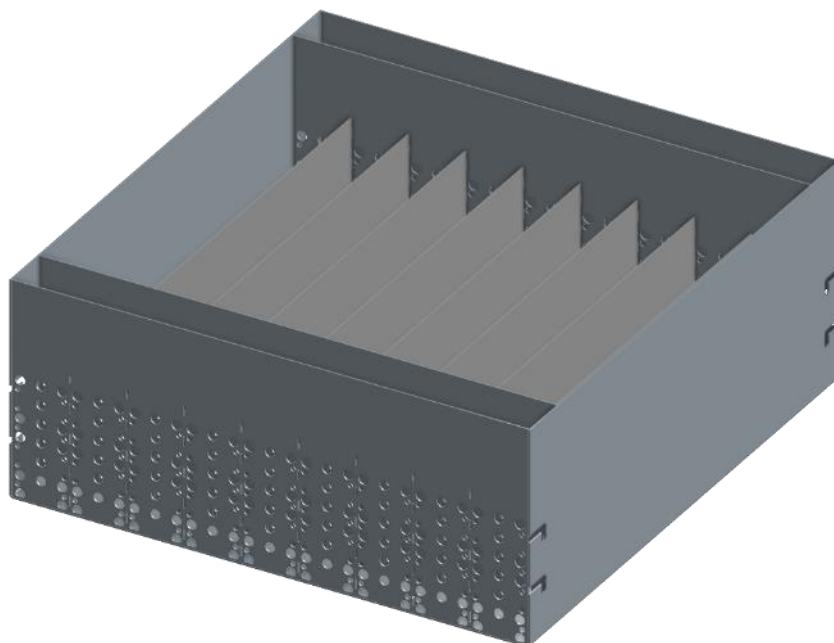


Figure 29 Accumulator container that would require additional walls to house internal fans.

Not only would this increase the weight of the accumulator, but it would also increase the amount of welding. Given the thin 1.6 mm sheet metal used, extreme warping could occur, an issue that

EV.Three encountered to the extent where clearances between internal walls and segment housings became unserviceably tight. Despite NU23 accumulator container increasing in thickness from EV.Three's 1.2 mm to 1.6 mm, it was advisable to minimise the amount of welding to avoid similar issues. Such design decisions are discussed at greater length in Jacob Searle's report on Accumulator Structural.

Additionally, as stated previously, the accumulator container had to be finalised at a very early stage in the development process. Housing the fans without the container restricted the flexibility in the final solution in comparison to housing fans on the exterior of the container. Without the time to thoroughly develop an internal fan design prior to committing to an accumulator container design, there were concerns the team would be unable to create a serviceable subsystem, while being committed to this design path.

As a result, work begun designing a method of attaching the fans to the exterior of the accumulator container. Initially, the idea of a two piece manifold each side was discussed. The first piece would be permanently attached to the accumulator container through a strong adhesive, and a second larger piece would house the fans and index onto the piece attached to the container. This method was used to conduct the axial fan test, detailed in the "Axial Exhaust Fan" section of this report.

However, with help from Jye Hollier and Jacob Searle, this idea was abandoned in favour of a solution that utilised the structure of the container for mounting. While the principles of the design and manufacturing process of the container are discussed in Jacob Searles' report, the folded base and walls allowed for an extension of this part to act as a flange. This design can be seen in Figure 30.

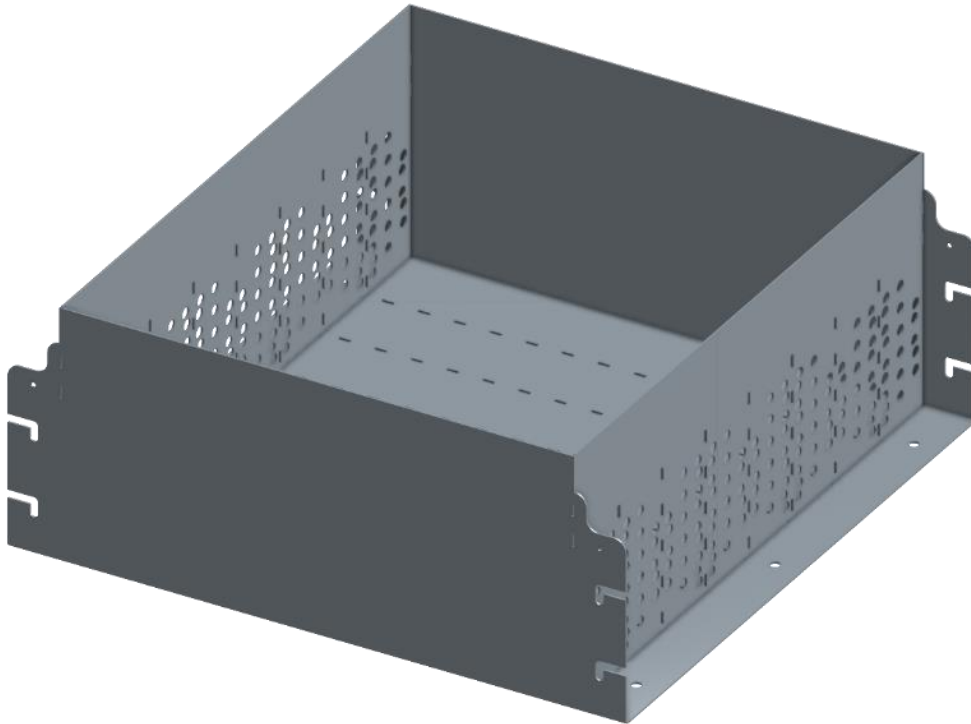


Figure 30 Flange on each side of the eight-segment accumulator to support fan manifolds.

Several methods of fastening the manifold to the container were implemented. As seen in Figure 30, slots at each side of the container allowed for pins to slide in horizontally, and then vertically downwards to lock the manifold in place. At the base of the flange, there were three more holes to locate the bottom edge of the manifold. These slots and pins were designed to ensure even pressure across the edges of the manifold, to make an airtight seal against the container wall. There were two additional holes seen at the top of the flange. These were to accommodate a threaded fastener for extra security, to ensure the manifold did not separate from the container during driving. These attachment methods are shown in detail in Figure 31.

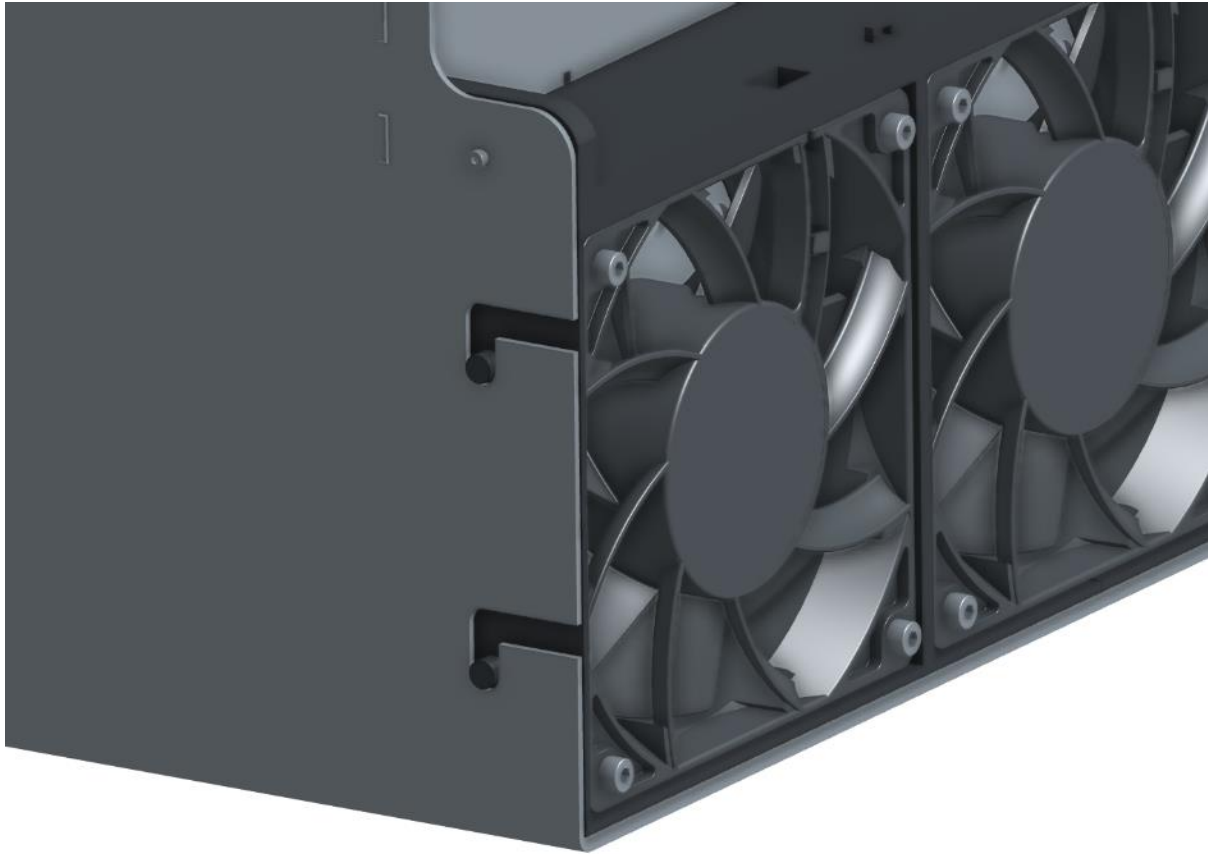


Figure 31 Fastening mechanisms for eight segment accumulator fan manifold, with slots for the pins, and threaded fastener.

With these attachment details, the accumulator container was finalised and detailed design of the fan manifolds begun. The goal of these manifolds was to provide a single piece that housed all the fans so they could be removed easily and quickly in one part. As discussed in the “Flow Analysis Through Pressure Testing” section of this report, 120 x 38 mm fans had been selected due to their optimal flow performance. Four of these fans on each face would be used, with each fan servicing two segments.

The exact model of fan selected was the Sanyo Denki 9GV1212P1J01. These were extremely similar to the Orion axial fans used in the pressure test. However, coming in slightly more expensive at \$82.35 per unit, it added the functionality of Pulse Width Modulation (PWM) speed control. The full data sheet for this fan can be seen in Appendix 3. Thanks to the work of Jed Reeves, this functionality provided the ability to lower the fan speed during charging, which is justified and detailed in his report.

The fans and manifold were designed to not extend beyond the flange of the accumulator container, which provided protection to the plastic manifold and fans. As a result, the manifold was designed so that it would press hard up against the face of the accumulator wall. In the face located against the wall, heat-set threaded inserts were used to fasten the fans to the manifold with M4 bolts. This is seen in Figure 32, which displays the final manifold design for the eight-segment accumulator.

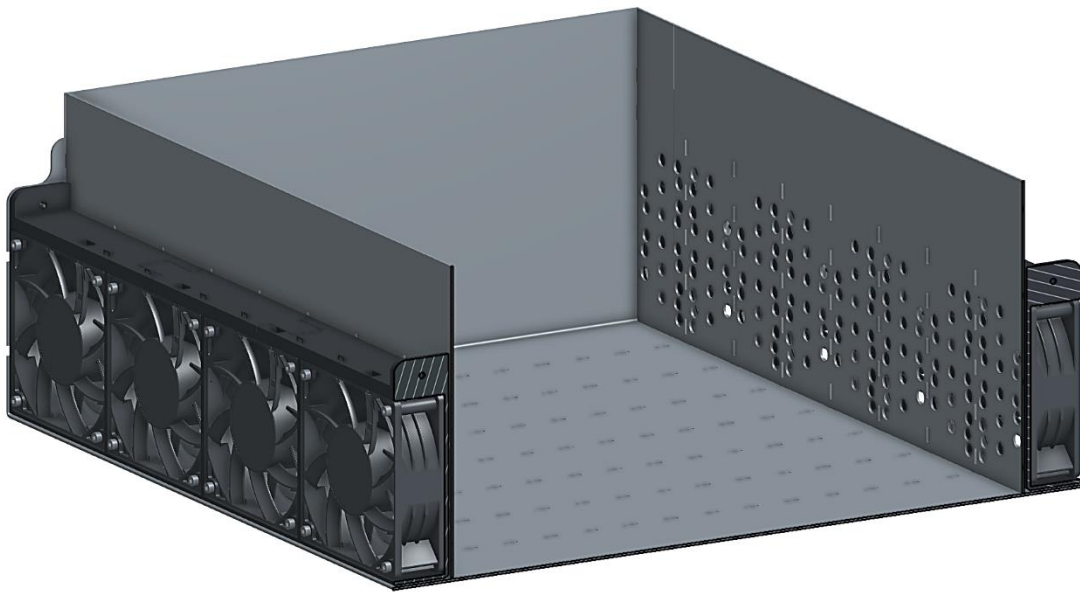


Figure 32 Section view of eight segment accumulator with fan manifolds.

One major design challenge related to the 3D printing manufacturing of the manifolds. Due to bed size limitations of the 3D printers, these fan manifolds had to be separated into several pieces. With the total length of the manifold over 500 mm, and a maximum linear dimension of the printers being 250 mm, the part had to be separated.

Ensuring the stiffness of the manifold remained was one of the challenges associated with splitting it into parts. This eventuated by having three parts that the fans mounted to, a “spine”, and two plates to secure mesh over the face of the fans. These parts are shown in Figure 33, where it can be seen that the splits in the main body of the manifold occurred across the position the fans would be mounted. This was an intentional decision, as the structural rigidity of the fan would transfer to the manifold as well. The spine was an additional piece that bridged both the aforementioned splits in the part. This was to further increase the stiffness across the weaker areas created by the splits.

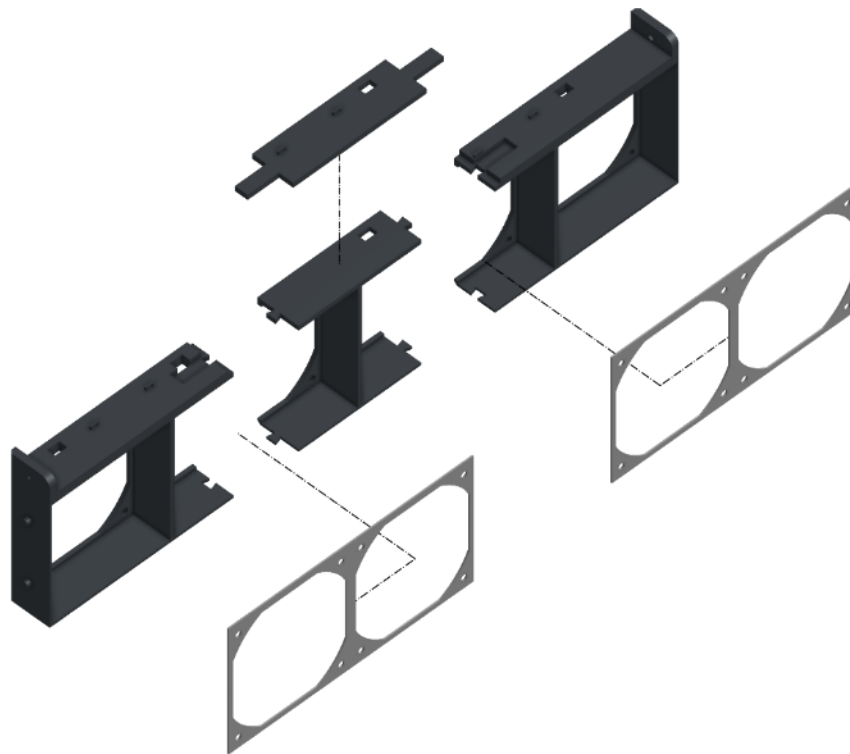


Figure 33 Exploded view of eight segment fan manifold.

What can also be seen in Figure 33 is the light-grey parts used to secure flyscreen in place. After early track testing, the abundance of rocks and other debris entering the manifolds required a solution to prevent them entering the fans. Stainless steel flyscreen was held in place using these parts, to prevent the debris entering the fans, as shown in Figure 34.

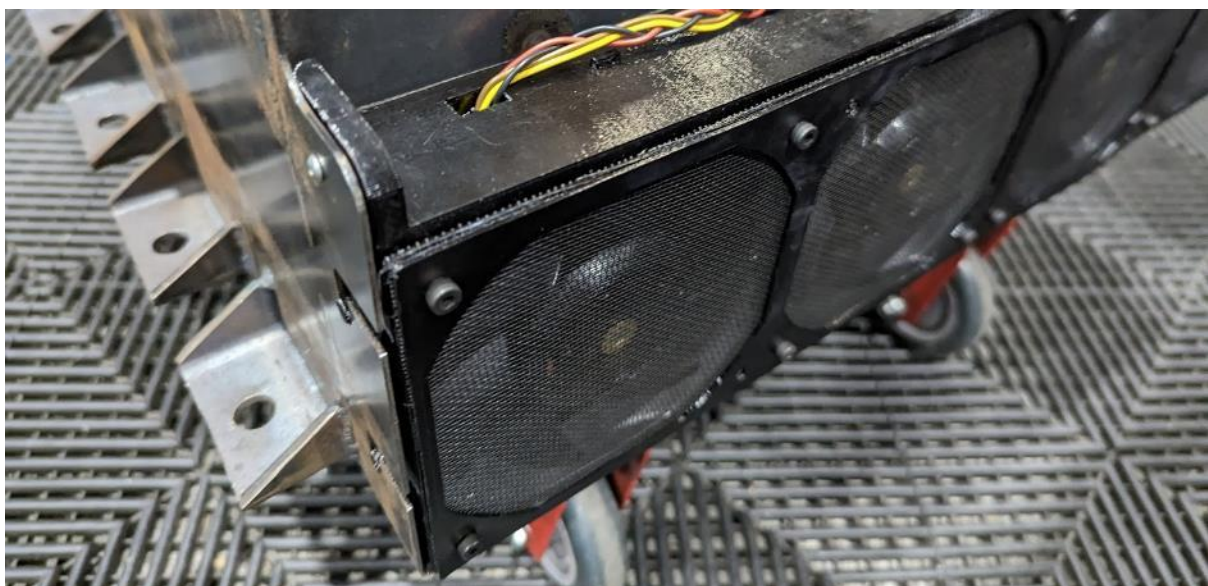


Figure 34 Stainless steel fly screen installed on fan manifold to prevent ingress of debris.

To improve the fit of the manifolds against the accumulator, the final revision of the eight-segment accumulator manifold included clearances for foam tape. A 1 mm gap was added between the manifold and the accumulator wall, with 3 mm foam tape applied to the manifold as shown in Figure 35, providing a good level of compression of the tap to create a better seal. Previously, some airflow was observed to be escaping around the edges of the manifolds, however this modification noticeably reduced this.



Figure 35 Foam tape location on fan manifold.

Nine Segment Accumulator

While many of the attributes of the fan manifold stayed the same for the nine-segment accumulator, there were necessary changes, as well as various improvements. The foremost issue that arose from adding a segment, was the location of the fans relative to the segments. In the eight-segment accumulator, fans were aligned to service two segments each, with the fans pressed directly against the container external wall. With the additional segment, there was no optimal way to distribute the segments and fans. As the fans didn't line up the same for each segment, the frames would cover some segment's airflow more than others. Thus, they wouldn't receive even airflow with the fans pressed directly against the wall.

It was therefore decided to try and create a gap between the fans and the wall. This attempted to create an even, high pressure "volume" across the face on the intake side, and low-pressure volume on the exhaust side. This was theorised to create a volume of even pressure across the whole wall, therefore providing even airflow to all the segments.

To achieve this, the mounting direction of the manifolds was required to change. In the eight-segment manifold, the manifold had a face pressed flat against the external of the accumulator. This face is where a heat set insert was placed, and an M4 bolt fastening the fan. Conversely, in the nine-segment used a nut placed on the other side of the fan, with a bolt inserted the opposite way. This difference is shown in Figure 36.

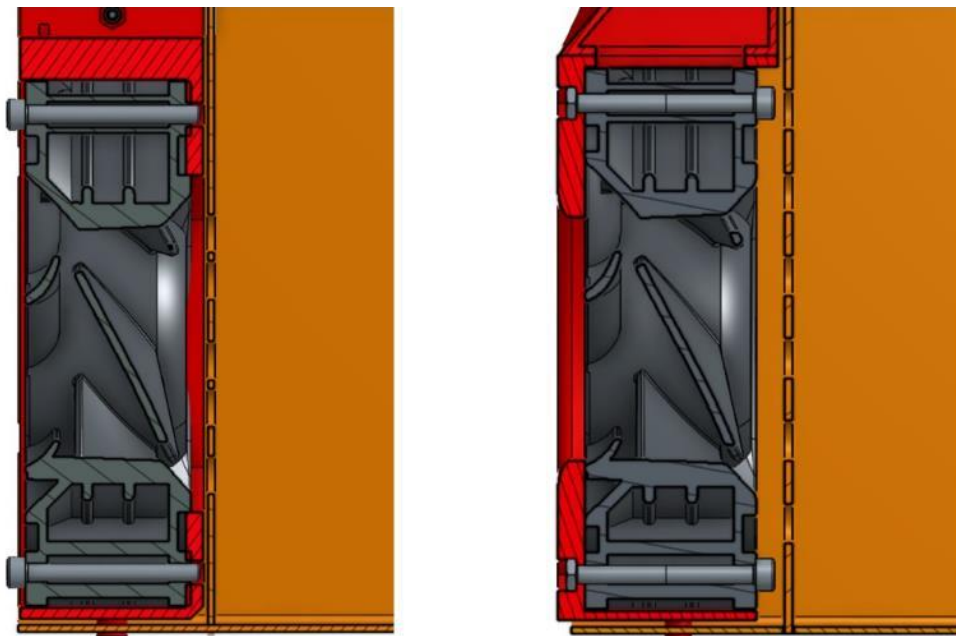


Figure 36 Section view of the accumulator container (red), fans (black) and manifold (red) showing fan spacing and orientation of fasteners for eight-segments left and nine-segment, right.

Additionally, what can be seen in Figure 36 is the large gap created between the wall and the fans afforded by the change in orientation. This displays the pressure “volume”, and the effectiveness of this design feature is discussed in the “Commissioning/Validation” section of this report.

This design feature also had a flow on affect to the holes in the accumulator container. The eight-segment accumulator had cooling holes that aligned with the blades of each fan, rather than the entire face of the segments. The introduction of the pressure zones in the nine-segment design meant that the frame of the fans no longer covered sections of the container’s wall, as was the case in the eight-segment accumulator. This resulted in far more holes being added to the accumulator container, shown in Figure 37. By decreasing the width of the segments, many of the cooling holes had to reduce in diameter. However, the area taken up by holes increased by 2% thanks to the additional holes. This would increase the channels for the air to flow through, theoretically reducing the impedance on the fans and increasing the airflow.

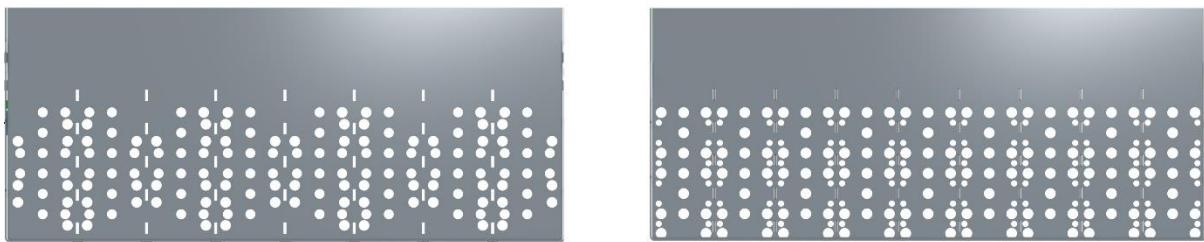


Figure 37 Eight-segment accumulator wall, left, and nine-segment accumulator wall, right.

During the eight-segment accumulator design stages, no thought was given to the cable management of the fans. The team’s priority was getting the fans serviceable to enable the maximum amount of track testing as early as possible. As seen in Figure 38, wires were secured with cable ties loosely above the fan manifold.



Figure 38 First fan manifold without improvements to wiring.

While this enabled the fans to function, several improvements were made to the integration of the wiring. The nine-segment accumulator brought updates to the Low Voltage Distribution (LVD) board, developed by Jed Reeves, including moving to Deutsch DT connectors, rather than auto-sport connectors. These two six pin connectors provided two positive, two grounds and two control pins per connector. Given that each fan required one of each of these, it was decided to insert an intermediate loom, shown in Figure 39.

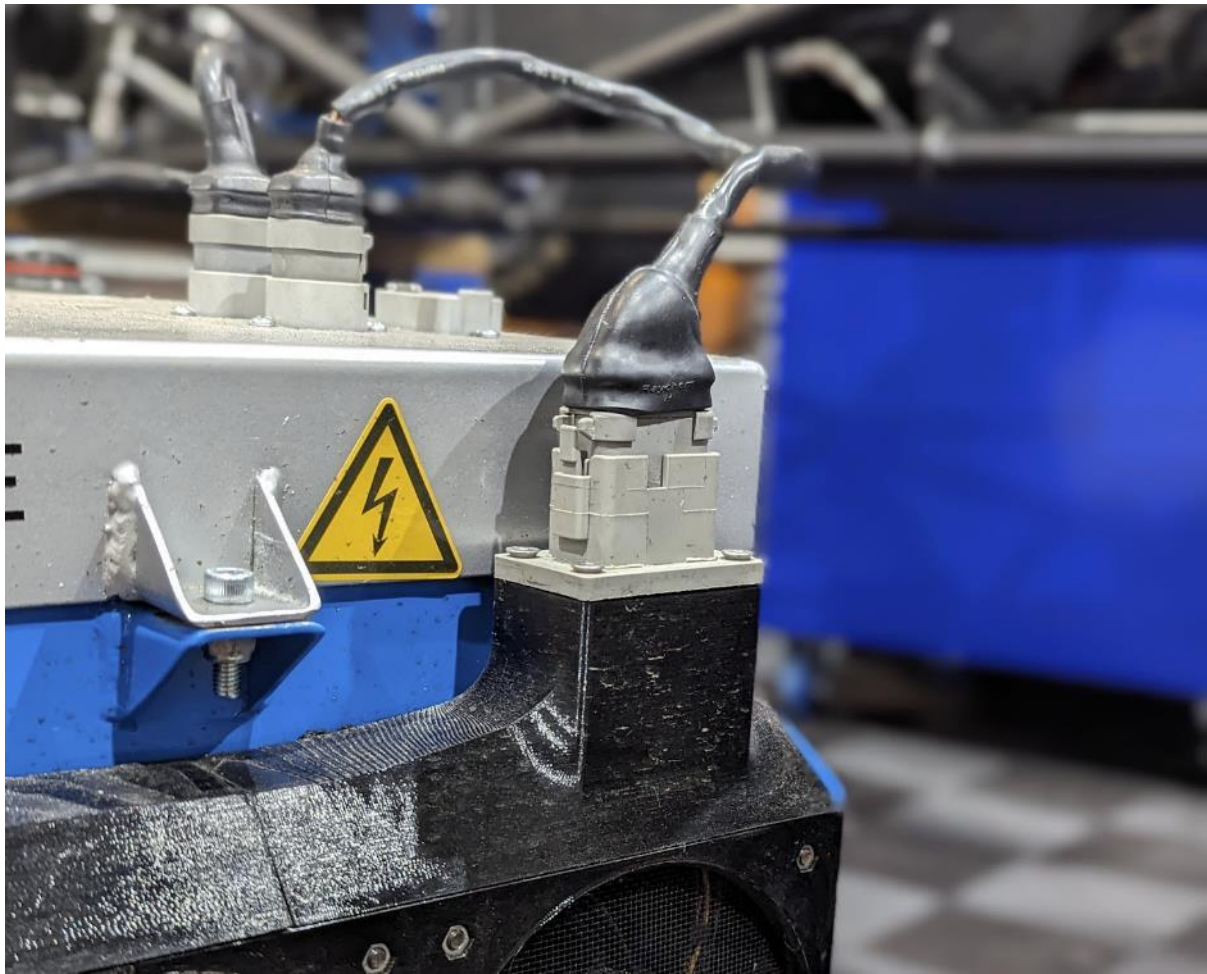


Figure 39 12 Pin Deutsch DT connector for fan power, ground and PWM wires, connected via the intermediate loom to the LVD on the lid.

This intermediate loom took the four positive, four ground and four PWM control wires from one fan manifold, and reduced them to the six pins found on the LVD. At the manifold end of the loom, a 12-pin female Deutsch DT connector was used to connect to the fan manifold, as shown in Figure 39. This means each fan connected directly to the 12-pin male Deutsch DT connector on their respective fan manifold, providing a key serviceability trait.

This enabled a faulty fan to be removed from the fan manifold by simply de-pinning the required fan, re-pinning another and attaching it to a manifold. By having the junction of wires in the intermediate loom, no loom deconstruction, soldering or crimping, except the DT pins, was required for a fan replacement. This was a key design element to maximise track testing and reduce downtime should a fan failure occur.

Further improvements to cable management were made through internal cable routing of the fan manifolds. Given the emphasis on easy removal of the manifolds, and their tight proximity to chassis mounts and other components when inserting/removing the accumulator, time was taken to create

a cable “tunnel” to protect the fan cables. This is shown in Figure 40. This cable tunnel provided protection to the fans cabling, preventing the cables being snagged or accidentally cut during removal of the accumulator, or just the fan manifolds. Additionally, it was made an extremely sleek, and tidy unit.



Figure 40 Section view showing the cable tunnel running along the top of the fan manifold.

With the addition of this cable tunnel, the decision was made to omit the fastened connections from the manifold to the accumulator container. While in the eight-segment accumulator manifold had bolts to ensure the manifold would remain in place, the slot and pin connection was deemed adequate. As the team no longer using the bolted connection at the end of the eight-segment accumulator’s service life, the increased serviceability of a tool-less manifold removal was considered advantageous, and therefore no fastened connection was included. The final fan manifold design used at competition is shown in the “Drawings” section of this report.

Very late in the commissioning process, another fan manifold was designed. Concerns outlined in the “Compliance Report” section resulted in a contingency plan in the form of an alternative fan manifold. This alternative manifold included a thinner profile at the back of the accumulator to maintain a 25 mm gap between the chassis and the accumulator. In the case these concerns did materialise into a failed technical inspection, the alternate fan manifold could be installed.

After much deliberation, it was concluded that this couldn’t be achieved with the 38 mm fans already being used in the manifolds. This resulted in an investigation into either removing the fan at the back of the accumulator closest to the chassis member or replacing it with a thinner alternative. Given that the next standard size down in 120 mm fans were 25 mm thick, it was decided to try and incorporate this type of fan into the manifold.

This however wasn't as simple as swapping the 38 mm fan and making the manifold thinner. The chassis member close to the accumulator was not parallel to the wall, and therefore the 25 mm fan had to be angled within the manifold to maintain the 25 mm gap to the chassis member. This can be seen in the final design, shown in Figure 41. With this angle, the fan still protruded slightly from the manifold. This required some material removal from the frame of the fan manifold to ensure the manifold seated properly against the accumulator wall.



Figure 41 Top view of the final design for fan manifold containing 25 mm fan.

Figure 41 displays the only key difference between this manifold, and the one that was used throughout testing and eventually at competition. As there were no glaring issues with the manifold already in service, and very little time until the competition, all other design aspects were carried over. While the manifold was manufactured, limited testing time, and no requirement to use it during competition saw it never be used in service.

Manufacturing

CNC Manufacturing

Computer numerical control (CNC) manufacturing was the primary form of manufacturing for many components of the accumulator. The segment housings, service handle, and top plate of the accumulator were all manufactured on campus with the use of CNC router. It was selected due to the precision, speed and safety that CNC routers provide, and the nature of the parts being manufactured. While each component followed similar steps, there were additional complexities with the top plate involving additional steps.

Components to be manufactured required some considerations during the design phase. These included tool geometry, access, and capabilities. These design decisions had to be documented to ensure that the tooling paths created matched those in deliberate design decisions.

The service handle and accumulator segment housings were essentially two-dimensional shapes, making them easier to manufacture. Because the design was oriented around the supplied thickness of these materials, only perimeter cuts to full depth were required. The sheet that the parts were cut from was glued to an MDF sheet. This served two purposes: one was to prevent the FR4 from lifting from the bed, and the second to provide a sacrificial layer, so when the tool plunged completely through the FR4, it would hit the MDF rather than the CNC bed. The FR4 and MDF was then clamped to the bed and cut as shown in Figure 42.

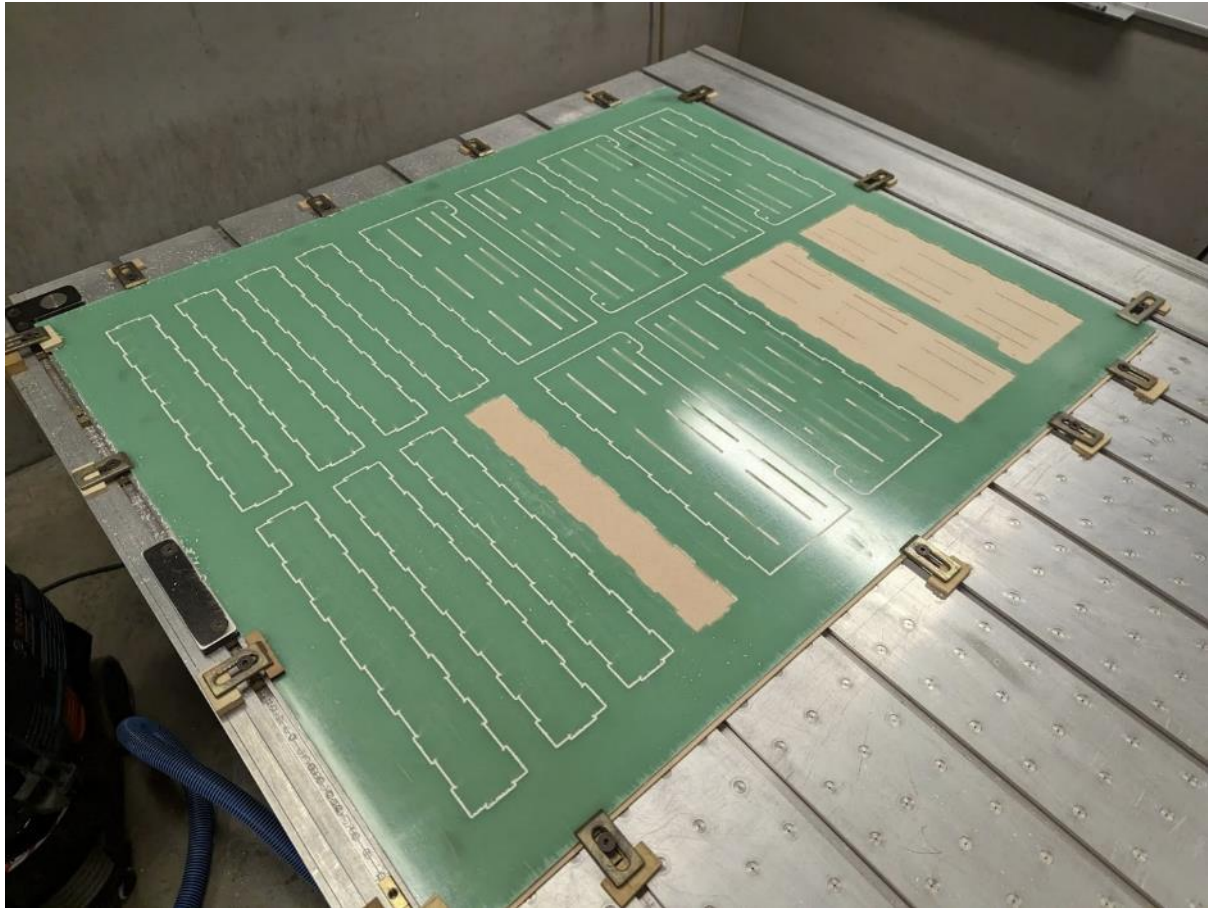


Figure 42 FR4 sheet after being cut on CNC router bed.

The size of tool needed to be decided based on several factors. The minimum internal corner radius on a workpiece was dictated by the tool diameter. While a smaller diameter allowed for a smaller internal radii, the feed rate reduced with radius of the cutter. An example of the internal radii requiring consideration was on the service handle. The layers of the service handle created a void for the busbars, shown in Figure 43, where the internal radii of a corner had to be considered, and tool selected to ensure adequate clearance was provided. Additionally, a tool with enough length to be able to plunge the full depth of the material was required.

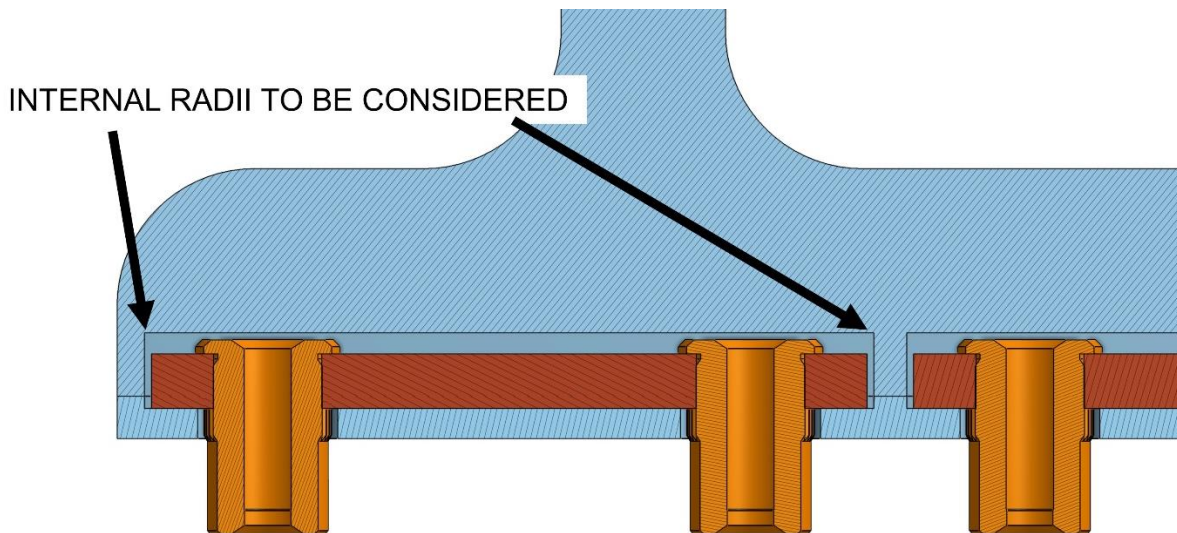


Figure 43 Section view of service handle showing internal radii to be considered in CNC manufacturing.

End mills were used for these types of cuts, and in the case of composites such as FR4, corn teeth end mill pieces were particularly well suited, due to how hard and abrasive FR4 is. An example of such a tool is shown in Figure 44.

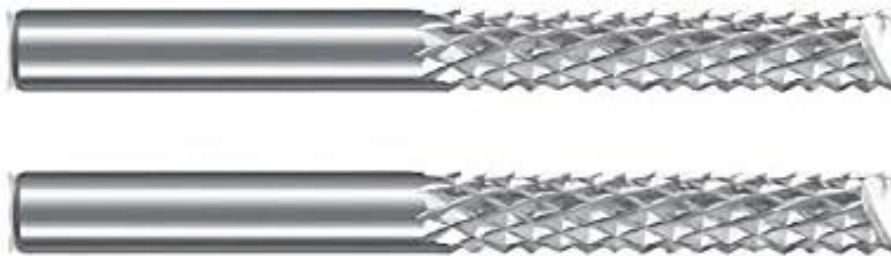


Figure 44 Corn End Mill CNC router tool (Cluster, 2024).

Once tool selection was completed, DXF's were loaded, with tool paths defined in Computer Aided Manufacturing (CAM) software. During this stage, feeds, speeds and depths of each cut were selected, including whether material was to be removed internal or external of DXF lines. This process was done by the staff in the Architecture and Built Environment Workshop (ABEW), with clarifications provided on operations required.

The top-plate of the accumulator that served several purposes. Its main purpose was to provide mounting area for electrical components including high and low Voltage, and the BMS. The walls located on the top plate served to separate high and low Voltage as well as transfer the compressive

force from the lid of the accumulator container, and evenly distribute it onto the segments. Jacob Searle completed the design, with the details contained in his report. However the manufacturing processes are covered in this report.

The top plate was to be manufactured from GPO-3, a glass fibre composite. This material is extremely similar to FR4, with similar tooling and the same precautions required during CNC manufacturing. However, the geometry of the top plate did differ from other pieces manufactured due to features that were not full depth. Slots in the upper surface of the top plate were used to locate the vertical walls, as seen in Figure 45.

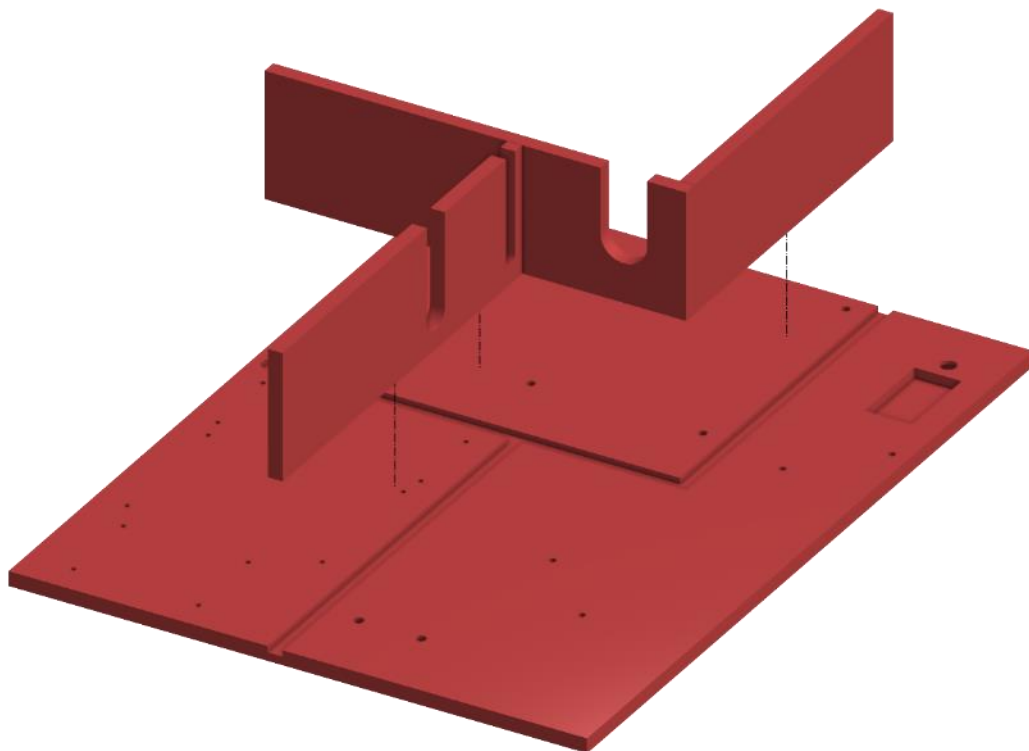


Figure 45 Top plate exploded view showing vertical walls and their locating slots.

Additionally, there were features on the bottom of the top plate that were not full depth either. These are shown in Figure 46, which are required to line up with the full-depth holes to provide positive locking on the fasteners that attached components to the top plate.

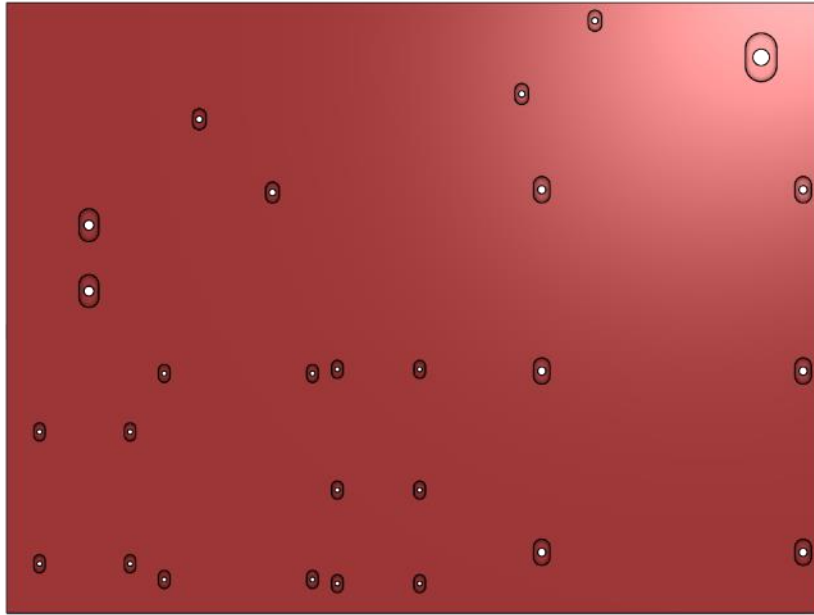


Figure 46 Non-full depth features on the underside of the top plate.

This posed the challenge of ensuring that the workpiece was located so that the router's zero position was correct for both sides, despite having to flip the workpiece over. To achieve this, an MDF jig was used that provided a press-fit around the exterior of the top plate. This jig remained clamped to the bed throughout the manufacturing process, with the GPO-3 of the top plate being removed only once from the jig to flip it.

Once the perimeter cuts of the walls and the base of the top plate were done, the jig was loaded on the bed, along with the base of the top plate. The first pass was done on the base side, completing all of the non-full depth features, as well as the holes for the fasteners. The workpiece was then flipped, taking care to ensure it was flipped along the correct axis, which was dictated by the orientation of the DXF's loaded to the machine.

This left only the slots for the vertical walls to be routed, which was done iteratively to ensure the best possible fit on the walls. The width of the slot was slowly increased, with the walls placed in the slot after each pass to check the fit. When a satisfactory fit was found, the CNC aspect of the top plate manufacturing was complete.

3D Printing

3D printing was used in various cooling related components for the accumulator. Two different kinds of filaments, PETG for the fan manifolds and jigs, and flame-retardant ABS for the ridges used in the accumulator segments.

The basic process for 3D printing took a 3D model, generally developed in CAD software, exported as an STL or STP file. This was then loaded into software referred to as a “slicer”. In the slicer, settings were chosen relating to infill, which dictated the style and percentage, as well as layer height and settings related to supports. It was also in the slicer that the filament being used was specified. The slicer then exported all this information as a gcode file, which was then loaded to the 3D printer via an SD card.

In the case of the fan manifolds, PETG was used. As there was no significant temperature resistance or flame-retardant properties required, the ease of access made this the obvious choice. As detailed in the “Initial System Design” section of this report, the design of the fan manifolds was required to be split into several pieces, due to limited bed size on the 3D printers available. This meant some consideration had to be taken orienting the parts on the bed in the slicing software.

While PETG possesses positive traits such as relatively good heat resistance and strength, it does suffer from lower quality surface finish. This is especially the case in areas where support material must be removed. When printing multiple parts that had to be joined, it became apparent that the support material affected the fitment, to the point of being unusable in some cases. This meant that in the case of the fan manifolds, orientation prioritised key fitting surfaces, rather than reducing print time or material.

However, steps were taken to reduce the support material where possible. Through investigation of the types of support material, it was found that significant time and material could be saved by changing to organic type supports rather than grid type, shown in Figure 47.

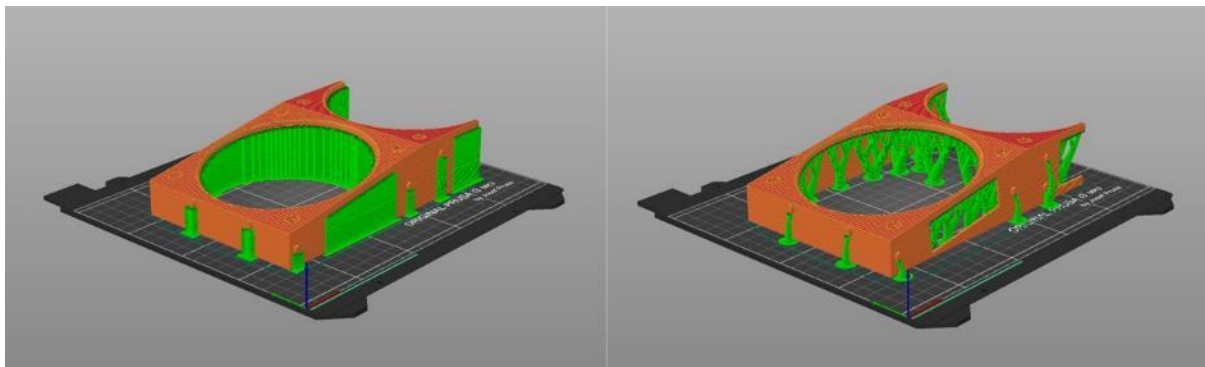


Figure 47 Grid type support, left and organic support, right.



In this example, using one piece of a fan manifold, it was found that organic used 101.45 g and took 7 hours 53 minutes. In comparison, grid type supports used 139.14 g of filament and took 10 hours and 35 minutes to complete the print. This small improvement helped reduce the manufacturing time and cost, to help get the car on track quicker, as the savings in time and cost stacked up significantly over many manifold iterations, as well as the other 3D printed parts.

Adhesives and epoxies

Various components associated with the accumulator required adhesives. Adhesives provide a strong connection that does not use potentially conductive fasteners. This is particularly beneficial in structural housings that contain electrical components, where the risk of an electrical short from something such as an exposed bolt or loose nut could result in catastrophic failure.

Several requirements from FSAE had to be considered when selecting an epoxy to be used. Given that the epoxy would be used on the insulating components, such as the segment housings, it was required to have a minimum temperature rating of 90 °C. It also had to meet the flammability rating stipulated in the rules of UL94-V0. The full data sheet for the epoxy used throughout the manufacturing process can be found in Appendix 8.

Eight Segment Accumulator Segments

The segment housings required a connection between the long vertical walls and the base. As detailed in the “Initial System Design” section of this report, the design revolved around epoxy as the mechanism of attachment.

However, given the use of a butt joint, considerations were made to ensure that the final segment walls would be perpendicular to the base. This was done through the use of a jig to hold the vertical walls at the desired distance. If the walls were not perpendicular to the base, it could result in additional stress being placed on the epoxied connection, resulting in a premature failure of the segment housings. These jigs were 3D printed from PETG.

In 3D printed components, some level of dimensional inaccuracy is to be expected. As a result, testing was done to ensure the spacing of the walls was appropriate. This was done by printing multiple blocks with the same internal width of a segment housing, but each with a slightly different variance smaller and larger than the housings. They were printed in the same layer orientation as the jigs would be to ensure that the effect of layer direction would be replicated. These blocks are shown in Figure 48, and were placed with a segment base and walls within the accumulator container to test the fitment. The markings on the top of the blocks were used to identify the variance from the actual CAD model, with a range of -0.2 mm to + 0.2 mm.

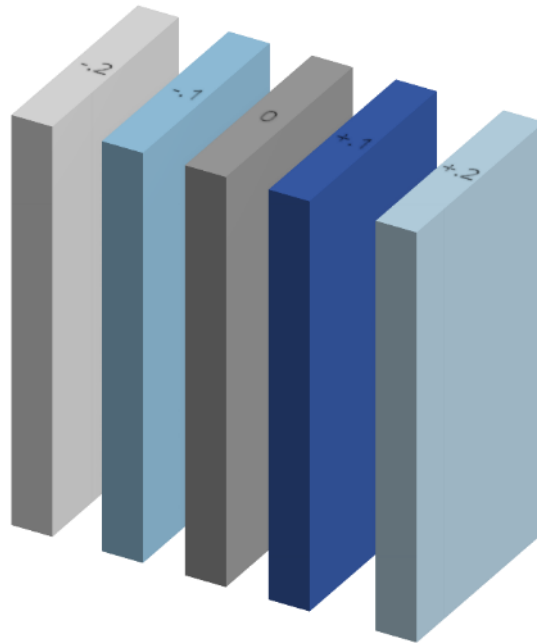


Figure 48 Segment spacing blocks.

It was found that the -0.1 mm variance provided the best fitment with the accumulator container. This resulted in the design of the jigs being completed shown in Figure 49.

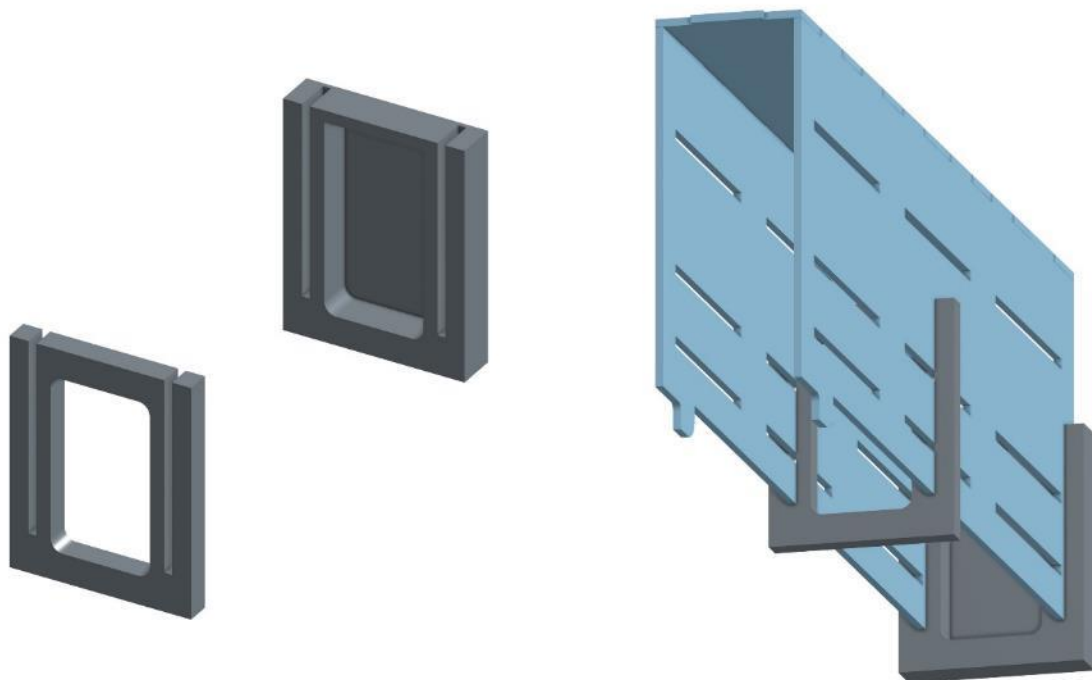


Figure 49 Eight segment housing jigs, showing the central and end jigs.

It can be seen that there were two different types of jigs: an end piece and a central piece. The end piece was essentially the same as the middle, with the addition of a backing plate to push the ends of the walls into. This ensured the walls were perfectly aligned with one another, before applying the base to the walls.

With the jigs finalised, assembling of the segments begun. Early into the process, it was found that the teeth of the base caused the walls to flare out slightly, creating a poor contact between the two parts. To reduce the risk of a poor final connection of the wall and base, cable ties were used to provide greater clamping force on the side walls. The final method of epoxying the segment housings is shown in Figure 50, with cable ties running through the unpopulated ridge slots.



Figure 50 Segment epoxying assembly.

This method was used to successfully manufacture all eight of the segments for the eight segment accumulator. No housing failures occurred during the service life of the eight-segment accumulator, however improvements were made to the process detailed in the following section.

Nine Segment Accumulator Segments

Several issues were noted with the jigs used in the eight-segment housing jigs. These issues were rectified in the jigs used for the nine-segment accumulator, shown in Figure 51.

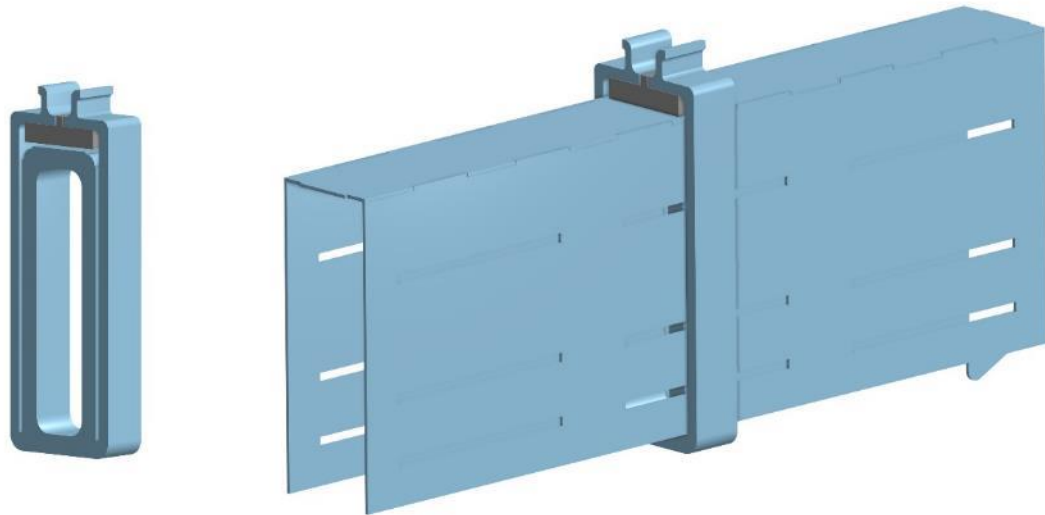


Figure 51 Nine segment housing jig, with second clamping piece shown in grey.

Foremost, the eight segment jigs didn't place any pressure on the base, pushing it into the vertical walls. While this didn't result in any segment failures, there was some degree of "float" between the base and walls noted. This was rectified through the use of a second piece in the jig on the top, that could be slid into the jig to compress the base into the vertical walls. This was required as a separate piece, to allow clearance for the base to be inserted initially.

Additionally, the overall stiffness of the jig was improved. By extending the walls of the jig the full vertical height of the wall, the stiffness of the jig was greatly increased, removing the need for the cable ties used in the eight segment accumulator housing jigs. Eyelets are also seen at the top in Figure 51. These are to supply additional clamping with cable ties if required, rather than using cable ties on the segment itself.

The jig being used is shown in Figure 52. There was a gap left in the jig allow the necessary corner pieces to be placed, while being held in the correct position during the drying process. From this point, the base could be inserted, and the second piece of the jig put in place to apply pressure the epoxy sections.



Figure 52 Nine segment housing with the corner pieces inserted into jig, ready for base to be placed.

These jigs were ultimately much more difficult to handle than those used in the eight-segment accumulator housings. However, this was more a product of the addition of the corner pieces, that were not a feature of the eight segment accumulator housings. However, the jigs were much more robust than the previous ones, and there were no fears of transporting the housings during the drying process, as the jigs restricted all movement in the pieces.

One aspect of the process that could definitely be improved, was the order of operations. As can be seen in Figure 52, the housings were assembled prior to the ridges being epoxied in. This resulted in a difficult process of epoxying in the ridges of the housing, something that was never done for the eight-segment accumulator housings. The limited clearance within the housings to locate and insert the ridges, leads to the recommendation that in future, the ridges should be epoxied in prior to the housings being assembled. This would require a slightly modified jig design, but would ultimately vastly improve the ease and speed of the process.

Top Plate

The four pieces cut via CNC router to make the top plate were joined via the same epoxy detailed in the “Eight Segment Accumulator” manufacturing section of this report. Due to the requirement for the vertical walls to transfer the full clamping force of the lid to the segments, it was important that the adhesion of the walls to the base was carefully completed.

Consequently, a simple jig was designed and 3D printed to ensure that the walls were epoxied perpendicular to the base of the top plate. This simple design is shown in Figure 53. The base of the jig would rest on the base of the top plate to ensure the walls remained at 90° angles to the base. Additionally, the circular cut outs at the base ensured that any seepage from the walls being pressed into their slots would not cause the jigs to be epoxied in place.

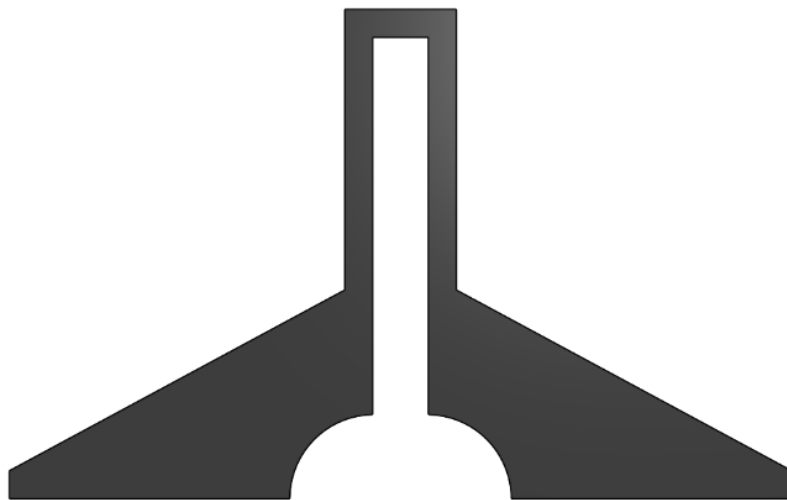


Figure 53 Jig to ensure perpendicular epoxy of top plate walls.

The top plate can be seen drying in Figure 54, with the jigs holding the walls in place. This method was ultimately deemed very successful, as no issues relating to the strength of the top plate arose throughout the commissioning and testing of NU23.



Figure 54 Top plate with jigs holding vertical walls while drying.

Integration of Electrical Systems

The eight axial fans used in the fan manifolds required power and PWM signal from the accumulator during operation. Jedd Reeves, designer of the LVD board provided two, six-pin Deutsch DT connectors on the accumulator lid for these connections.

As discussed in the “Initial System Design” section of this report, an intermediate loom connected the LVD to a 12 pin Deutsch DT connector on each fan manifold. In order to wire the fans to the DT connector, the pre-supplied cables had to be extended. This was done via a soldered and heat shrunk connection, with the wires then concentrically twisted as shown in Figure 55. To further protect these cables, Raychem heat shrink was used.

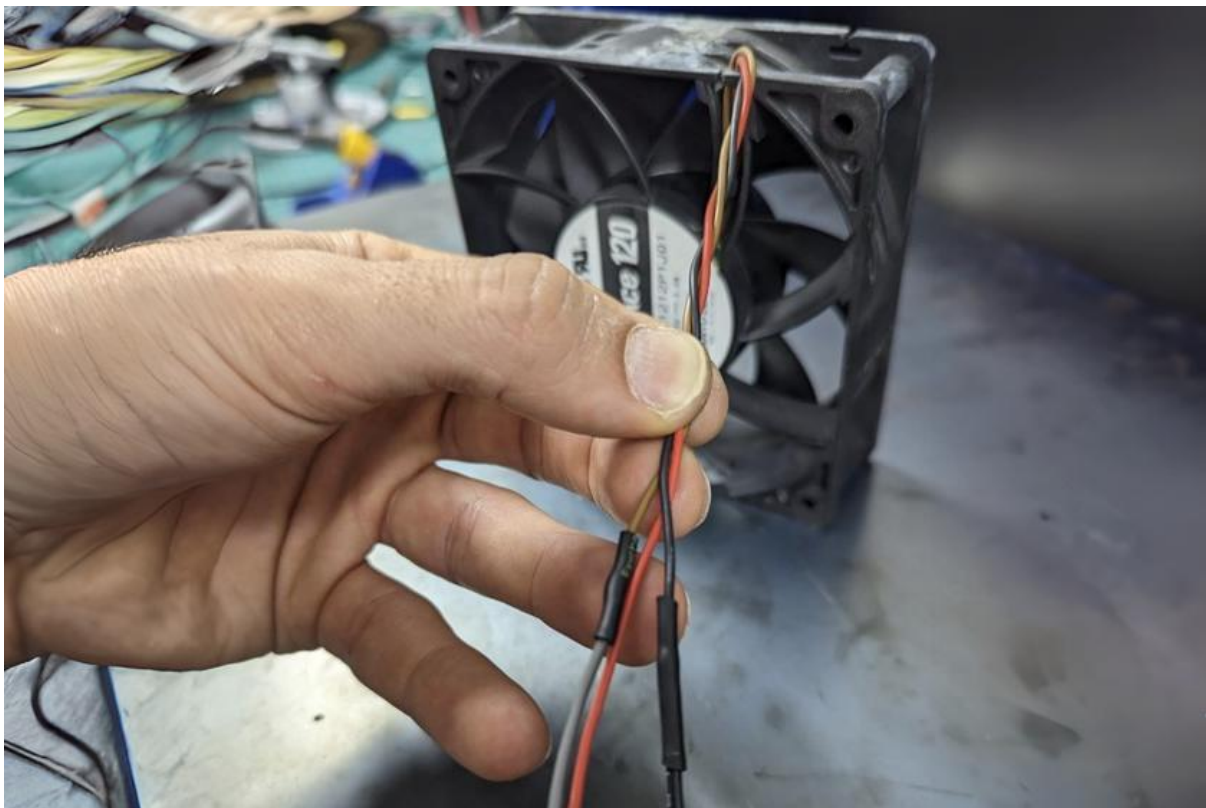


Figure 55 Concentrically twisting of fan wires, with soldered connections.

The ends of the fan wires were crimped with a DT connector pin and inserted in the 12-pin DT connector. This required the wires to run through the cable tunnel in the manifold prior to being pinned into the DT connector. This was somewhat of a design flaw, as it made the install and removal process of the fans far more difficult. It is recommended that should this manifold design be continued, a method that allows fans and DT connector to be removed all together from the manifold, without the need to de-pin the fans should be included.

Additional Work

FR4 Accumulator Lining

FSAE rules require that all conductive surfaces within the accumulator container are insulated from electrical components. As the material chosen to line the accumulator was FR4, given the experience of working with segment design and manufacture, the lining of the accumulator walls was included in the scope of this work.

Design was done by simply matching the features of the accumulator container. Finer details, such as the cooling holes in the accumulator walls required some degree of offset, to ensure there was no obstruction to airflow. The designed pieces can be seen in Figure 56.

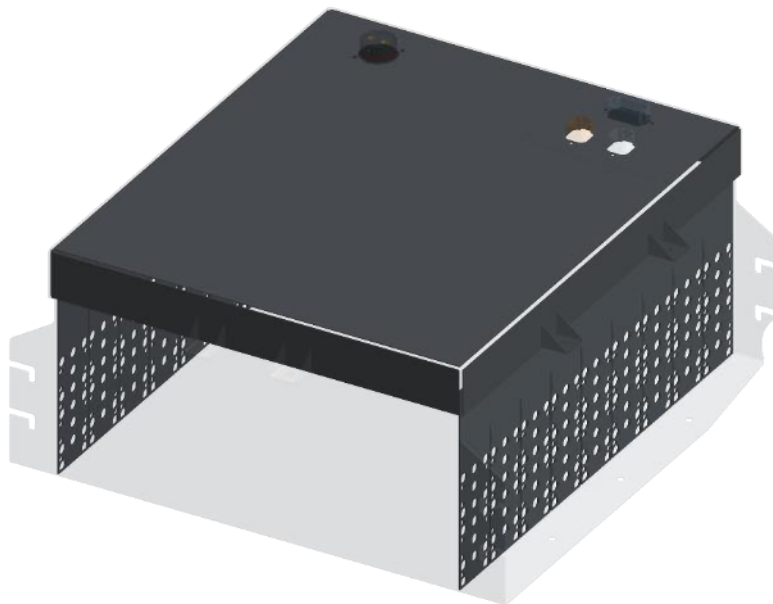


Figure 56 FR4 linings for the internals of the accumulator.

It is worth noting that no lining is shown on the bases and the two of the vertical walls. The base, as well as the internal dividing walls did not require an insulative material, as these surfaces were already completely covered by the FR4 of the segment housings. However, two of the vertical walls did not have pieces designed, as it was found that the clearance of the segments did not allow for full length FR4 on these walls. Instead, a small piece was hand-cut to cover the exposed steel wall between the top of the segment housings and the lid.

The other pieces, shown in Figure 56, were all modelled and exported as DXF's. These were then used to order the pieces pre-cut through an offsite PCB manufacturing service, PCB GOGO. This was a time saving measure, and also allowed the much more economical option of using 1.6 mm thick sheet, which was unavailable to buy within Australia.



The FR4 lining was secured in place, using the flame-retardant epoxy previously mentioned in the “Manufacturing” section of this report. Minimal epoxy was applied to reduce the thickness added to the container, affecting the already tight clearances.

A recommendation to the methodology is to ensure an even pressure is applied to the FR4 during the curing process. During the application in the nine-segment accumulator, little care was taken to ensure even pressure across the FR4 pieces, and as a result, some sections were elevated off the surface of the steel wall. Using a clamped jig, or even also a thinner sheet of FR4, would ensure the pieces better conform to the walls of the accumulator. These higher spots resulted in tighter clearances in the select locations of the accumulator, causing wear spots on the accumulator segments and the top plate.

Firewall FR4 Lining

As previously stated, given the work already completed with FR4, the lining of the firewall was also included in this work. The firewall was required to separate the driver from all tractive system components, with a non-permeable, rigid, non-flammable material, as per the FSAE rules (Formula SAE Australasia, 2024).

The main structure of the firewall was made from 1 mm steel sheet, which was designed and manufactured by Jacob Bush. It was required to be lined with FR4 to be rules compliant. This process was done purely by hand, taking 1 mm FR4 and using composite scissors and guillotine to cut it to shape. The pieces required can be seen in Figure 57, which were simple shapes defined by the pre-existing sheet metal firewall.



Figure 57 The seven FR4 firewall pieces required to be made.

This proved somewhat challenging, as the upper firewall included Zeus clips to hold it in place. This finer geometry is shown in Figure 58.



Figure 58 Zeus clips in upper firewall.

With the shapes cut out, the pieces were attached to the steel of the firewall with the flame-retardant epoxy detailed in the “Manufacturing” section of this report. Due to the high viscosity of the epoxy, no clamping was required, which would have posed quite a challenge given the geometry of the fixed firewall. The final result of the FR4 lined firewall is shown in Figure 59.

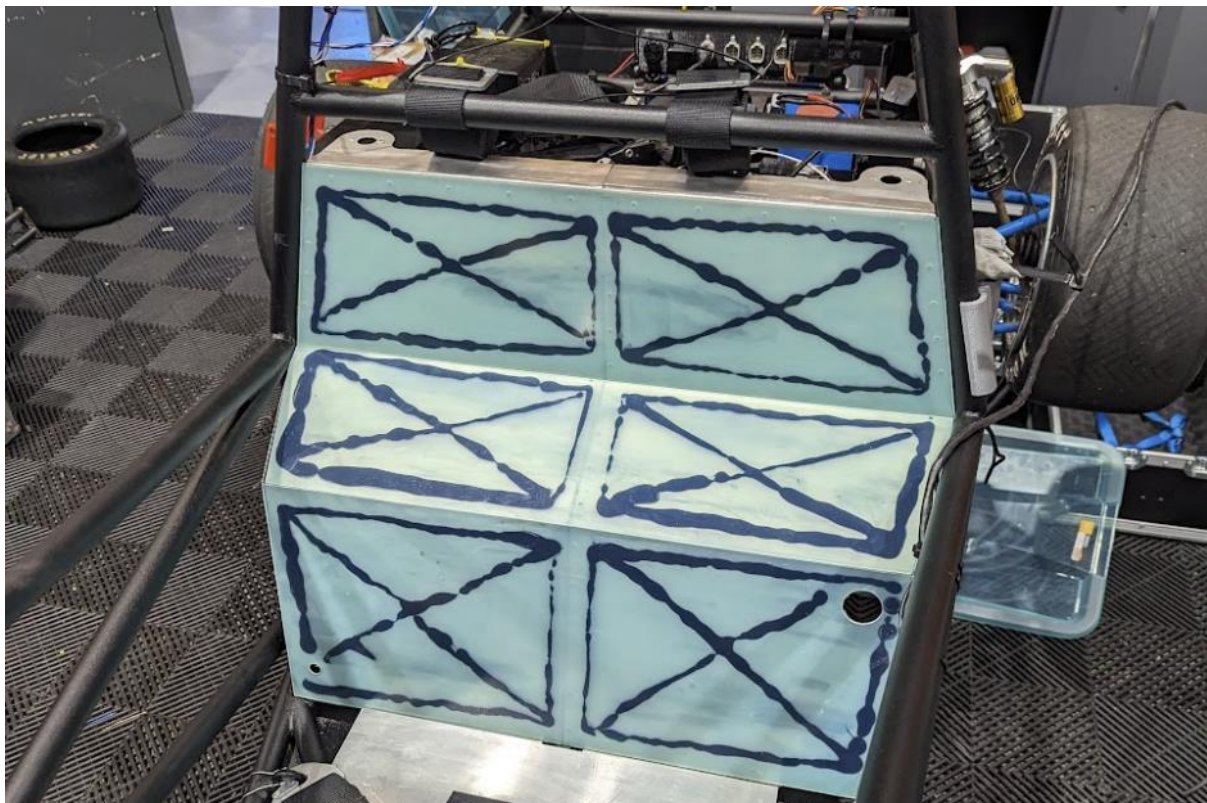


Figure 59 FR4 epoxied to firewall.

Voltage Separation on Top Plate

The top plate, as previously mentioned, was a component in the accumulator that served several purposes relating to electrical component mounting, as well being a structural component. Given the presence of both low and high Voltage electrical systems, the rules required that they be separated either by 30 mm or a physical, insulating barrier (SAE International, 2023).

Given the high density of components on the top plate, in most cases a physical barrier was selected. 3D printed enclosures were designed, in collaboration with Jye Hollier, to enclose both the IMD and Pre-Charge boards. This was required as both these high Voltage boards were within close proximity to low Voltage cables either on the top plate or extending to the lid. These enclosures can be seen in Figure 60, with the low voltage cables clearly visible over the top of the enclosures.

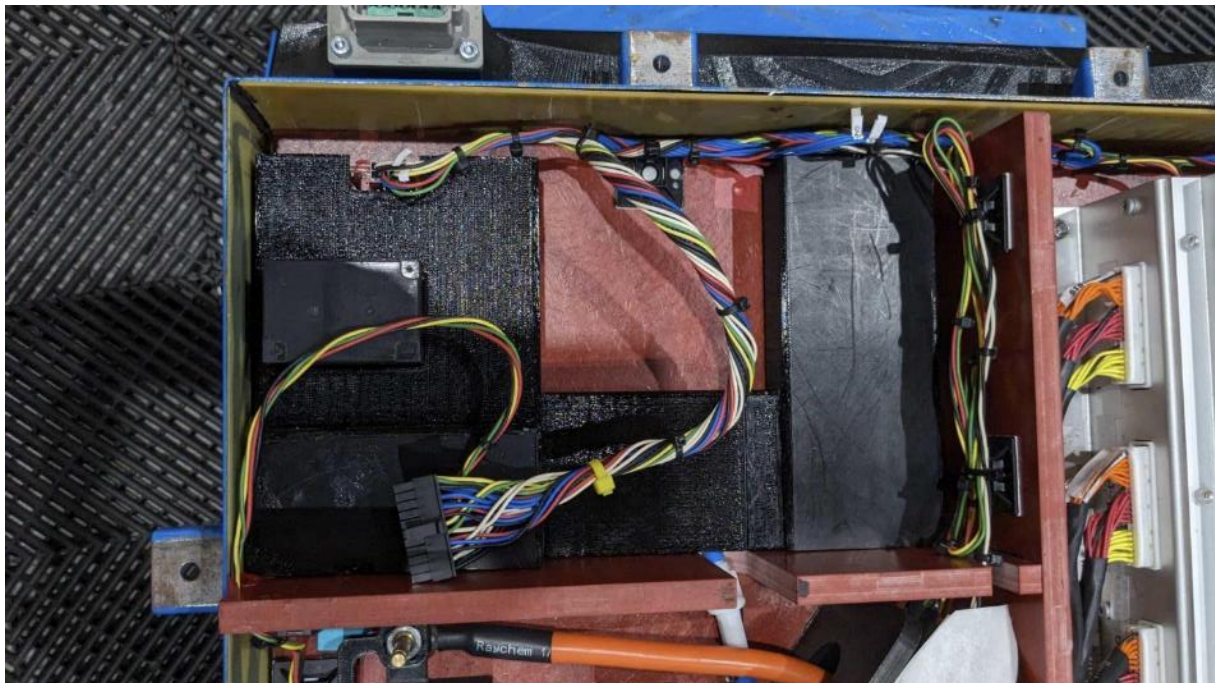


Figure 60 Enclosures to separate high and low voltage in the top plate of the accumulator.

This was an ideal method of separation, due to the durability and rigidity of the flame-retardant ABS. However, this method could not be used in other areas requiring the same separation. The AIR's had two low Voltage cables required to control the opening and closing mechanism. These passed over various high voltage connections, like the information cables that connected to the current sensor. In this case, a rigid structure was not feasible, given the complicated cable run, and the requirement for movement in surrounding cables with opening the lid.

As a result, Nomex sheet was used to separate these cables from the high Voltage areas of the top plate. Nomex is a paper like material that possesses electrically insulative and temperature resistant properties, suitable for such an application. The full data sheet is located in Appendix 9. Several pieces

of this sheet were used, to create a “sandwich” with the low Voltage cables for the AIR’s and current sensor inside, shown in Figure 61.

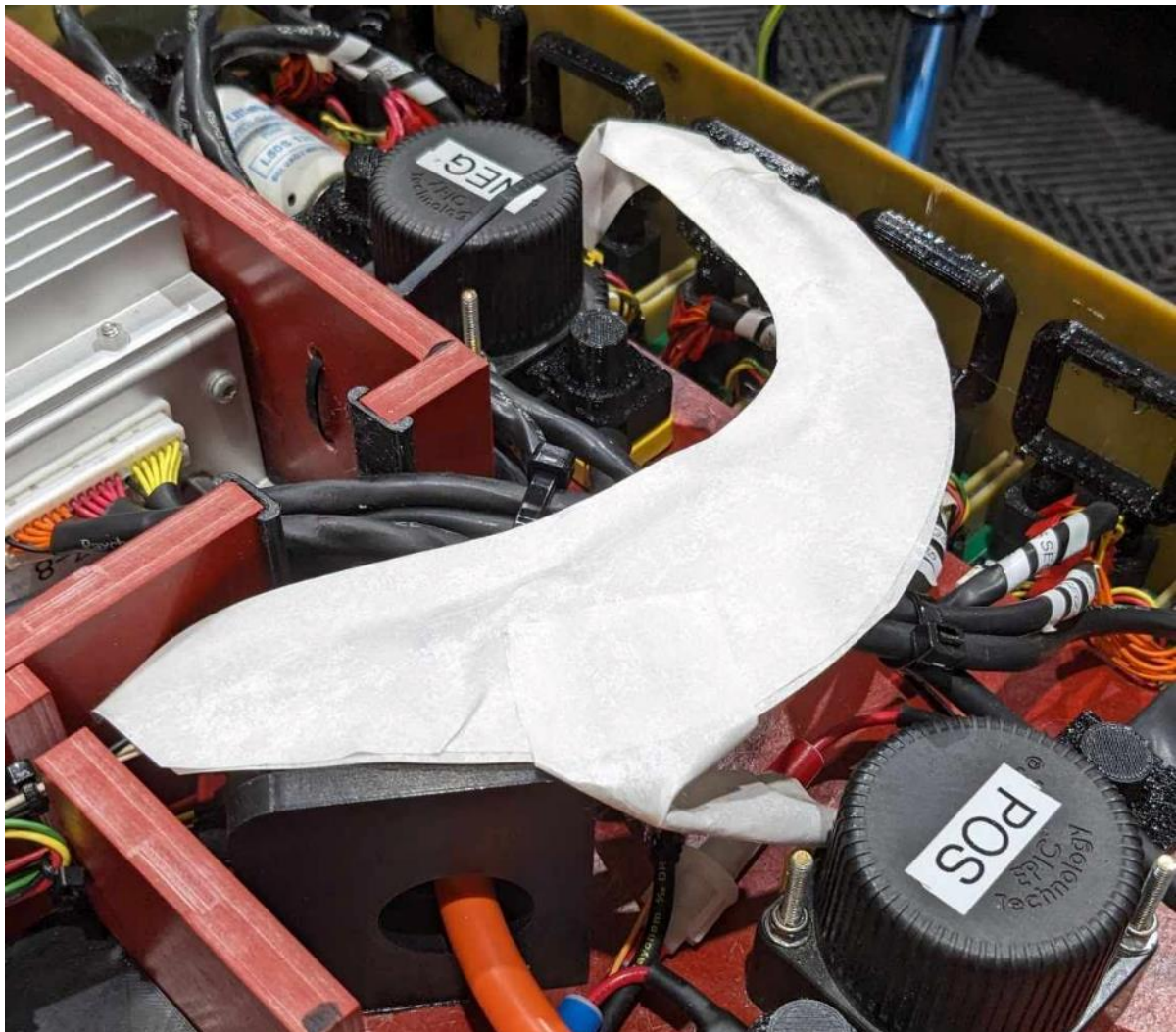


Figure 61 Nomex sheet used to separate low Voltage cables in high Voltage areas.

This provided a flexible solution, that could be manipulated easily to access cables that needed regular connection/disconnection, such as the BMS wires. While not as durable as the 3D printed enclosures seen in other areas, it provided a rules compliant solution that maintained the serviceability of the top plate.

Methode Pin Protectors

An improvement made during the service life of the nine-segment accumulator was to the Methode pin protectors. The Methode pins, located on each segment, indexed on the Methode buds in the service handle's busbars, to connect the segments together. When working within the accumulator, the service handle was removed in order to reduce the maximum potential in the accumulator to 50 V. However, this left the Methode pins exposed, posing a risk to short circuit the segments. These pins and the previously used covers can be seen in Figure 62.

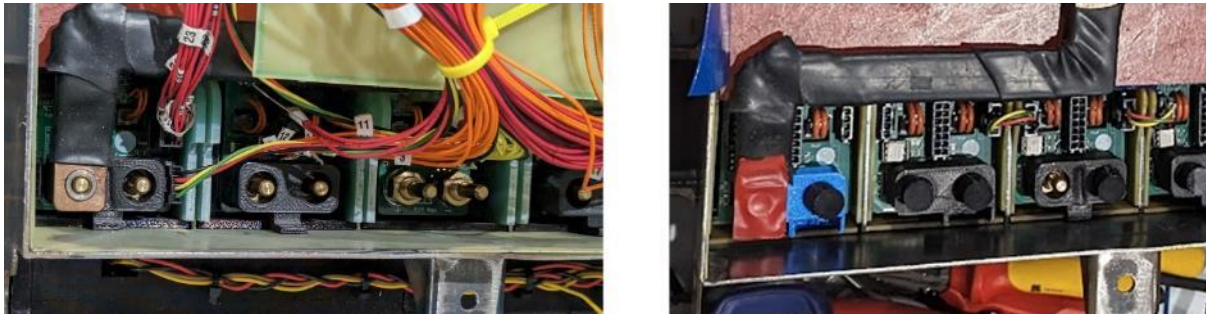


Figure 62 Methode pins without covers, left, and with the previous covers, right.

These covers were designed for individual pins and were extremely easy to lose. This often led to serviceability issues as time was lost finding the covers to continue work on the accumulator. Not only did this cause disruption to work, but also posed a safety risk.

Another concern with this design, was the optics during technical inspection. It was paramount to avoid a scenario where the team was struggling with small Methode pin covers during the technical inspection. This was particularly pertinent, as it was a requirement to complete the accumulator inspection wearing high Voltage protective gloves, reducing dexterity.

The combination of these factors resulted in the implementation of a more ergonomic and larger solution. Initially, it was proposed to design a service handle style solution, that would take the same form as the service handle, without the busbars included. This would provide a solution requiring only one part installed to isolate the Methode Buds, as opposed to 18 individual covers for each pin.

However, this kind of design posed one major issue. If a segment were to be removed, a single cover would no longer be feasible, as the removed segment would not be protected. As a result, it was decided to make a single part per segment, allowing each segment to be removed, with the Methode covers still intact, shown in Figure 63.

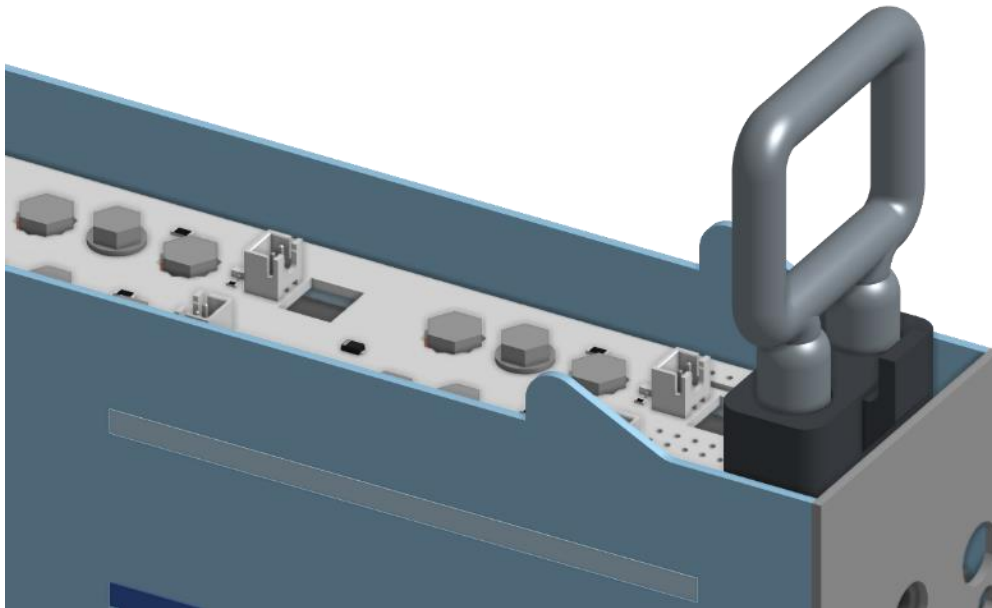


Figure 63 New Methode pin cover design.

This design was much more ergonomic, and easily used even with high Voltage PPE on. During technical inspection, the scrutineers did commend the organisation and safety measures the team had implemented with the Methode pin covers. However, the development of a single service handle style Methode pin cover is still recommended, for the simple instance of tech inspection, or when working on just the top plate of the accumulator. The simplicity and speed would still be considered advantageous in these scenarios, the latter happening regularly.

Track Day and Competition Support

While this report focusses on the design aspects of this project, track day and competition contributions are also worthy of mention. The team completed 33 track days throughout 2023, each one requiring the efforts and time of all team members to make them possible.

While credit for organisation, strategy and scheduling is duly given to the team leader Marisa Mclean, many hours were taken packing and transporting every track day. Particularly on track days where both NU23 and NU16 were taken, the amount of equipment took several hours every track day to pack. Tools, spares parts, driver gear, marshalling equipment and gazebos made the two-hour trip to Sydney Motor Sport Park (SMSP) and back, with one such day shown in Figure 64.



Figure 64 NU Racing setup at SMSP for a track day in August 2023.

Of the 12 SMSP track days the team held, nine were attended by the author, providing support in set up and pack up, as well as driving university vehicles to and from Sydney to transport team members and equipment. This included attending every track day on campus assisting in setting up the test track, marshalling and transporting gear to and from the TA building on campus, to the carpark for testing.

Additionally at competition, the team leader delegated responsibility to the author to supervise the accumulator removal and install. While many team members were more than capable of this task, there were many risks, both to personnel and to the vehicle if the job were not completed correctly. With NU23 on stands, and the accumulator being lifted via a trolley with a scissor lift, it was important that care was taken to prevent a serious accident while fastening the accumulator bolts to the chassis mounts. A photo midway through this process is shown in Figure 65.



Figure 65 Accumulator being installed at competition, with the car on stands.

Care had to be taken to keep track of all accumulator fasteners. During competition, the accumulator, as well as the accumulator lid, had to be removed on several occasions for technical inspections and charging. It was the responsibility of the author to ensure these parts were accounted for and not lost during installing and removal of these components.

Furthermore, at competition the author was tasked with assisting during the accumulator technical inspection, shown in Figure 66. This involved minor disassembling of the accumulator while observing strict high Voltage safety rules and adhering to the required personal protective equipment, as shown in Figure 66. The author was also the designated representative for answering any mechanical related questions the scrutineers had regarding the accumulator. This required preparation and several hours of multiple simulated technical inspections to ensure thorough knowledge and coherent responses could be delivered during the inspection.



Figure 66 Accumulator technical inspection at the 2023 FSAE-A competition.

Commissioning and Validation

Eight Segment Accumulator

On July 30, NU23 embarked on its first track test on the university campus, complete with the eight-segment accumulator. This was simply a shakedown of all the systems on the car, many of which were completely overhauled from EV.Three, and ensuring they were integrated appropriately. This resulted in extremely mellow driving conditions, not placing the car under extreme load, and thus generating little to no heat in the accumulator. Unsurprisingly, no issues related to the accumulator cooling were experienced, and the car was considered ready for its first day at Sydney Motorsport Park.

NU23 was taken to SMSP on the 2nd of August for its first track testing. Shown in Figure 67, the car was in a very early stage of the development process, with only absolutely necessary systems, the biggest omission being the body kit which was yet to be completed.



Figure 67 NU23 at its first Sydney Motorsport Park track day on August 2nd, 2023.

NU23 exhibited good, fault free performance in the morning, which resulted in the decision to complete a full endurance in the second stint of testing after charging the accumulator. An ambient temperature of under 20 °C meant that conditions were ideal for maintaining cool temperatures. Additionally on this occasion, the team was allocated the figure 8 track at SMSP, which does not allow

for long straights. This reduces the amount of sustained, high current draw sections per lap, further reducing the likely overall temperatures of the accumulator.

The car managed to complete the full 20 km of endurance on its first attempt, without exceeding the temperature limit in the accumulator. As can be seen in Figure 68, temperatures did not exceed 33°C. This was an extremely positive result for the car as a whole, showing strong reliability so early in the testing phase and in the year.

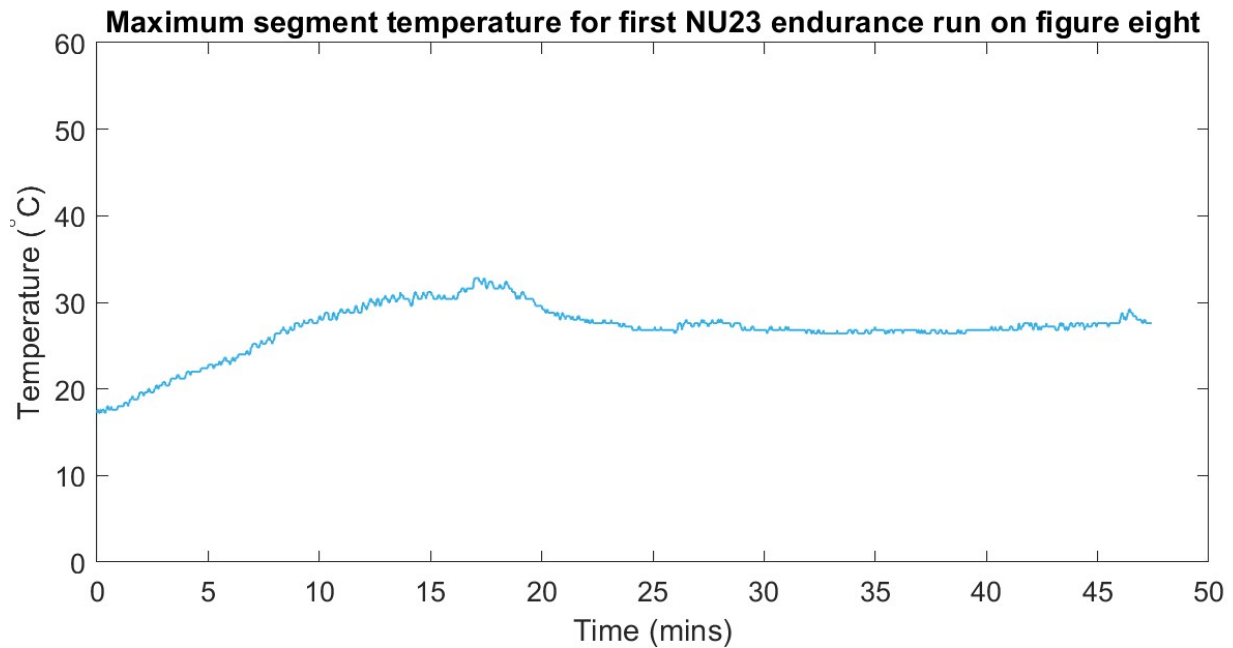


Figure 68 Maximum segment temperatures for first NU23 endurance run on figure eight.

While there was no immediate limiting factor, the car was at 4% state of charge at the closing of endurance. Given a more open, faster track, it was clear that state of charge could become a factor limiting the cars performance in endurance, more than the temperature of the accumulator currently appeared to be.

Furthermore, this low maximum temperature was reached despite the drivers reaching 100% throttle on multiple occasions throughout the lap denoted by Apps out gated, shown in Figure 69. It is also clearly noticeable that in the second stint, throttle input was significantly lower than the first stint. The second driver, Robert Stanley, confirmed this was not due to a temperature monitoring, but rather limitations in the track.

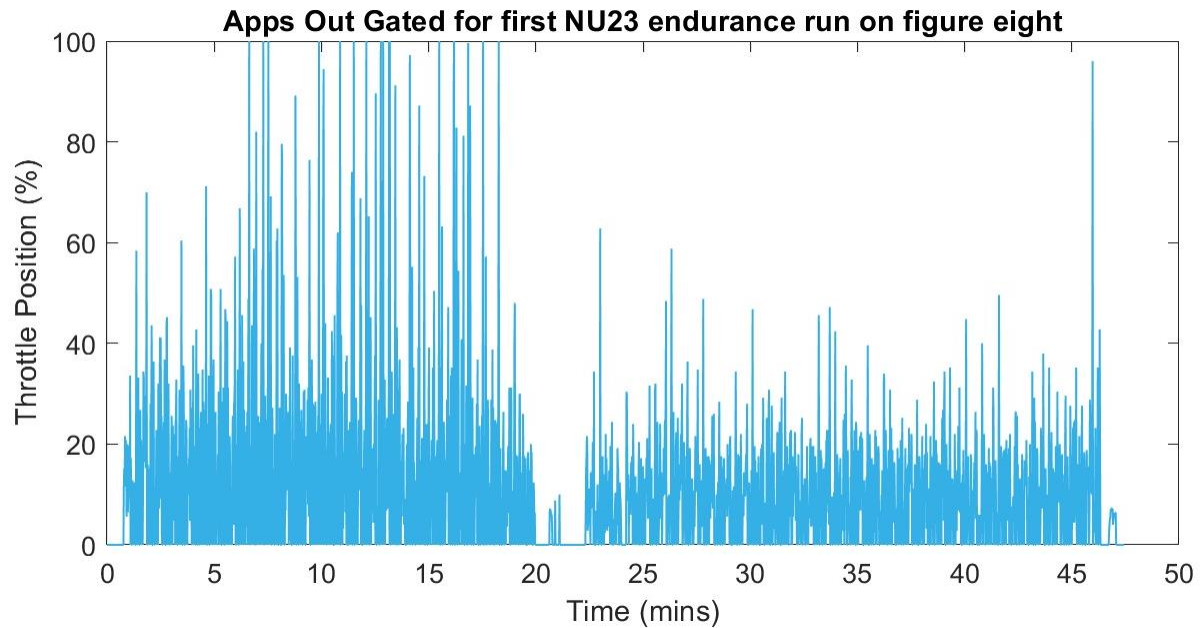


Figure 69 Throttle input during first endurance run.

However, the following visit to SMSP allowed for full endurance testing on a more indicative track. The SMSP skid pad allowed for a track with longer straights for more sustained current draw in comparison to the figure eight track used for the first test. This can clearly be seen when examining the current draw, shown in Figure 70, with more sustained peaks well above 150 A in the skid pad endurance.

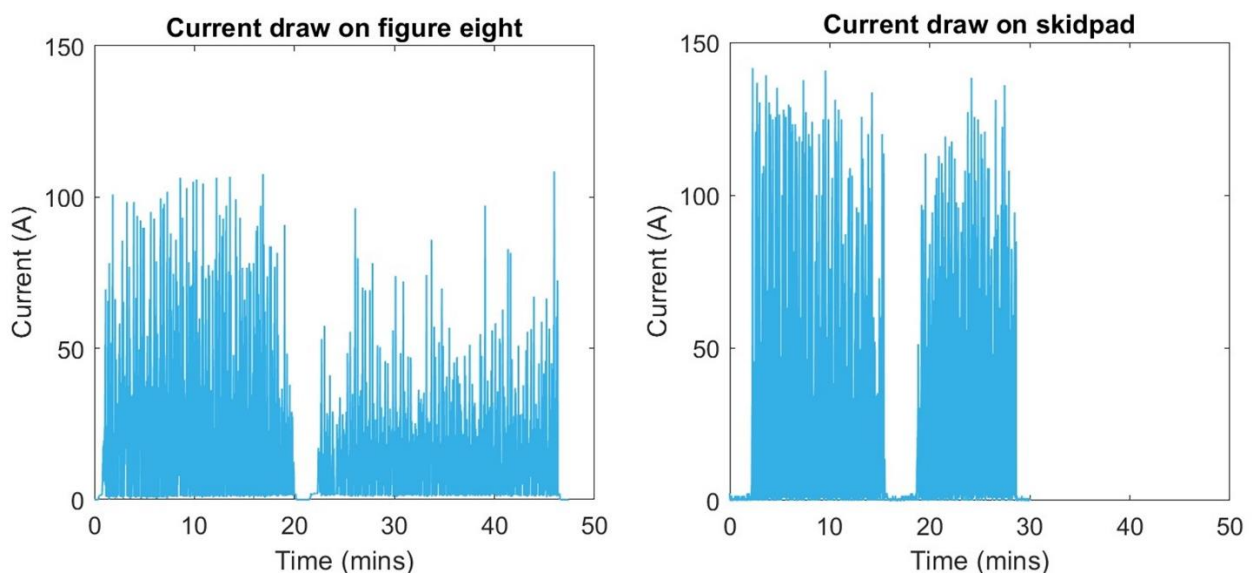


Figure 70 Current draw from figure eight track and skid pad endurance.

This resulted in significantly higher temperatures being generated in the accumulator, despite a lower ambient temperature. On the first track day, an ambient temperature of 17° C was measured, while

on the second, skid pad track day it was 12° C. Despite this, temperatures of up to 55° C were seen on the skid pad track day, as shown in Figure 71.

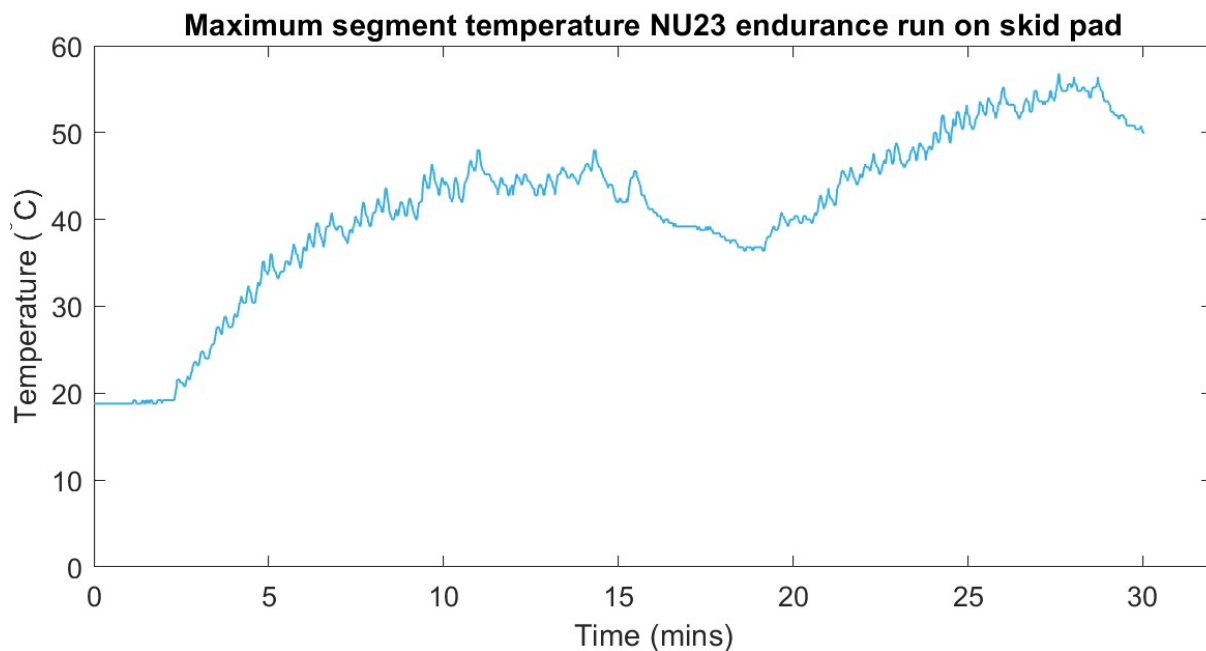


Figure 71 Maximum segment temperature NU23 endurance run on skid pad.

While this data implied that there may little to no improvement in the cooling performance, driver feedback suggested otherwise. Jye Hollier reported that through the assistance of the Motec dashboard, he was able to actively monitor and control the temperature of the accumulator, via the maximum segment temperature read out on the display. As temperatures rose, by moderating the duration of extended throttle inputs, the temperature began to plateau or even drop.

This can be seen in the final couple laps, where temperatures did spike above 55° C. Following this, no throttle input above 50% was registered on apps out gated. This resulted in the car's temperature plateauing, and not continuing to increase as it had done.

This quick drop in temperature can also be seen at the drivers swap at the middle of the run. Maximum segment temperature was at 46° C when NU23 stopped and after two minutes, had dropped to 38° C. This is the mandatory stop duration in competition and shows that heat was actively being dissipated from the cells.

In stark contrast, EV.Three's cooling set-up saw it drop from 34° C to 33° C in max segment temperature during its two-minute driver swap at the 2022 competition. This can be seen in Figure 72, where temperature seemingly increases consistently throughout the event, with a minor plateau at the driver swap around the 20 minute mark, with no discernible decrease in the segment

temperatures. It is also immediately clear just how much faster NU23 completed the 20 km, finishing over 10 minutes quicker than EV.Three did.

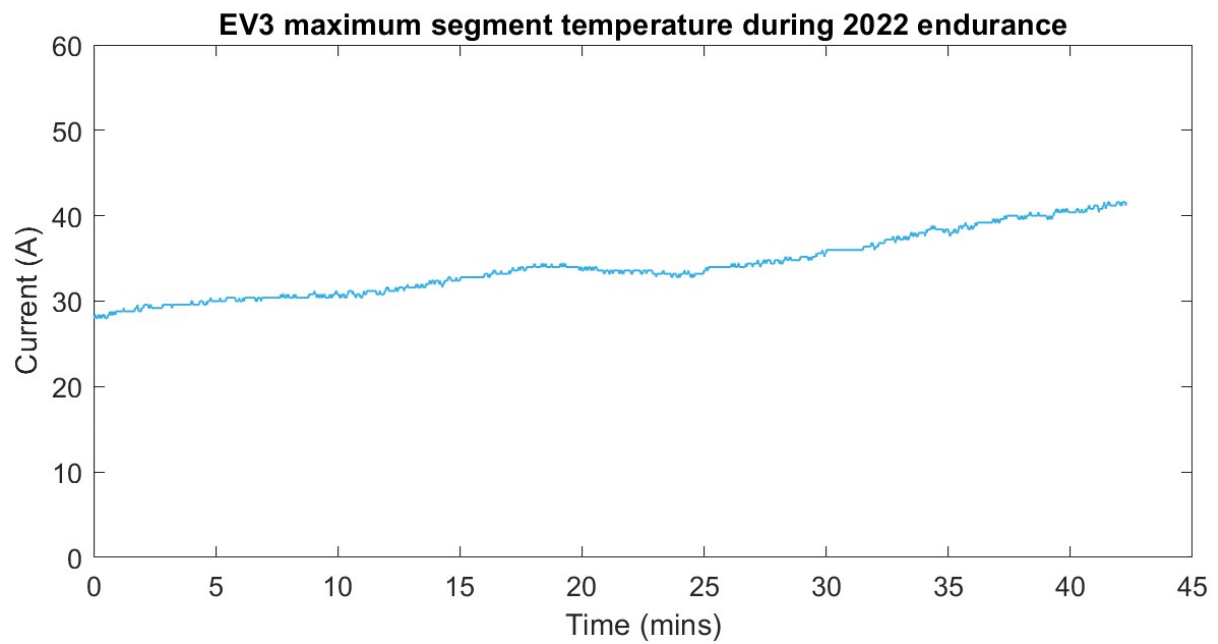


Figure 72 EV.Three maximum segment temperature during 2022 endurance.

Nine Segment Accumulator

Validation of a nine-segment accumulator began even before the design stages. When initial discussions began about the challenges associated with adding a segment, decreased cooling capacity was flagged. The addition of another segment would reduce the width of the segment housings, as discussed extensively in the “Initial System Design” section of this report.

Consequently, a test was developed whereby it could be verified that the improved cooling performance in NU23’s accumulator was not solely due to the increased width of the segments, but also a function of the new fan orientation. This was done by having four of the segments’ fans depinned on both the intake and exhaust side of the accumulator, leaving four segments without cooling from fans. This was to compare whether the fans made a difference to the temperature of the segments, or whether simply the wider segments were creating the better cooling performance. Following this, NU23 completed approximately 30 laps to generate heat in the segments. The results can be seen in Figure 73.

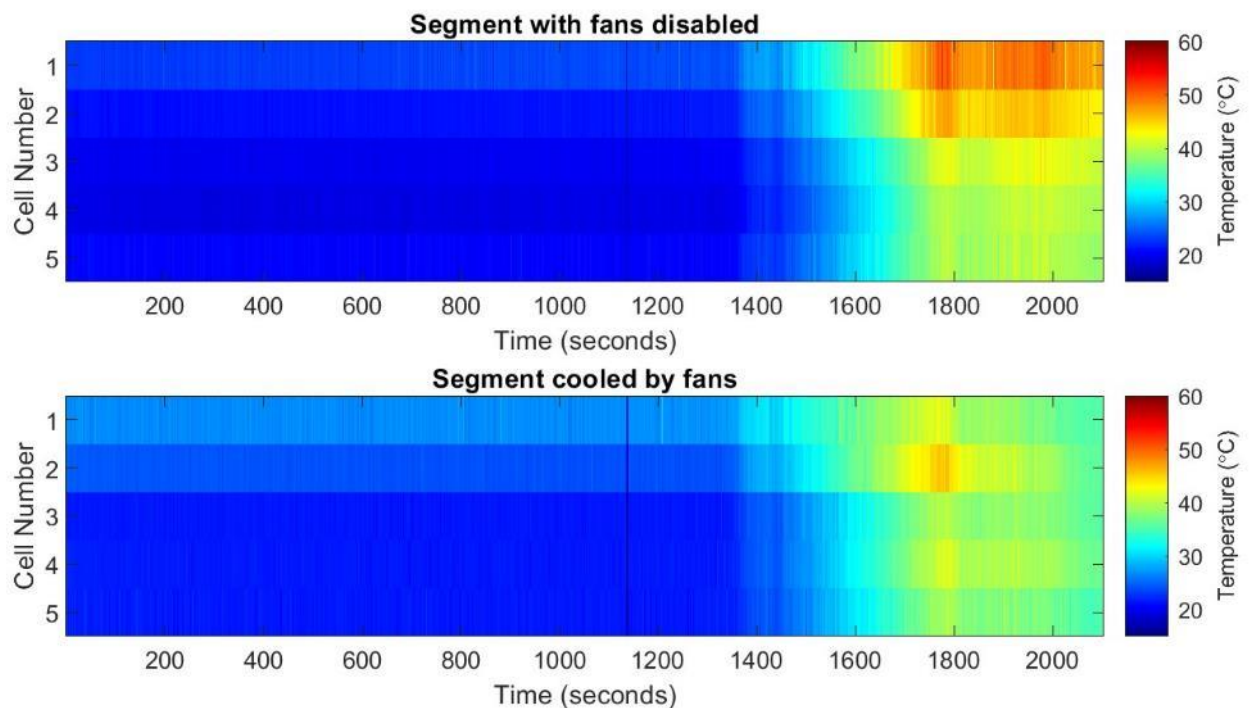


Figure 73 Passively cooled segment (top) versus actively cooled segment (bottom) in NU23.

Figure 73 shows that once NU23 started driving, the temperatures slowly began to increase on both the actively and passively cooled cells. However, one end of the passively cooled segment began heating up significantly higher than the rest of the segment, to the point where it reached 10° C higher than the hottest cell in the actively cooled segment.

Another interesting feature about the data is the increased temperature at one end of the segment for the passively cooled segment. It was theorised that the service handle might create additional heat at its end of the segment, due to the large copper busbars housed within it to connect the segments. This was considered when choosing the respective sides for the intake and exhaust fans, as it was suggested that the hotter end of the segment should receive fresh air, therefore having the intake on that side. This testing seemingly proved that the service-handle-end of the segment was indeed the hottest end of the segment, although it wasn't proven whether it was indeed the service handle increasing the temperature.

While only two segments are examined in Figure 73, other segments told a similar story, with maximum temperatures reaching 51° C and 41°C for the passively and actively cooled segments respectively. It was therefore concluded that the increased cooling performance was not solely a result of the wider segments, but that the fans were making a significant contribution. This therefore gave the team the confidence to pursue a nine-segment design knowing that the increased cooling performance was thanks to the fans cooling configuration.

Commissioning of the nine-segment accumulator was much the same as the eight-segment with some additional challenges. Given that this accumulator was going to be taken to comp, additional work had to be done that was never completed on the eight-segment accumulator.

FR4 lining of all the interior sides of the exterior walls was required, as discussed in the “Additional Work” section of this report. This step which was not completed on the eight-segment accumulator, as the commitment to a nine-segment accumulator was made prior to its completion. This caused significant issues with clearances, as the designed clearances did not translate to the physical container. Pieces affecting the clearances at the end of the segment are shown in Figure 74, where large clearances were included in an attempt to ensure the segments fit in the container easily.

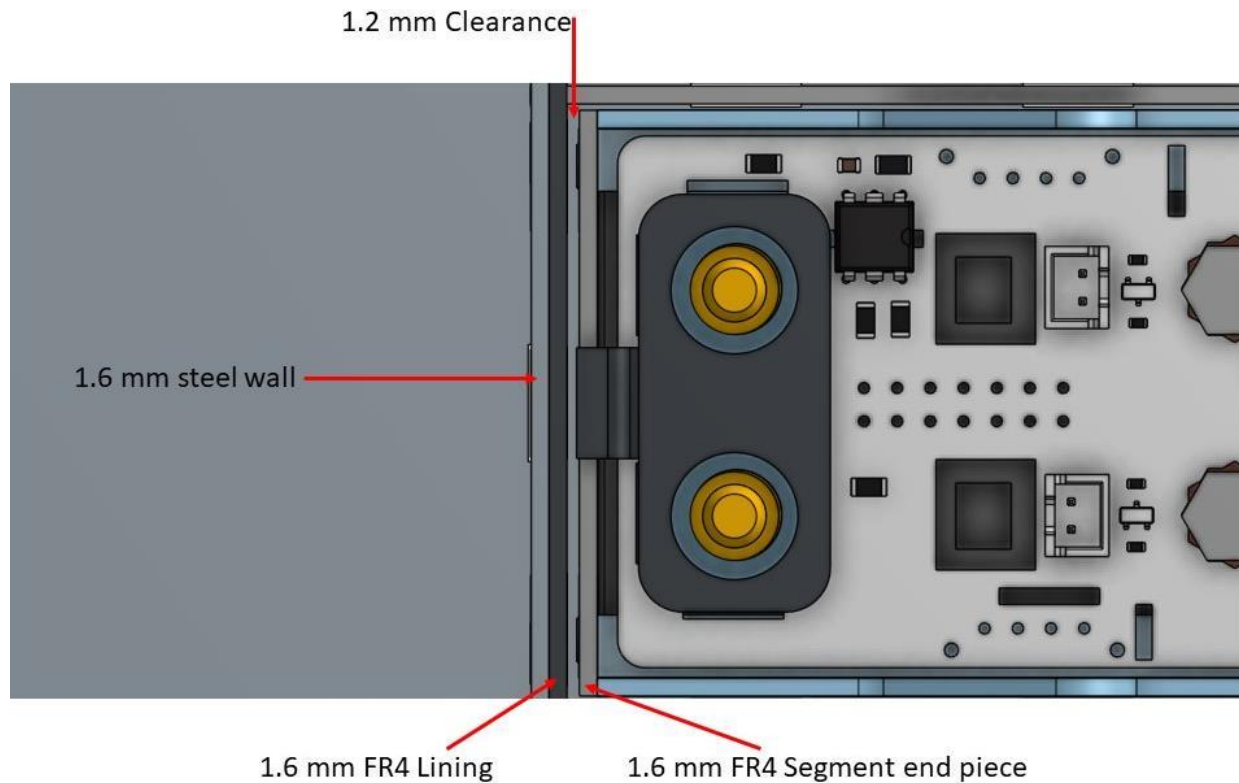


Figure 74 Clearances on end of segments.

However, due to variations in manufacturing, the segments simply could not fit with the 1.6 mm lining installed. This meant modifications had to be made. Discussions started with creating thinner end pieces for the segments. This could have been achieved through 3D printed end pieces, however it would require the use of flame-retardant ABS, which would take more time than was available to the team. It was ultimately decided to remove the end piece from the segment all together, as the FR4 lining on the wall of the accumulator would provide the required insulation, even with the segment end pieces removed.

This required some cutting of the segment housings. The pins on the ridges of the segment, used to locate the end pieces, and the tab on the base of the segment housing were all removed to enable the segments to fit. The segment after having these modifications can be seen in Figure 75. These modifications however did provide near perfect fit in terms of the length of the accumulator, which was fortunate as the team was left with little time and resources to rectify the problem.



Figure 75 Modified segment housing with pins and tab removed.

However, clearances along the sides of the segments were variable at best, with some segments being extremely tight in comparison to others. This was likely due to slight variances in manufacturing, both in the accumulator container, and the segment housings. Consequently, some segments required extreme force to install and remove.

This large amount of force often saw the tape used to remove the segments tearing, shown in Figure 75, resulting in an extremely challenging removal process. Left with no other option, this often led to the use of insulated pliers to grip the walls of the segments housings to pull them out of the container. This loaded the segment dangerously, as the tape wrapped the entire segment around the base, lifting it from the bottom, whereas the pliers essentially lifted the segment via the wall. This caused all the load from lifting the segments to be transferred through the epoxied connection between the wall and base, rather than directly into the base like it was with the tape.

This is likely what caused the segment failure, shown in Figure 76. On one occasion when removing segments from the accumulator, it was found that the housing on segment five, a particularly tight segment, had its walls separated from the base. However, on inspection, it was also clear that less than optimal amounts of epoxy were found in the area that had failed. Likely a combination of these two factors, the housing was repaired with additional epoxy, and swapped to a segment that didn't suffer from such a tight fit reducing the amount of force required to remove it from the container.



Figure 76 Failure in segment with base separating from walls.

Although this failure was likely more a result of poor epoxy application, work begun remedying the tight clearances on the sides of the segments. As the internal dividing walls were 1.6 mm steel, there was little that could be done in terms of modifications to the container. Some material was removed from the housings in what was considered the tighter spots which yielded some success, but the biggest improvement came through changing the type of tape used.

The original tape was a PVC Duct Tape, designed for being torn without scissors. This was extremely undesirable in the case of segment removal, as when any lateral force was applied to the tape, it often tore. It was therefore decided to purchase a Fibreglass tape, that advertised being “non-tear able” thanks to Fibreglass filament running lengthwise through the tape. The two different tapes can be seen in Figure 77.

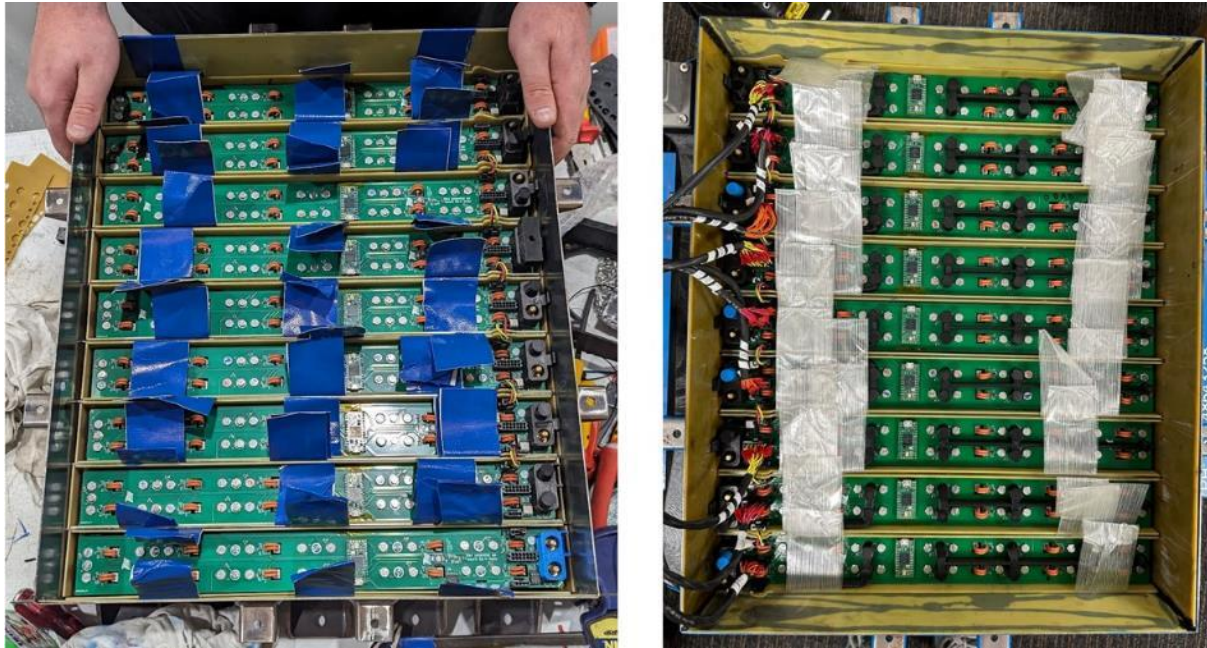


Figure 77 PVC Duct tape originally used, left, and Fibreglass tape that replaced it, right.

While the change in tapes was ultimately made due to the added strength, an unexpected benefit was the reduced thickness of the Fibreglass tape. Upon installation, it became clear that the fibreglass tape was significantly thinner, resulting in a looser fit for the segments. Previously difficult to remove segments became easier, without the need for excessive force. This was a substantial improvement, as it significantly decreased the removal and installation time of segments, increasing the serviceability of the accumulator, and lowering the risk of a critical housing failure.

The cooling performance of the nine-segment accumulator was also a big question during the commissioning stage. Interestingly, the first full endurance run indicated that state of charge would be the limiting factor, rather than accumulator temperatures. Figure 78 shows that hard driving in the first stint of endurance left less than 40% state of charge for the second stint, without even exceeding 55° C.

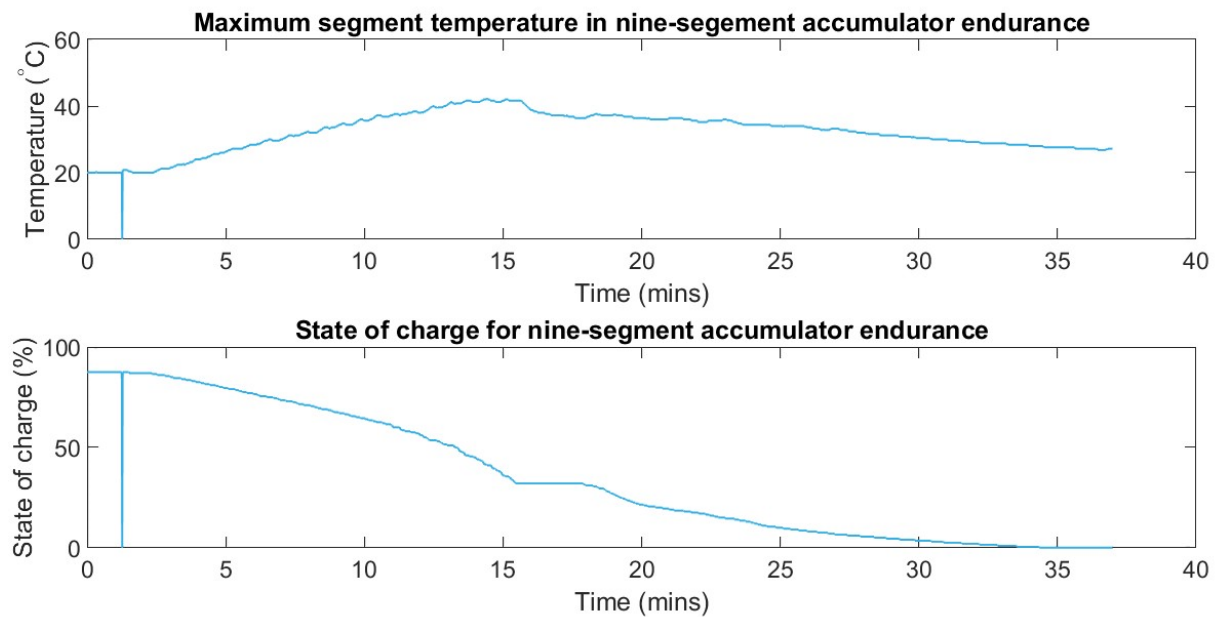


Figure 78 NU23's first complete endurance run with nine segment accumulator.

When compared to an eight-segment accumulator run on the exact same course, with similar ambient temperatures, it appeared that nine segment accumulator was able to discharge more energy without generating any more temperature in the segments. This eight-segment accumulator endurance run is shown in Figure 79.

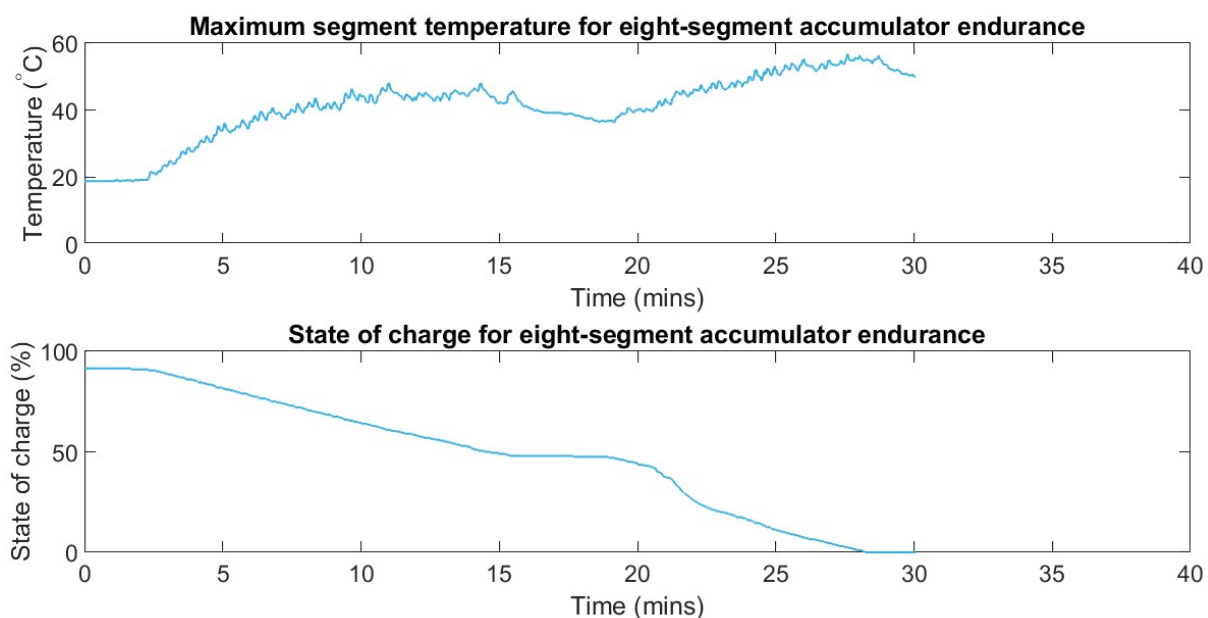


Figure 79 NU23 endurance run with eight segment accumulator.

When comparing these two sets of data, it is clear that the nine-segment accumulator was able to discharge more energy, while maintaining the same maximum temperatures as the eight segment

accumulator at 45° C. However, driver error meant that too much charge was used, resulting in extremely moderated driving in the second stint, so much so that the overall endurance time in the nine-segment accumulator was far greater than the eight-segment endurance run.

This improvement was likely a function of the reduced current draw required for a given power output. This can be seen in Equation 8, where for a constant power, increasing the Voltage would decrease the current. This decreased current draw for the same power would result in less temperature generation within the accumulator and is likely the main factor in the perceived cooling improvement.

Equation 8 Power equation, where P = power in Watts, V = Voltage, I = current in Amps.

$$P = VI$$

Another factor however that may have affected the cooling performance was the redesigned manifolds. As discussed in detail in the “Initial System Design” section of this report, the relocation of the fans to create high and low pressure zones within the manifold may have led to the detectable improvement in cooling performance. While less likely to be the main contributing factor compared to the aforementioned current draw reduction, further testing could have been completed to confirm. Two different manifolds, one in the eight-segment accumulator configuration, and one in the nine-segment accumulator configuration, on the same accumulator could have proved whether the manifold re-design provided any added performance. With more time and resources, this could have a valuable test to confirm the superior manifold design.

Another step in the commissioning stage, was to test the thinner 25 mm axial fan performance. As discussed in both the “Initial System Design” and “Compliance Report” sections of this report, a manifold was designed to house a 25 mm fan at the back of the accumulator, due to a possible rules noncompliance. For more details on this, refer to the mentioned sections of this report.

During the final test day before competition, the 25 mm fan was substituted into the fan manifold, so its performance could be observed, in preparation for a possible last-minute change at the competition. Due to time constraints, rather than taking the time to relocate the fans to the slim fan manifold, the 25 mm fan simply replaced the 38 mm fan at the back of the current fan manifold.

Testing was completed on the University campus, around the short track used for on campus testing. With no exact event being simulated, just a final shake down of the vehicle, it can be seen there was no major delta in average temperature between the segments, shown in Figure 80.

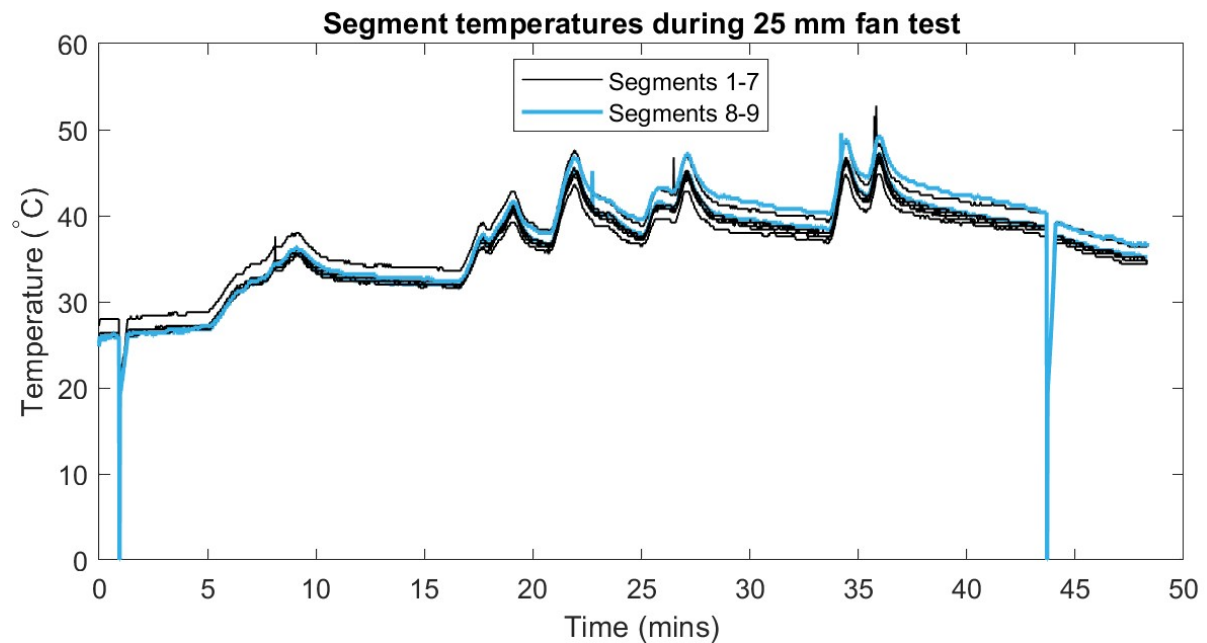


Figure 80 Segment temperatures with the thin fan at the segment nine end of the accumulator.

This was a positive result, as the team's confidence in the 25 mm fans was greatly improved, should the team be forced to swap them in. Despite this positive result, the team was still committed to trying to pass tech inspection without the 25 mm fans installed. Given the majority of testing was completed with all 38 mm fans, and the consequential driver's familiarity with their performance, it was preferable to be able to run this set up for competition. Ultimately, as discussed in the "Compliance Report" section, NU23 did pass technical inspection with the 38 mm fans and manifold installed.

Performance at Competition

NU23 achieved some of the highest results in various events for a NU Racing electric vehicle. The car showed competitive pace, as well as performing well in static events.

As expected, for skid pad, acceleration and autocross, the accumulator cooling was more than capable of keeping the accumulator within legal temperatures, shown in Figure 81. This enabled the car to achieve an impressive 2nd place in skid pad and autocross, as well as 7th in acceleration.

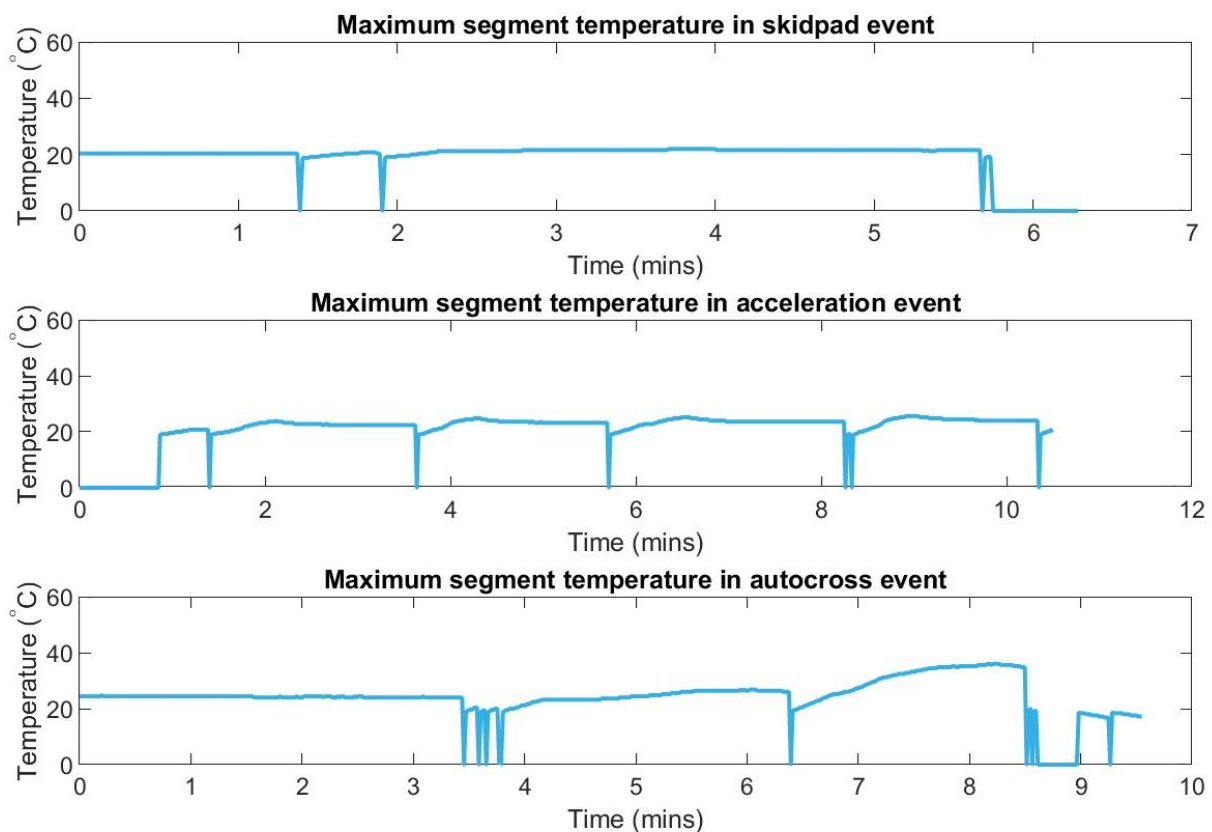


Figure 81 Accumulator temperatures for NU23 during the 2023 Skidpan, top, acceleration, middle, and autocross, bottom, events.

While it was expected that the car could complete the autocross event without temperature issues, it is worth noting EV.Three's performance in the event at the 2022 competition. Figure 82 shows EV.Three exceeding the temperature limit at 62° C in the second lap attempt of autocross.

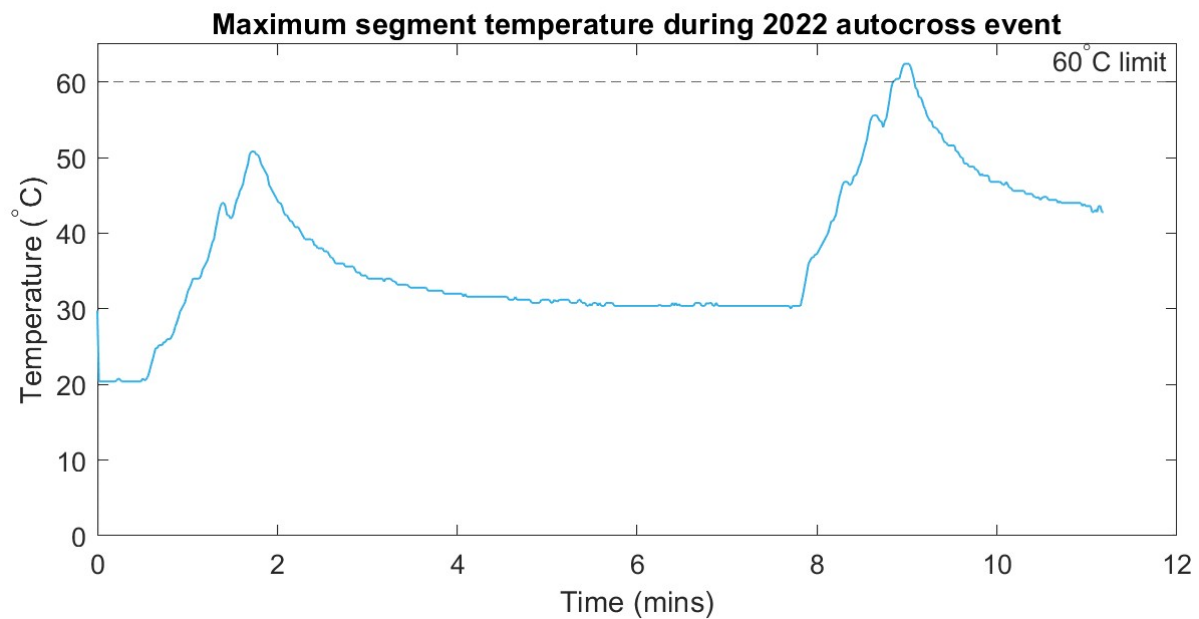


Figure 82 EV. Three segment temperatures in the autocross event.

However, the true test of NU23's cooling performance was in the endurance event. As discussed in the "Commissioning and Validation" section of this report, NU23 was limited by state of charge or temperature depending on the driver and conditions. Given the endurance event was taking place at an unfamiliar track, where no previous testing had taken place, a conservative driving approach was taken.

Tim Kerr, driving the first stint of endurance, was requested to maintain temperatures around 40° C, and to finish the first stint with more than 50% state of charge. As can be seen in Figure 83, driver swap occurred just after the completion of lap 8, around the 17-minute mark, with a maximum temperature in the accumulator of 43° C reached, and 59% state of charge left.

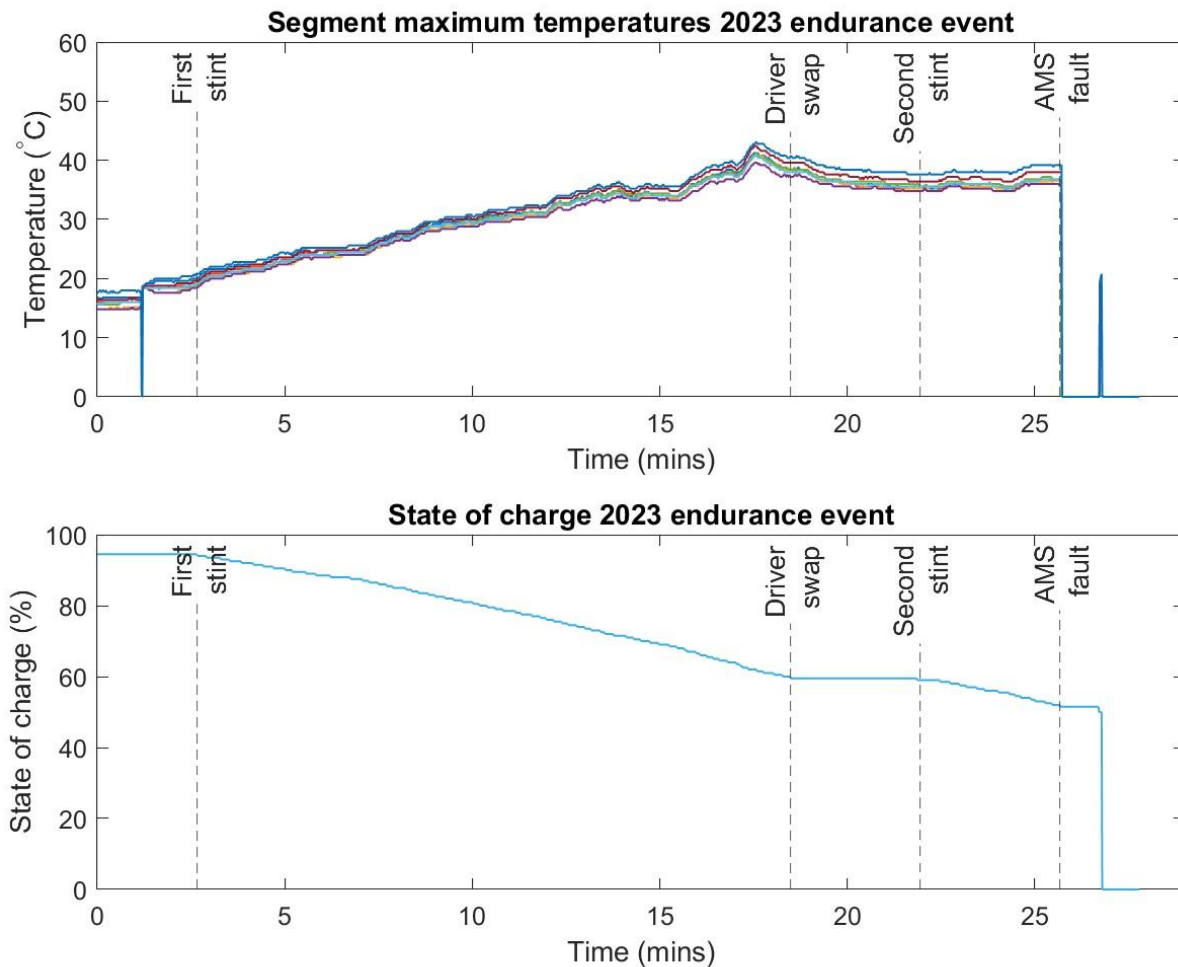


Figure 83 Temperatures and state of charge of NU23's accumulator during the 2023 endurance event.

Unfortunately, shortly after the driver swap, NU23 encountered an AMS fault, with all reported temperatures going to zero, and ending NU23's endurance run. The error was attributed to an Arduino board failing to report the temperature data on the LVD temporarily. While this resulted in an extremely poor endurance event score, the team was confident that without this fault, the cooling would have maintained temperatures and enabled the car to finish the event at the same pace.

To try and compare the cooling performance of NU23 to EV.Three, the difference in average lap time between both cars, and the respective winners for endurance in both years was compared. As can be seen in Table 4, in the first stint of endurance, EV.Three was 8.03% slower on average than Wollongong, the winner of the 2022 endurance event. In comparison, NU23 was faster in the first stint than the eventual winner in 2023, Adelaide, managing to be 6.92% quicker on average.

Table 4 Comparison between EV.Three and NU23's first stint lap times to the respective winners of endurance in 2022 and 2023 (NATSOFT, 2023).

Year of competition	NU Racing average lap time (m:s)	Leaders average Lap time (m:s)	Variance of NU Racing from leader (%)
EV.Three - 2022	1:38.35	1:31.04	+8.03
NU23 - 2023	1:48.63	1:56.71	-6.92

This is a significant improvement on EV.Three's pace in the first stint, which was entirely limited by temperature within the accumulator. However, it is by no means saying that without the fault that ended NU23's endurance, it would have won in 2023. Based on the temperature trend seen in the first stint, it isn't certain the average time set in the first stint would have been maintainable without exceeding 60° C. This, paired with the fact that Adelaide dropped their average time in the second stint to 1:41.01 seconds, would likely mean they would have still likely ended on top of the endurance event. Additionally with Monash maintaining an average lap time of 1:40.64 seconds and Auckland averaging 1:35.74 seconds before their respective DNF's, it would be mere speculation to draw conclusions beyond the comparison between EV.Three and NU23's relative performance.

Finally, the full results of the 2023 FSAE-A competition are shown in Figure 84. Notably shown is the fourth place, with 130.33 points, in the design event. While this event covered all design aspects of the car, and no specific feedback was given, the cooling design of the accumulator was discussed at length with the design judges. With various questions relating to the validation and testing stages, and the team as a whole preparing and presenting competently, the team achieved a deserved fourth place behind Auckland, Monash and Canterbury.



Formula SAE-A 2023 Final Results EV Class

Place	Car No.	Team Name	Presentation	Cost	Design	Skidpad	Acceleration	Autocross	Endurance	Efficiency	Total Score
1	E13	University of Canterbury Motorsport	59.51	55.95	138.05	64.84	62.85	99.94	267.68	97.02	845.84
2	E63	UNSW Redback Racing	71.74	58.92	116.89	42.93	68.9	112.87	269.3	54.82	796.37
3	E42	UQ Racing	55.43	99.25	106.85	57.85	18.02	99.74	237.44	100.00	774.58
4	E44	UTAS Motorsport	30.98	43.44	113	30.63	52.44	94.01	249.18	61.16	674.83
5	A65	Monash Motorsport	75	59.09	147.83	52.46	89.92	97.62	9	59.09	590.01
6	E47	University of Auckland Formula SAE Team	63.18	63.94	150	75	100	125.00	6	0.00	583.12
7	E03	University of Newcastle	72.55	51.06	130.33	65.77	50.92	120.99	9	52.52	553.14
8	E14	Curtin Motorsport Team	31.79	36.35	115.69	30.21	46.47	92.68	95.21	86.62	535.02
9	E8	Adelaide University Motorsport Team	54.02	11.38	115.4	0	8.42	0.00	275	54.05	518.27
10	E17	Team Swinburne	50.95	5.38	71	33.98	71.09	0.00	218.88	53.24	504.52
11	E88	RMIT Motorsport	26.09	3.25	75.37	15.65	46.69	39.56	155.48	87.03	449.13
12	E16	UniSA Motorsport	13.45	15.95	101.91	3.5	30.89	6.50	139.77	91.74	403.71
13	E85	UOW Motorsport	51.36	54.05	123.18	0	0	102.25	0	0.00	330.84
14	E46	QUT Motorsport	34.24	21.58	107.97	24.18	4.5	0.00	12	86.21	290.68
15	E20	UWA Motorsport Team	0	80.75	88.12	0	0	0.00	11	87.45	267.32
16	E71	Arjuna EV UGM	61.14	13.38	95.9	3.6	4.5	6.50	10	0.00	195.02
17	E111	Griffith Racing Team	25.68	37.49	53.07	0	0	0.00	12	0.00	128.24
18	E59	UTS Motorsports Electric	63.59	8.38	17	0	0	0.00	0	0.00	88.97
19	E301	Melbourne University Racing	39.13	14.58	32	0	0	0.00	0	0.00	85.71
21	E105	Taiwan Tech Racing	13.04	0	0	0	0	0.00	0	0.00	13.04
22	E15	Academy Racing	0	9.38	0	0	0	0.00	0	0.00	9.38
23	E106	NYCU Vulpes Racing	0	0	3.43	0	0	0.00	0	0.00	3.43

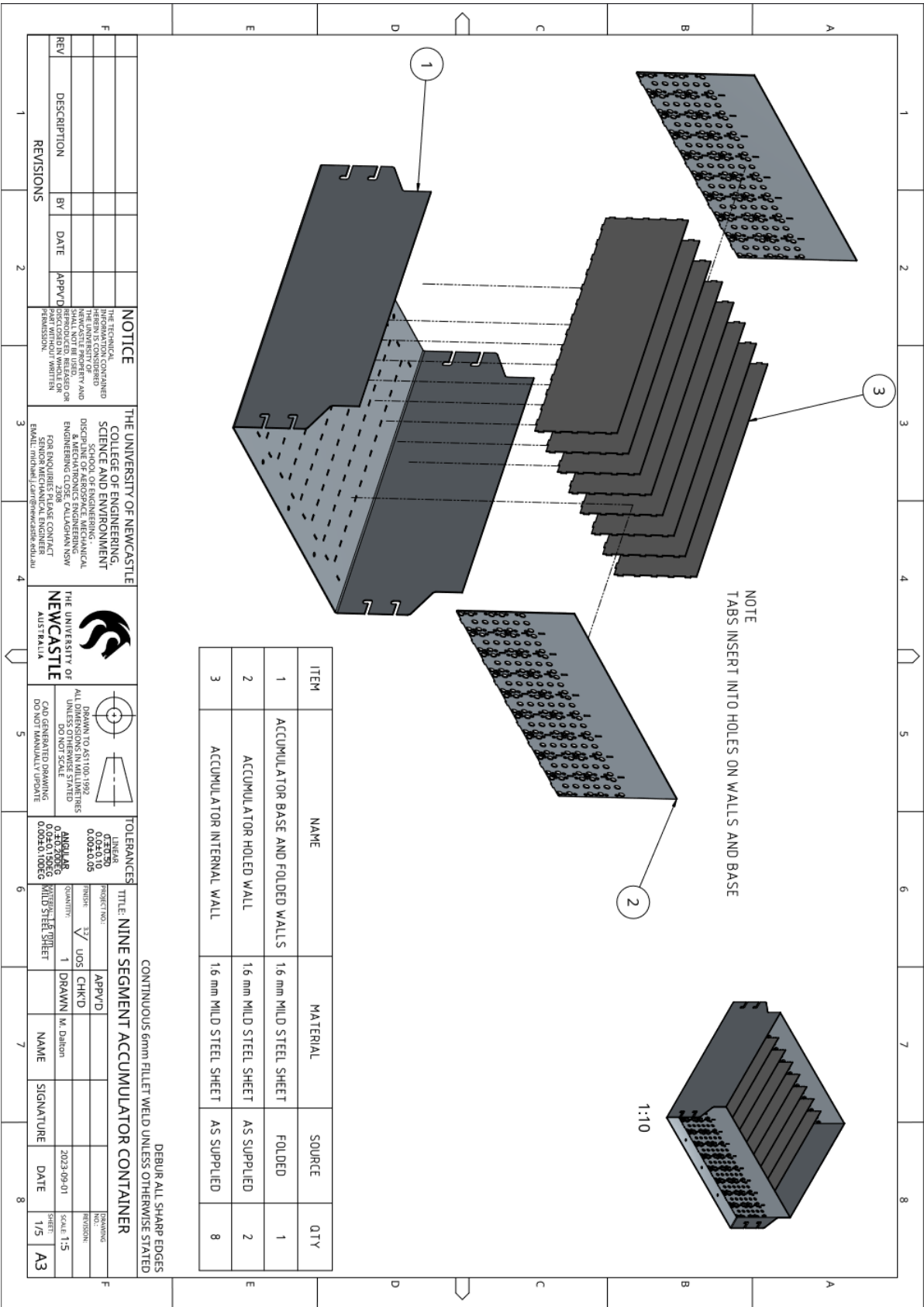
Figure 84 Complete results from FSAE-A 2023 competition.

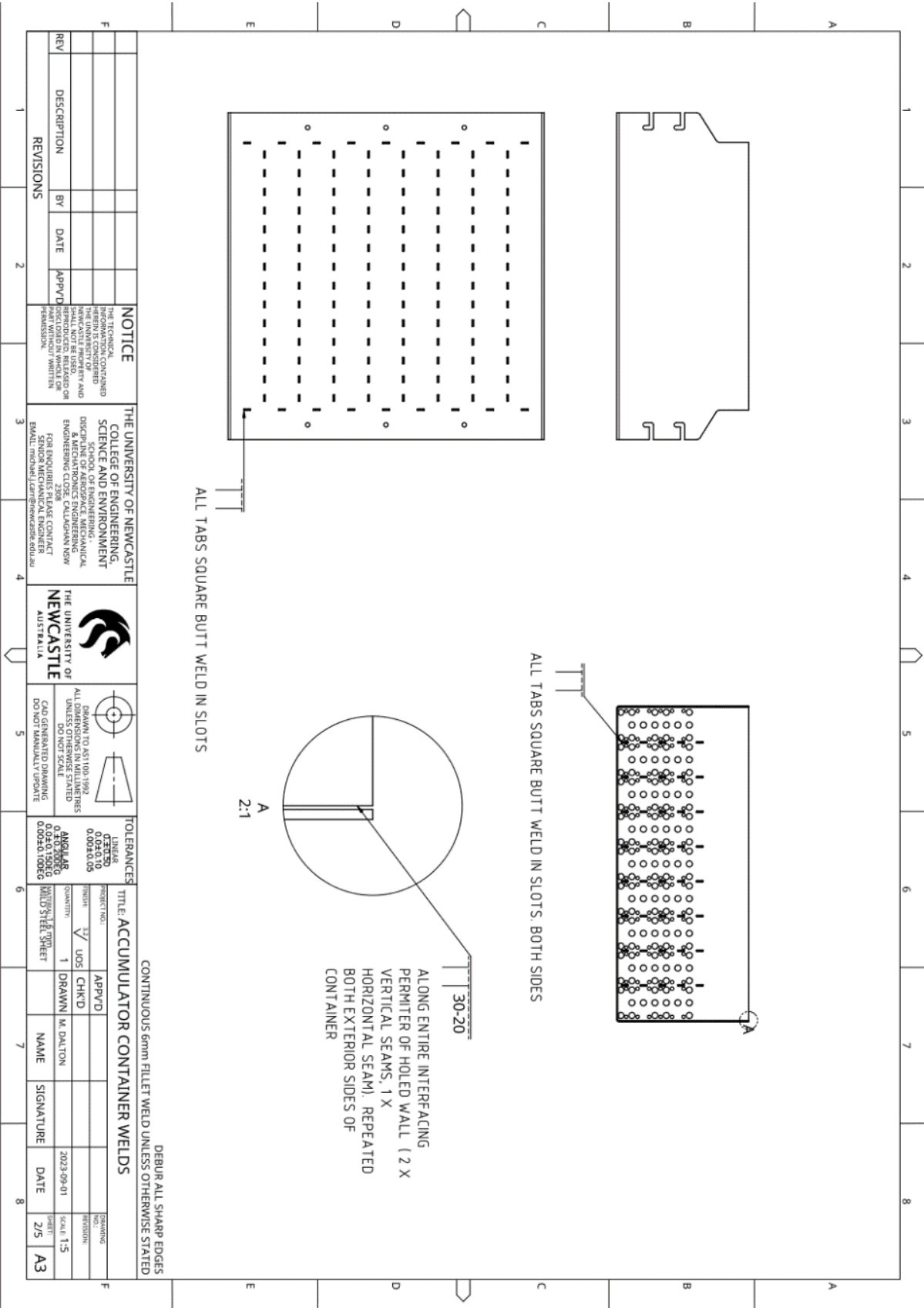
Bill of Materials

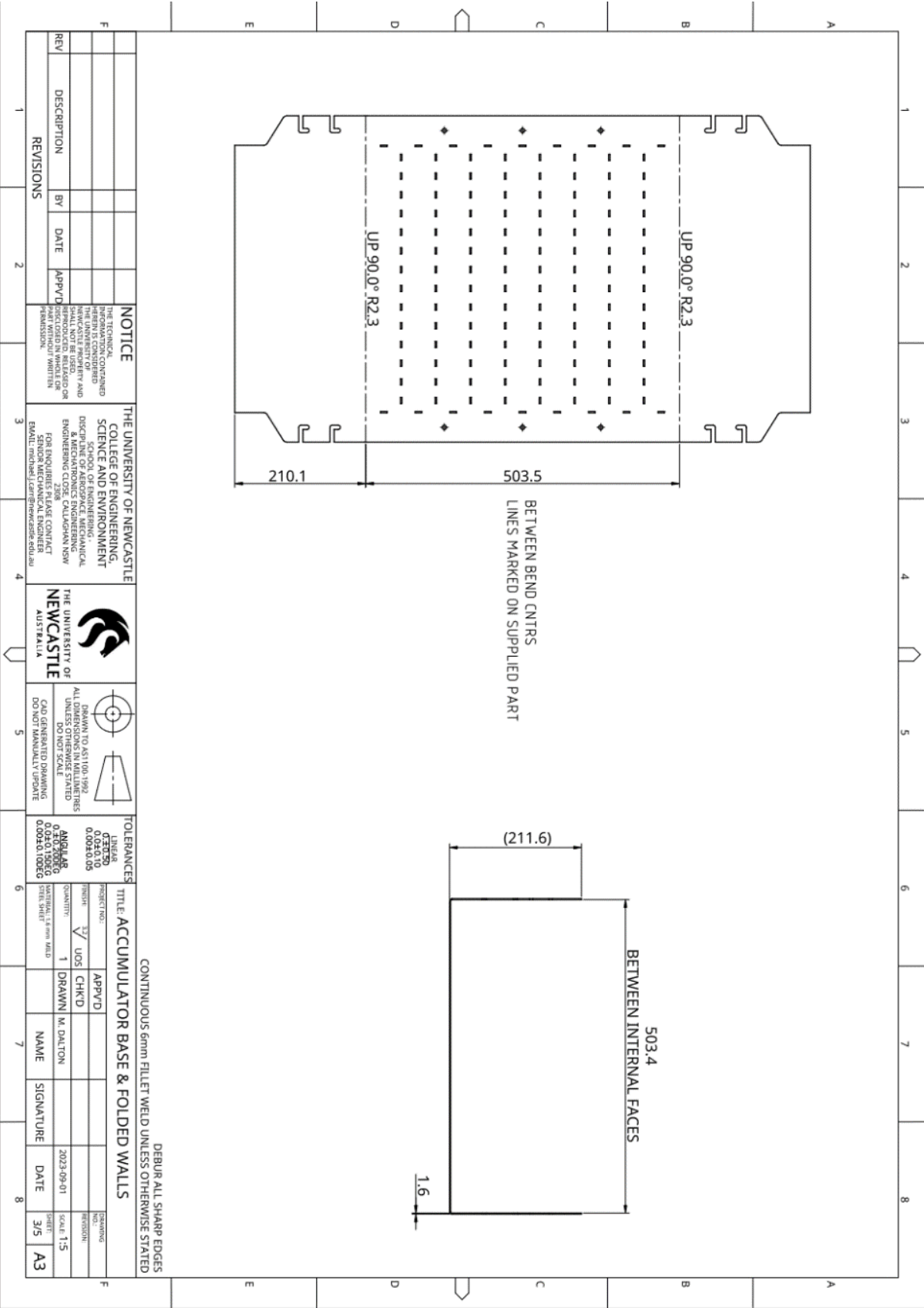
Table 5 Bill of materials.

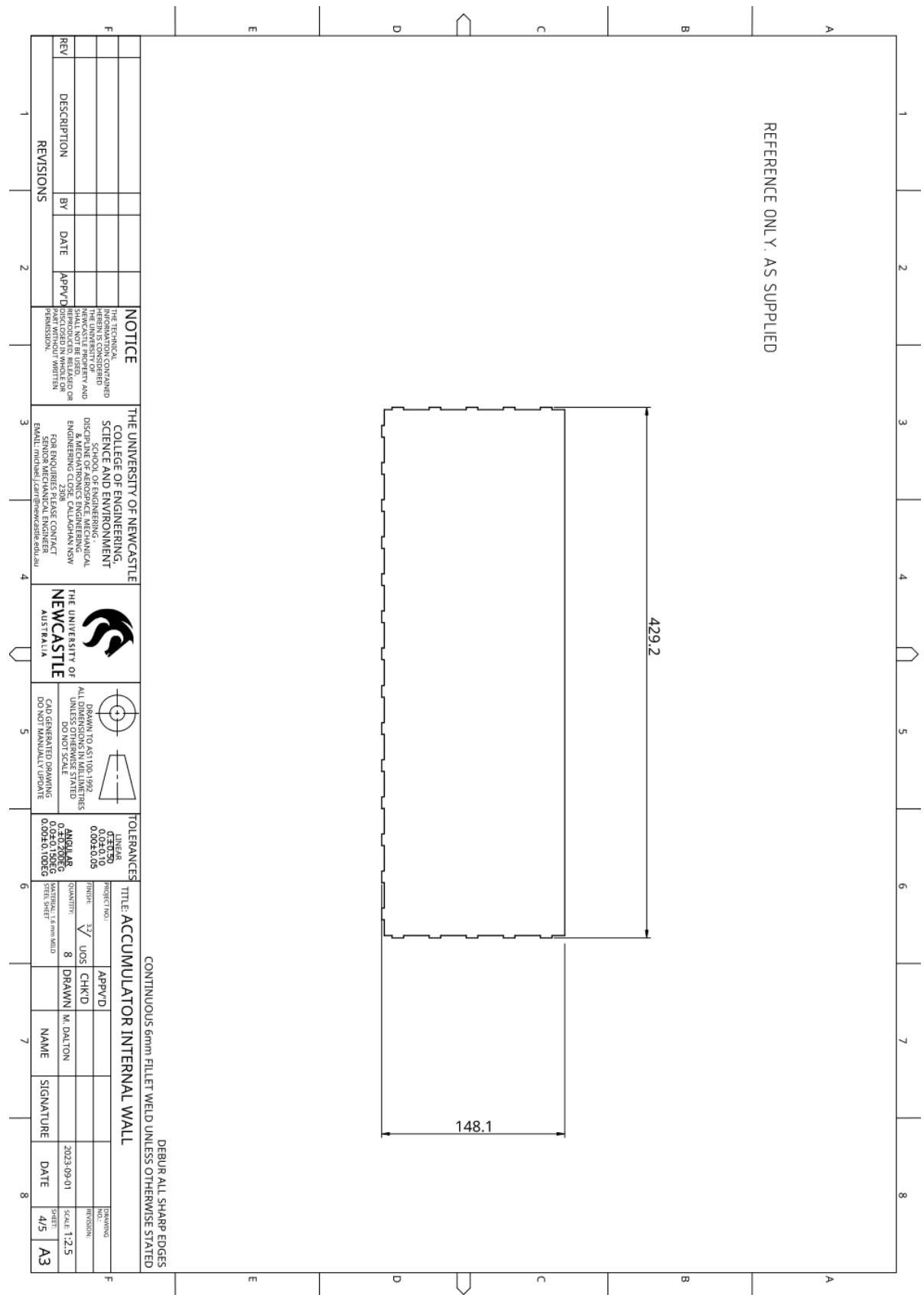
Part	Unit Cost	Quantity	Total Cost	Supplier
Sanyo Denki 9GV1212P1J01 120 x 38 mm axial fan	\$72.14	8	\$577.12	RS Pro Online
PETG Filament	\$29.95/kg	1.1 kg	\$29.95	Cubic Tech
ABS Flame Retardant Filament	\$74/0.75kg	1.5 kg	\$148	3DXTech
FR4 1.6 mm (segment housings, linings)	n/a	n/a	\$633	PCBGOGO
M4 x 40 mm	\$0.87	32	\$27.84	Newcastle High Tensile Bolts
M4 hex nut	\$0.18	32	\$5.76	Newcastle High Tensile Bolts
Nomex sheet	\$30.35/m	1 m	\$30.35	Swift Supplies
SS Flyscreen	\$19/1.6m ²	0.1m ²	\$1.20	Bunnings
Flame Retardant epoxy	\$151.90/375 ml	750 ml	\$303.80	MEKTRONICS
Fibreglass Tape	\$23.06/50 m	25 m	\$11.53	Blackwoods
Deutsch DT 12 Pin connector	\$17.00	2	\$34	Deutsch
Total			\$1,802.55	

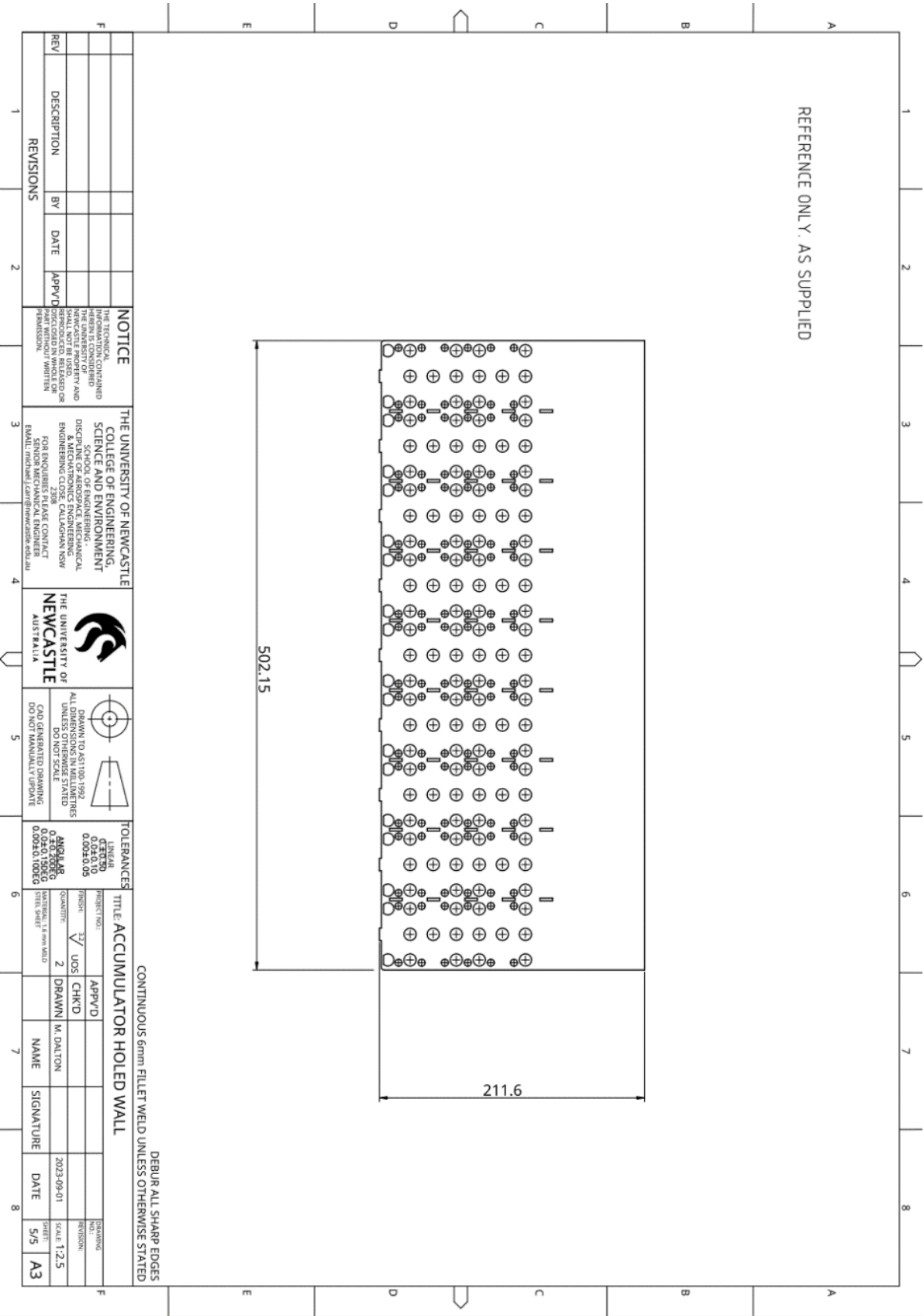
Drawings

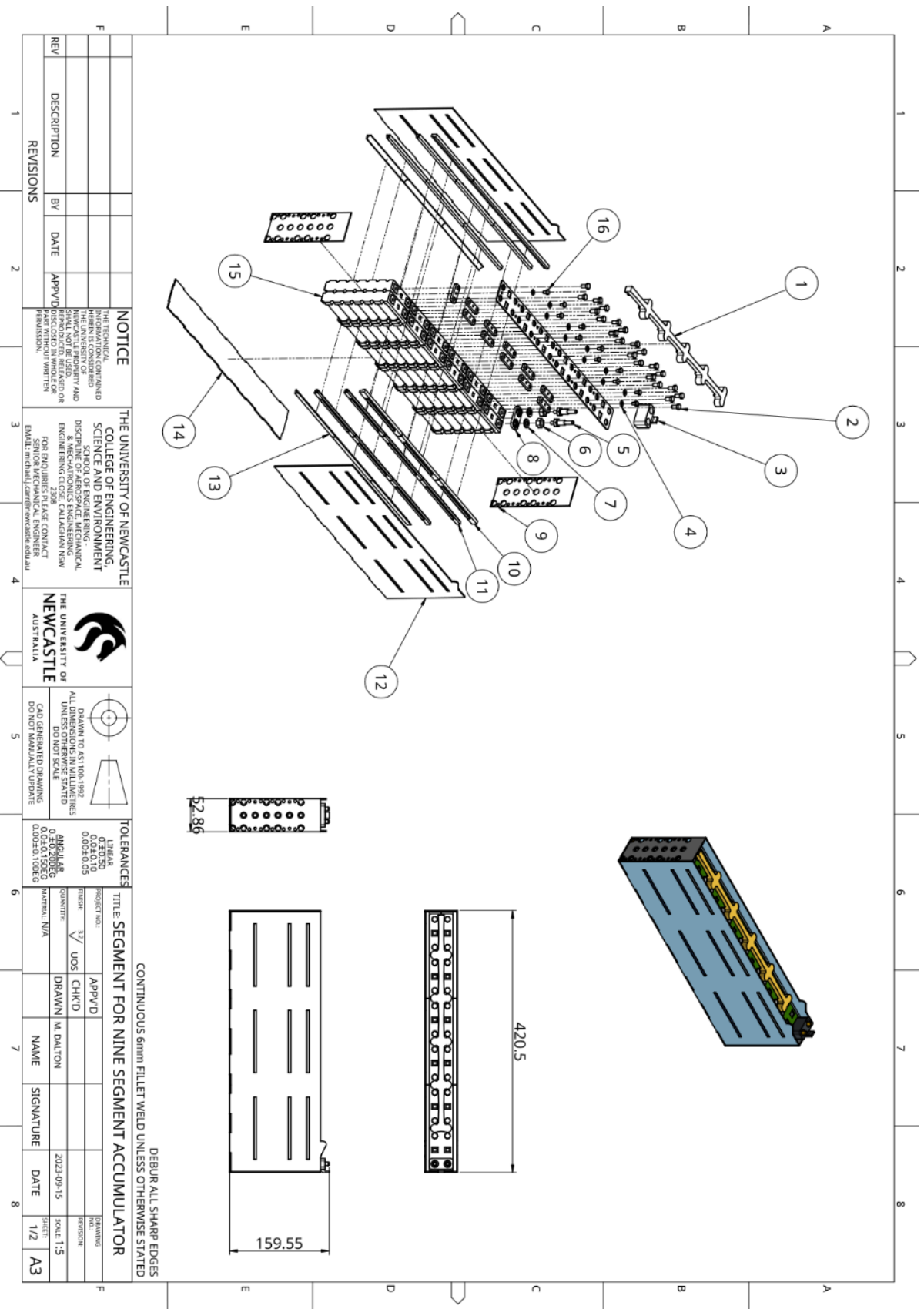




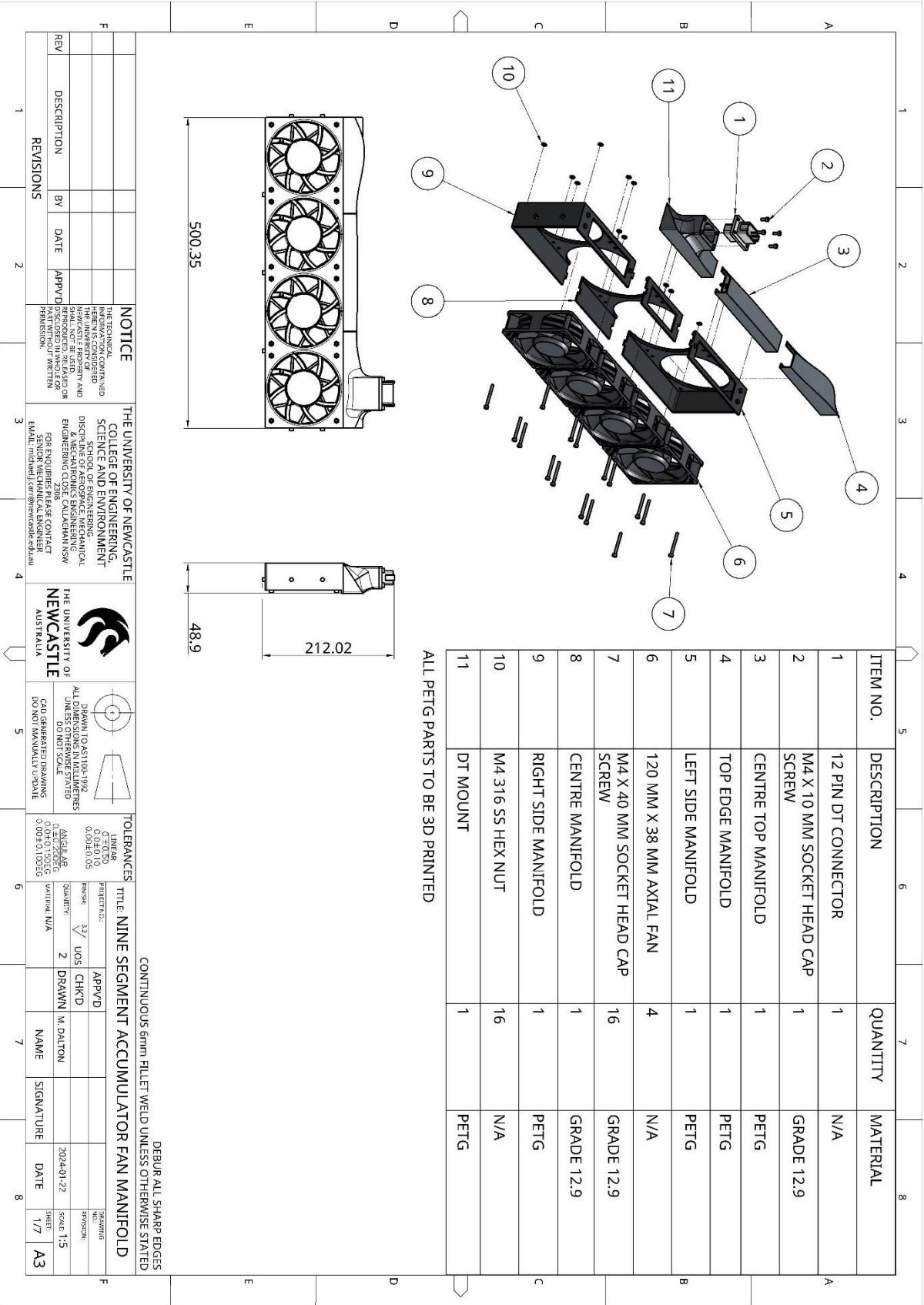


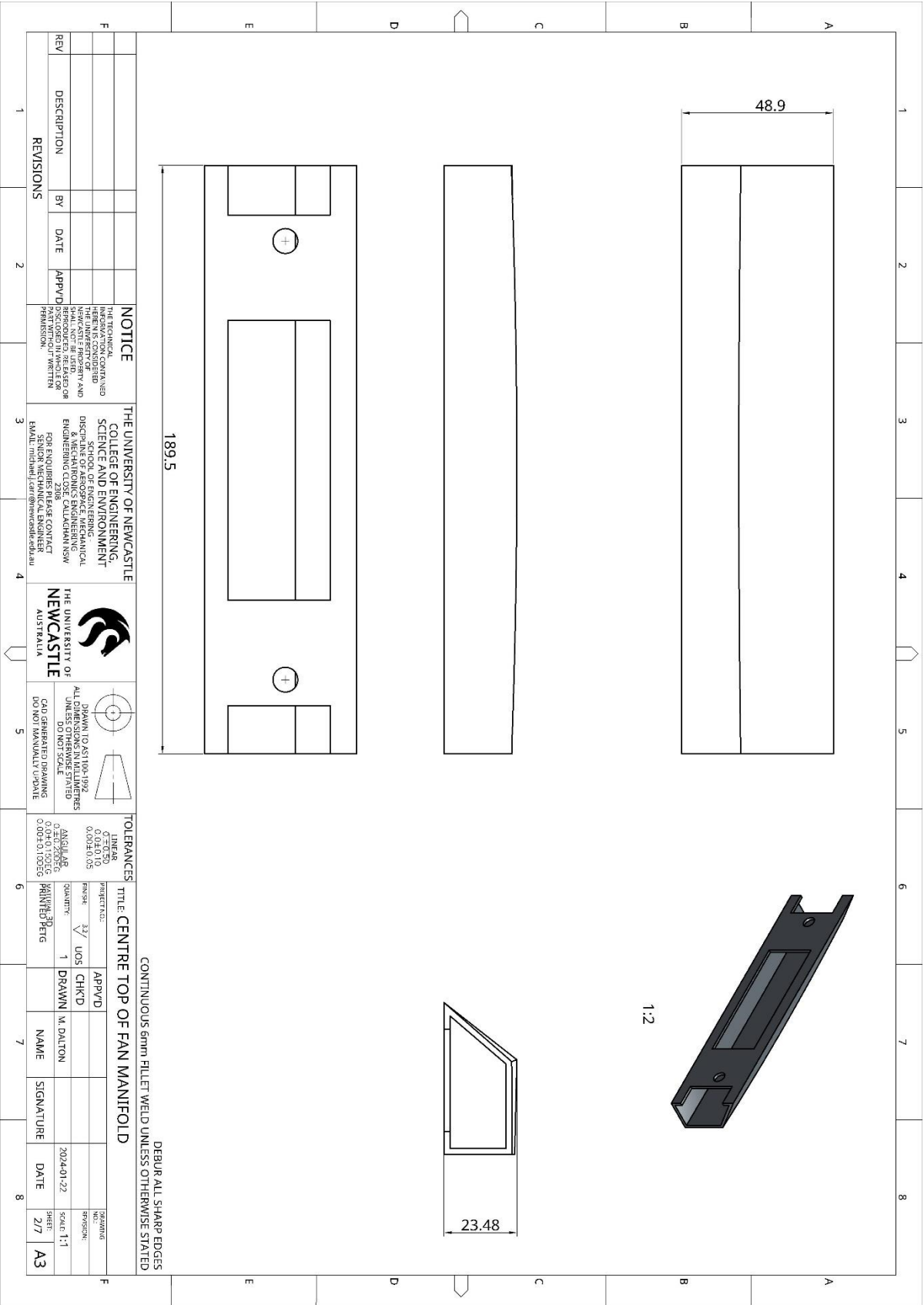


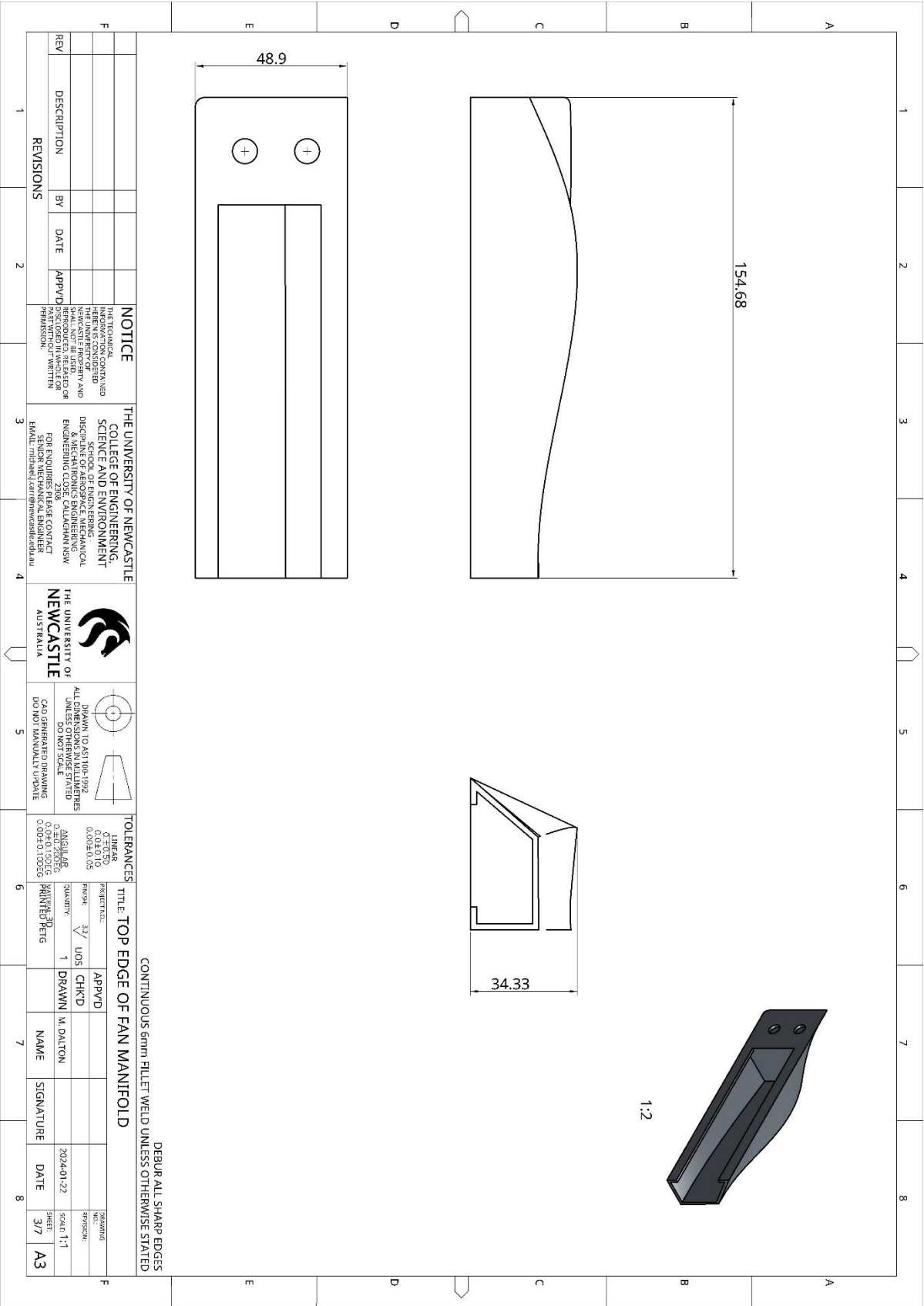


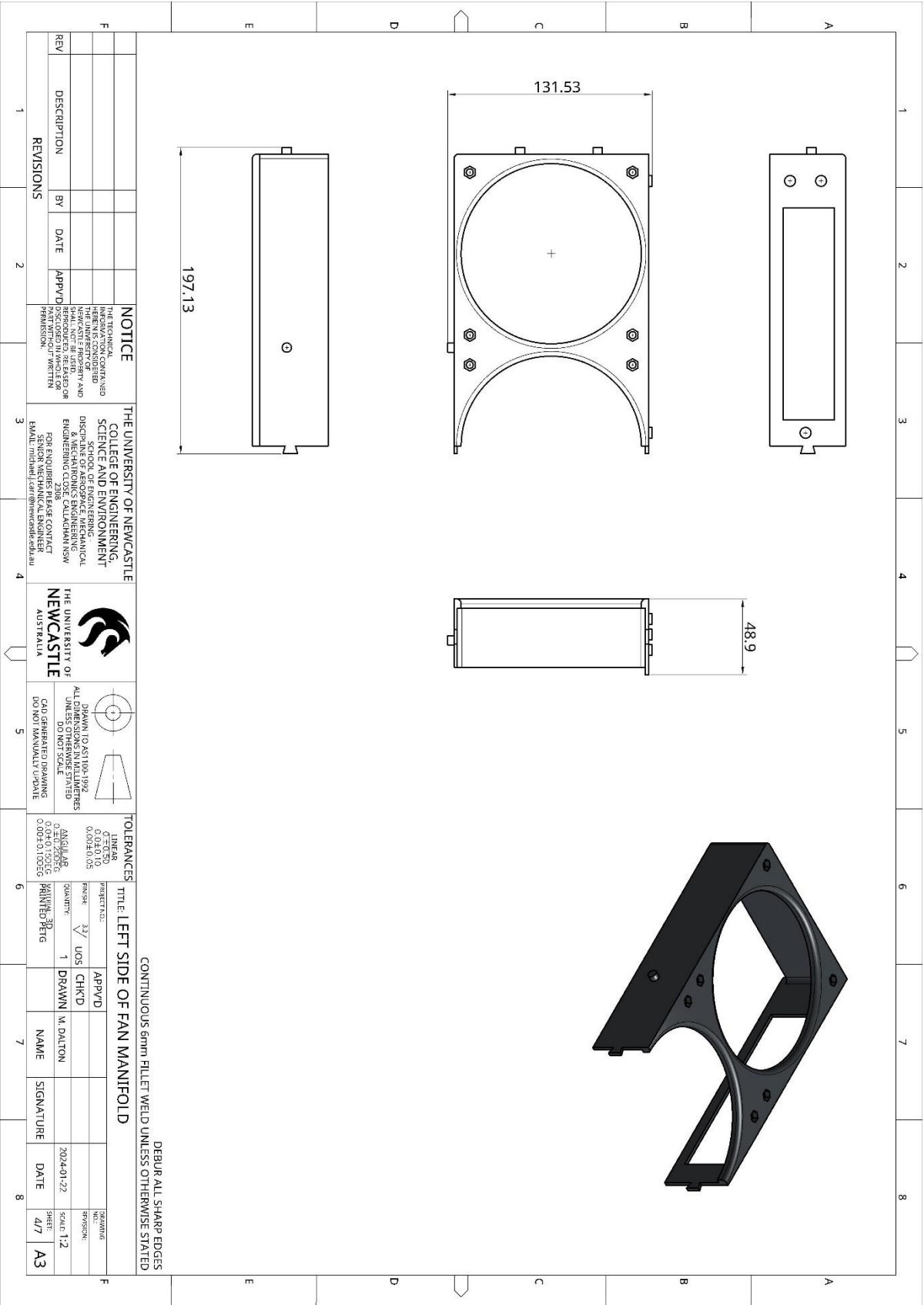


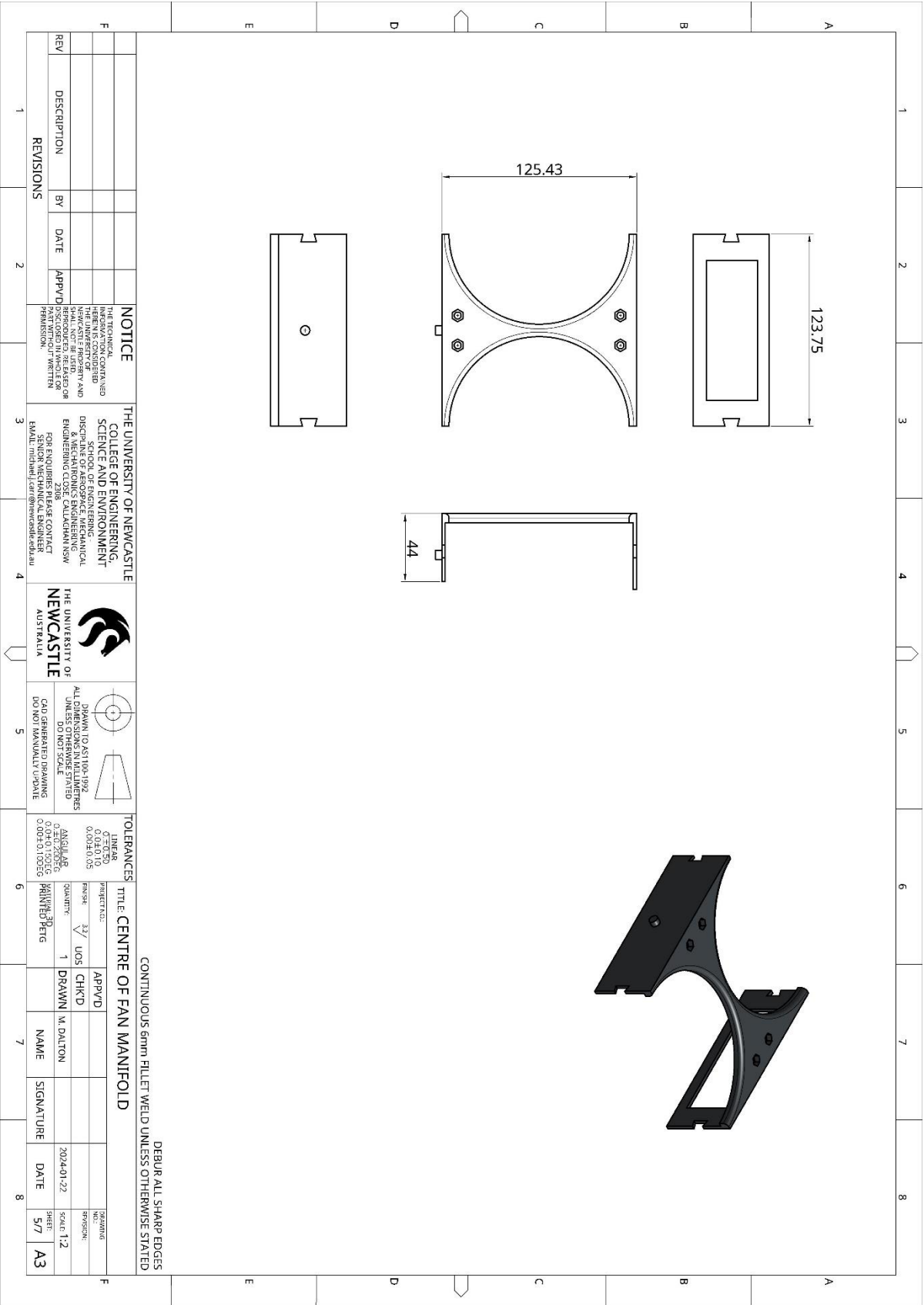
111



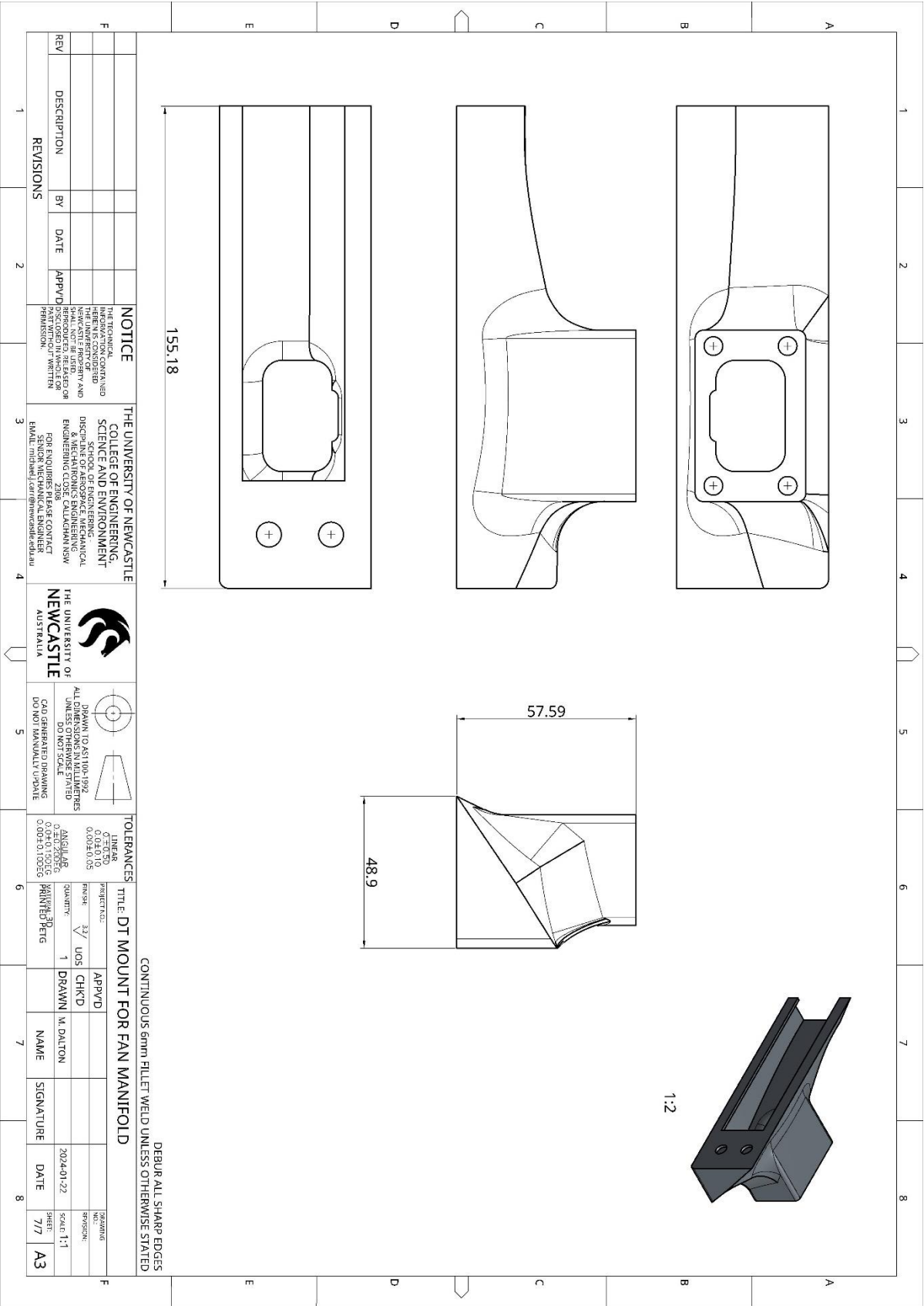












Compliance Report

NU23, as well as the accumulator, passed technical inspection at the 2023 FSAE competition first out of all other teams. However, this was not without some ambiguity and initial non-compliance during inspections.

Leading up to competition, there was some ambiguity as to the legality of the fan manifolds and their proximity to the side impact structure of the chassis. Rule F.11.2.3 states that “All Non-Crushable Items should have a minimum 25 mm clearance to the surface of the Accumulator Container” and goes on to mention side impact structures as one example of such non crushable items (SAE International, 2023). Given the design of the accumulator, the definition of the external wall becomes somewhat obscure due to the large flange that supports the fan manifolds. A rules enquiry was submitted on 28/11/2023, less than 3 weeks prior to the competition that included the photo found in Figure 85 that highlighted the presence of the accumulator’s external structure was not necessarily an external wall to the cells within the accumulator. The full rules enquiry can be found in Appendix 4.

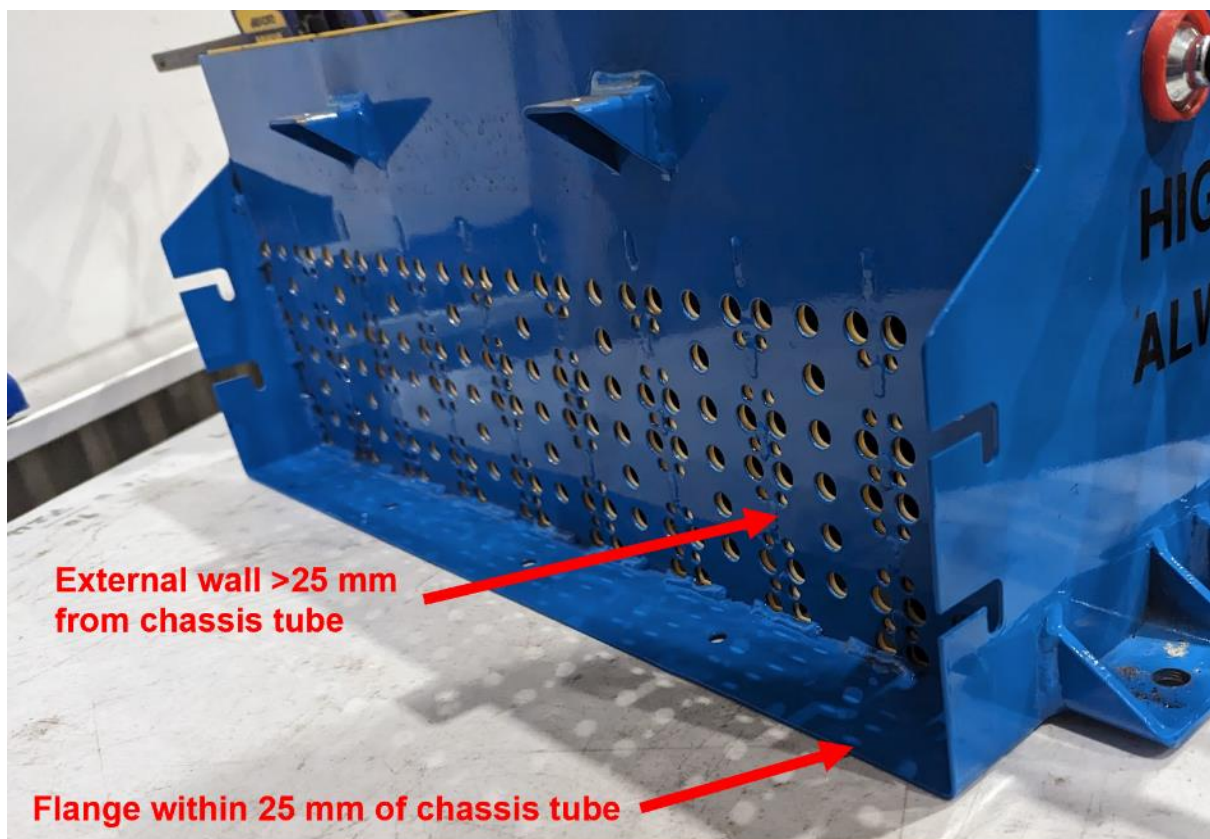


Figure 85 Photo from rules enquiry seeking clarification of accumulator external structure.

This also posed the question of whether the fan and the manifold were considered crushable objects or not. In a document posted by FSAE Australasia, titled “Rules Enquiries Q&A” a question asking

whether fans could occupy the 25 mm gap between chassis and accumulator was answered with “at least 25 mm must be provided between the fan exterior face and the chassis structure”. However, in this same response, the assumption was made that “it is almost certain that the majority of a fan would be metal” (Formula SAE Australasia, 2024). However, in another answer given to a similar question in the same document, it was stated that one way of seeking compliance with fans within 25 mm of a chassis member could be “the fan would have to be mounted on crushable mounts to the container”. It was considered that the fan manifold, made of plastic would be deemed a ‘crushable mount’.

Shortly after submission of the rule’s enquiry, it was decided to remove the container flange in the areas that failed to remain more than 25 mm from the side impact structure. This decision was a result of little time before competition, and the discovery it could be done with little to no impact on the mounting of the fan manifolds. An estimation for the material requiring removal was easily achieved using CAD, as shown in Figure 86. This indicated that the flange required removal only at the back of the accumulator and not so high that it would affect the mounting pins’ slots.

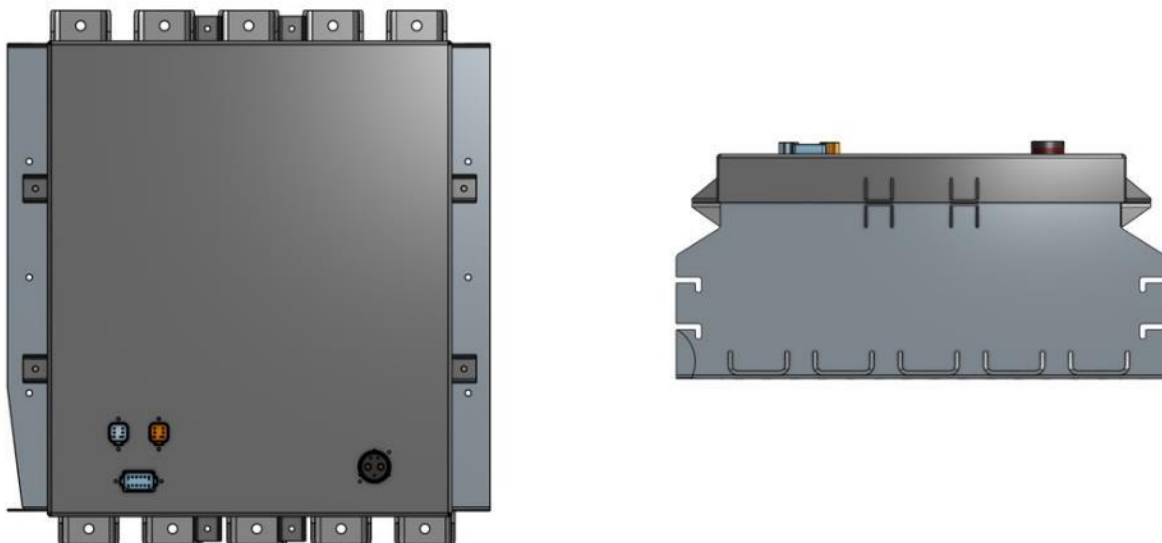


Figure 86 Estimated material removal required from container to comply with 25 mm chassis clearance, shown only on left side of the accumulator.

With this estimation made, the container was modified accordingly. To ensure the clearance was satisfactory, a semi-circular spacer was run along the chassis member, shown in Figure 87.



Figure 87 The resulting cut made to the container, with 27 mm spacer showing ample clearance from side impact structure.

With no response from FSAE within a week, a decision was made on how to address the fan and manifold concerns. Given the plastic construction of both the fans and manifolds, the team was confident these could be considered crushable based on the responses in the Q&A document previously discussed. Additionally, in case the scrutineers did not agree on the legality of the fans and manifolds, a substitute fan manifold was designed.

While this alternative fan manifold is discussed in greater detail in the “Initial System Design” section of this report, it managed to keep the fans and manifold more than 25 mm away from the side impact structure. In order to achieve this, a thinner 25 mm thick fan replaced the rear-most accumulator cooling fan where the side impact structure was closest to the accumulator. This enabled the manifold to become slimmer at the back as shown in Figure 88.

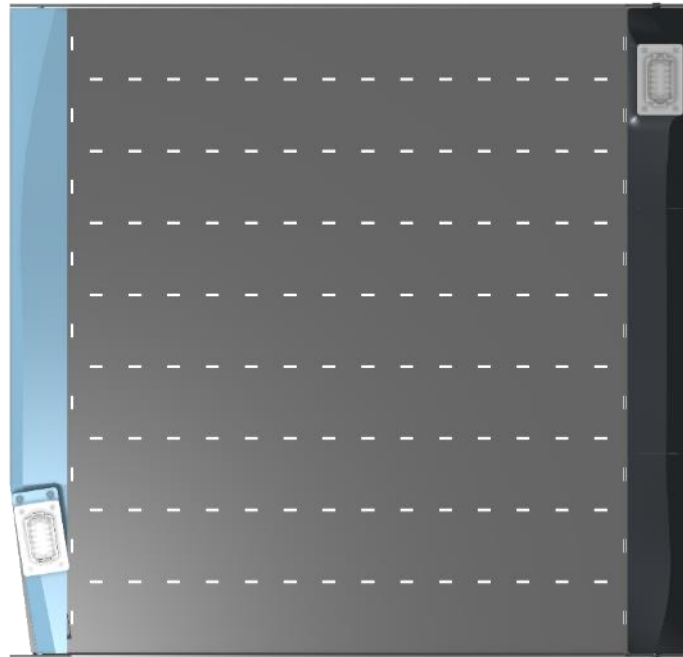


Figure 88 Top View of the accumulator external container with the final manifold on the right (black), and the thinner, compliant manifold on the left (blue).

Conclusions and Recommendations

The accumulator cooling for NU23 showed overall improvements from the previous design in EV.Three. The complete overhaul of the cooling configuration resulted in two key improvements. A more predictable, reactive cooling system, that drivers could moderate their driving to control cell temperatures, as well as being a more serviceable system. The former of these two helped align one of the cars biggest bottlenecks in EV.Three, while serviceability was key design goal for every system for NU23.

Some of the more minor improvements came from a maturing platform. Use of thinner, lighter FR4 was a small improvement, as a result of the team's confidence in the material from using it in EV.Three. Moving to offsite manufacturing for the CNC routing of FR4 components was also a decision formed from the knowledge of how time consuming, and sometimes difficult the process can be.

However, there are still many recommendations to be made. Foremost, clearances are something that simply did not translate from CAD design to real world manufacturing. Due to inaccuracies in manufacture, particularly welding processes such as the container, critical dimensions that depend on fractions of millimetres cannot rely on CAD. This is so evident in the fitment of the segments in the accumulator, where despite this knowledge from EV.Three, and ample clearance being added to NU23's design, there were still many irregularities in internal wall width leading to many segments not fitting, and the rest being far too tight.

It is recommended that in future the container is manufactured and fully assembled first before segment widths and lengths are finalised. Mock segments can then be used, an option could be laser cut MDF board, to test if the designed width is compatible and provides the desired fit.

Another key recommendation to the cooling performance is to determine the hottest point in the Energus modules. Some battery topologies primarily cool the busbars connected to the cells rather than the cells themselves. In NU23, the orientation of the Energus modules is such that the busbars receive little to no airflow, and the air predominantly flows over the cells themselves.

Time should be taken to confirm whether the busbar or the cell itself is generating the highest temperatures. This could be done by discharging a single Energus cell stack and finding the hottest part of the cell with the infrared heat camera. If it was indeed found that the busbar was the hottest component of the Energus module, a complete re-design of the Energus modules orientation in the segment should be considered.

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Appendix

Appendix 1: Data sheets for fans used in pressure testing.

OD1238 VXE Series with Alarm

DC Fans - 12, 24, 48V
120mm x 38mm (4.7" x 1.5")

Lead Wires:
UL 1007, 24 AWG
Red Wire: DC+
Black Wire: DC- / GND
Yellow Wire: Alarm Out

Frame	PBT, UL94 V-0 Plastic	Operating Temperature -20°C ~ +65°C	
Impeller	PBT, UL94 V-0 Plastic	Storage Temperature -25°C ~ +70°C	
Connection	3x Lead Wires, 24 AWG	Life Expectancy 70,000 Hours (L10 at 40°C)	
Motor	Brushless DC, Auto Cutoff, Auto Restart, Polarity Protected, Electronically Protected	Installed Options Open Collector Alarm Output	
Bearing System	Ball Bearings	Available Options TTL 5V Alarm Output	
Dielectric Strength	500VDC/min, <1mA	Inverted Alarm State	
Insulation Resistance	≥10MΩ @ 500 VDC	IP Ratings	

Part Number	Rated Voltage (Volts DC)	Voltage Range (Volts DC)	Input Power (Watts)	Rated Current (Amps)	Rated Speed (RPM)	Max Airflow (CFM)	Noise Level (dBA)	Max Static Pressure (\"H2O)
OD1238-12HBVXE02A	12	9 ~ 13.8	46.3	3.80	4925	235.5	60.6	1.6
OD1238-12MBVXE02A	12	9 ~ 13.8	34.2	2.80	4450	215.0	58.2	1.3
OD1238-24HBVXE02A	24	16 ~ 26	60.0	2.45	5475	258.8	62.3	1.9
OD1238-24MBVXE02A	24	16 ~ 26	46.3	1.90	4925	235.5	60.4	1.6
OD1238-24MBVXE02A	24	16 ~ 26	34.2	1.40	4450	215.0	58.2	1.3
OD1238-48HBVXE02A	48	30 ~ 52	77.7	1.60	5980	285.2	64.6	2.3
OD1238-48HBVXE02A	48	30 ~ 52	56.4	1.15	5475	258.8	62.3	1.9
OD1238-48HBVXE02A	48	30 ~ 52	45.1	0.90	4925	235.5	60.4	1.6
OD1238-48MBVXE02A	48	30 ~ 52	34.2	0.70	4450	215.0	58.2	1.3

Alarm Max Pullup Voltage: Fan Vcc
Alarm Max Sink Current: 5mA
Alarm Active Locked Rotor State: HIGH

Knight Electronics, Inc.
10557 Metric Drive
Dallas, TX 75243
214-340-0265

Revised 7/13/2023
m0

Orion Fans
Information and data is subject to
change without prior notification

Appendix 2: Data Sheet for blower fans used in EV.Three

SANYO DENKI

San Ace B97

9BMC type

Blower

Features

High Airflow

The maximum airflow has increased by 15% compared with our current model.*

High Static Pressure

The maximum static pressure has increased by 50% compared with our current model.*

Low Noise and High Energy Efficiency

The PWM control function enables the external control of fan speed, contributing to lower noise and higher energy efficiency of devices.

* New model 9BMC12P2G001 compared with our current blower model 9BMB12P2K01, 97 x 33 mm San Ace B97 9BMB type.



97 x 33 mm

Specifications

The following nos. have PWM controls, pulse sensors.

Model no.	Rated voltage [V]	Operating voltage range [V]	PWM duty cycle [%]	Rated current [A]	Rated input [W]	Rated speed [min ⁻¹]	Max. airflow [m ³ /min]	Max. static pressure [Pa]	SPL [dB(A)]	Operating temperature [°C]	Expected life [h]
9BMC12P2G001	12	10.8 to 13.2	100	6.2	74.4	8200	1.85 65.3	1950 7.83	69	-20 to +70	40000/60°C (70000/40°C)
			20	0.38	4.56	2800	0.58 20.4	121.0 0.48	44		
			20	0.19	4.56	2800	0.58 20.4	121.0 0.48	44		
9BMC24P2G001	24	21.6 to 26.4	100	3.1	74.4	8200	1.85 65.3	1950 7.83	69	-20 to +70	40000/60°C (70000/40°C)
			20	0.19	4.56	2800	0.58 20.4	121.0 0.48	44		
			20	0.19	4.56	2800	0.58 20.4	121.0 0.48	44		

* PWM frequency: 25 kHz. Fan does not rotate when PWM duty cycle is 0%.

Models with the following sensor specifications are also available as options: **Without sensor** **Pulse sensor**

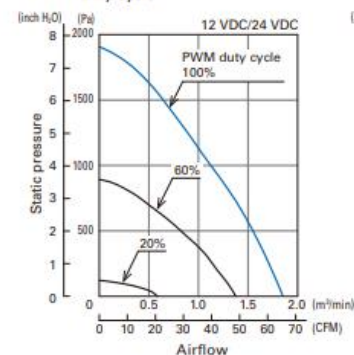
Please inquire as the availability of these options depends on the model: **Lock sensor**

Common Specifications

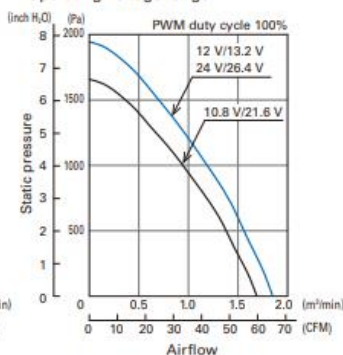
- ☐ Material Frame, Impeller: Plastics (Flammability: UL 94V-0)
- ☐ Expected life Refer to specifications
(L10: Survival rate: 90% at 60°C, rated voltage, and continuously run in a free air state)
- ☐ Motor protection system Current blocking function and reverse polarity protection
- ☐ Dielectric strength 50/60 Hz, 500 VAC, 1 minute (between lead conductor and frame)
- ☐ Sound pressure level (SPL) Expressed as the value at 1 m from air inlet side
- ☐ Operating temperature Refer to specifications (Non-condensing)
- ☐ Storage temperature -30°C to +70°C (Non-condensing)
- ☐ Lead wire ⊕Red ⊖Black Sensor: Yellow Control: Brown
- ☐ Mass Approx. 200 g

Airflow - Static Pressure Characteristics

• PWM duty cycle

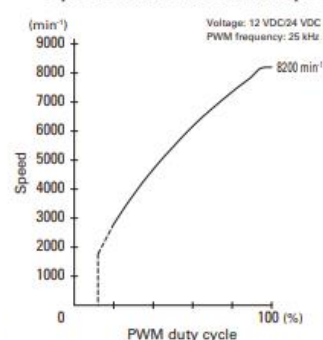


• Operating voltage range



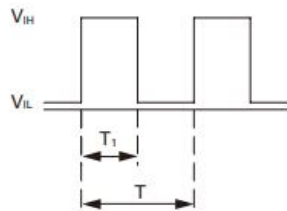
PWM Duty

- Speed Characteristics Example



PWM Input Signal Example

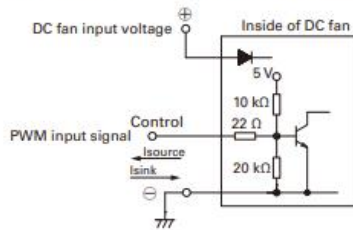
Input signal waveform



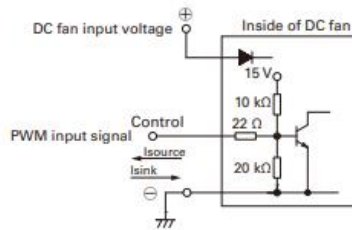
$V_{IH}=4.75$ to 5.25 V $V_{IL}=0$ to 0.4 V
 $\text{PWM duty cycle (\%)} = \frac{T_1}{T} \times 100$ PWM frequency 25 (kHz) $= \frac{1}{T}$
 Current source (I_{source}) ≈ 1 mA max. (when control voltage is 0 V)
 Current sink (I_{sink}) ≈ 1 mA max. (when control voltage is 5.25 V)
 Control terminal voltage $= 5.25$ V max. (when control terminal is open)

When the control terminal is open,
 fan speed is the same as when PWM duty cycle is 100% .
 Either TTL input, open collector or open drain can be used for
 PWM control input signal.

Example of Connection Schematic



Rated voltage 12 V fan

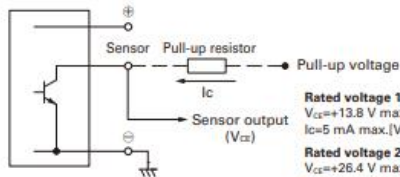


Rated voltage 24 V fan

Specifications for Pulse Sensors

Output circuit: Open collector

Inside of DC fan



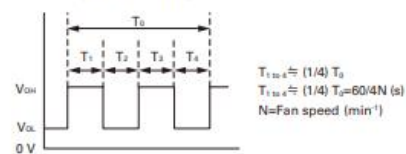
Rated voltage 12 V fan
 $V_{CE} = +13.8$ V max.
 $I_C = 5$ mA max. [$V_{OL} = V_{CE}$ (SAT) ≈ 0.6 V max.]

Rated voltage 24 V fan
 $V_{CE} = +26.4$ V max.
 $I_C = 10$ mA max. [$V_{OL} = V_{CE}$ (SAT) ≈ 1.0 V max.]

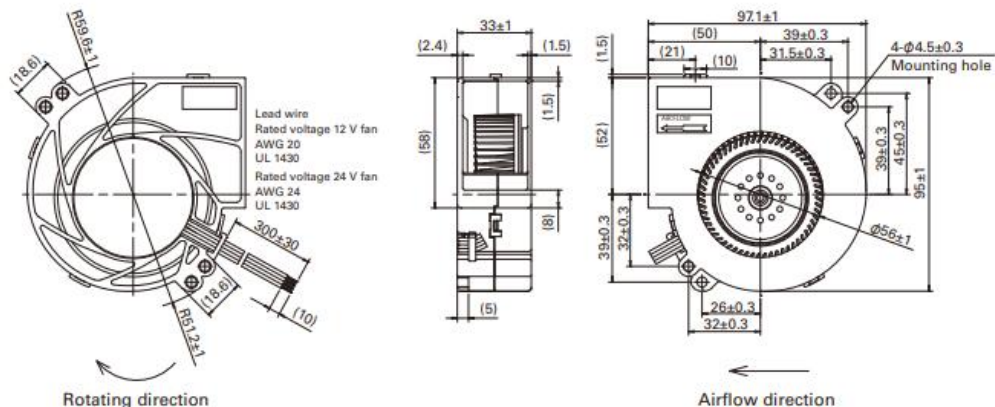
Output waveform (Need pull-up resistor)

In case of steady running

(One revolution)



Dimensions (unit: mm)



Notice

- Please read the "Safety Precautions" on our website before using the product.
- The products shown in this catalog are subject to Japanese Export Control Law. Diversion contrary to the law of exporting country is prohibited.
- For protecting fan bearings against electrolytic corrosion near strong electromagnetic noise sources, we provide effective countermeasures such as Electrolytic Corrosion Proof Fans and EMC guards. Contact us for details.

SANYO DENKI CO., LTD.

3-33-1, Minami-Otsuka, Toshima-ku, Tokyo 170-8451, Japan TEL: +81 3 5927 1020

<https://www.sanyodenki.com>

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Specifications are subject to change without notice.

CATALOG No. C1069B001 '17.5

Appendix 3: Datasheet for fans used for accumulator cooling

DC Fan

120×120×38 mm

San Ace 120 9GV type


DC

DC Fan 120 mm sq.

General Specifications

- Material Frame: Plastic (Flammability: UL 94V-0), Impeller: Plastic (Flammability: UL 94V-1)
- Expected life See the table below. (L10 life: 90% survival rate for continuous operation in free air at 60°C, rated voltage)
- Motor protection function Locked rotor burnout protection, Reverse polarity protection
For details, please refer to p. 529.
- Dielectric strength 50/60 Hz, 500 VAC, for 1 minute (between lead wire conductors and frame)
- Insulation resistance 10 MΩ or more with a 500 VDC megger (between lead wire conductors and frame)
- Sound pressure level (SPL) At 1 m away from the air inlet
- Storage temperature -30 to +70°C (Non-condensing)
- Lead wire Red Black (Sensor) Yellow (Control) Brown
- Mass 360 g

Specifications

The models listed below **have ribs and pulse sensors with PWM control function**. For models without ribs, append "1" to the end of model numbers.

Model no.	Rated voltage [V]	Operating voltage range [V]	PWM duty cycle* [%]	Rated current [A]	Rated input [W]	Rated speed [min ⁻¹]	Max. airflow [m ³ /min] [CFM]	Max. static pressure [Pa] [inchH ₂ O]	SPL [dB (A)]	Operating temperature [°C]	Expected life [h]
9GV1212P1J01	12	10.2 to 13.8	100	3.0	36.0	6400	6.35 224.0	360.0 1.45	64	-20 to +70	40000/60°C
			0	0.19	2.28	1500	1.49 52.6	19.8 0.08	33		
9GV1212P1G01	12	10.2 to 13.8	100	2.1	25.2	5500	5.45 192.6	265 1.06	60		
			0	0.19	2.28	1500	1.49 52.6	19.8 0.08	33		
9GV1224P1J01**	24	20.4 to 27.6	100	1.5	36.0	6400	6.35 224.0	360 1.45	64		
9GV1224P1H01**			100	0.8	19.2	5200	5.16 182.3	237 0.95	58		
9GV1248P1J01	48	40.8 to 55.2	100	0.75	36.0	6400	6.35 224.0	360.0 1.45	64		
			0	0.06	2.88	1500	1.49 52.6	26.1 0.106	33		

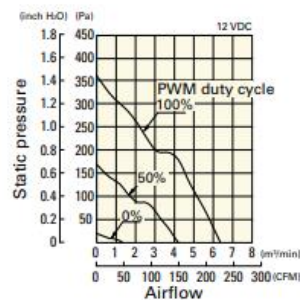
* PWM frequency: 25 kHz ** Fan does not rotate when PWM duty cycle is 0%.

Other sensor specifications are available as options. Refer to the index (p. 555).

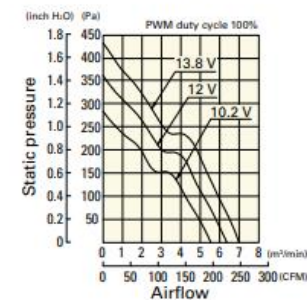
Airflow - Static Pressure Characteristics / PWM Duty - Speed Characteristics Example

9GV1212P1J01 With pulse sensor with PWM control function

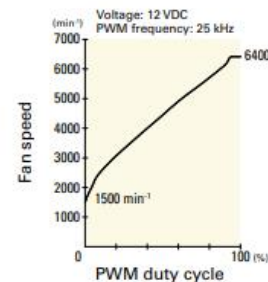
PWM duty cycle



Operating voltage range



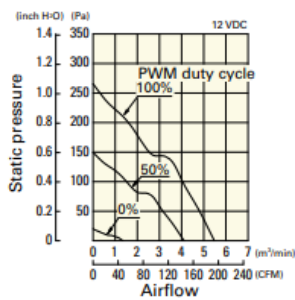
PWM duty - Speed characteristics example



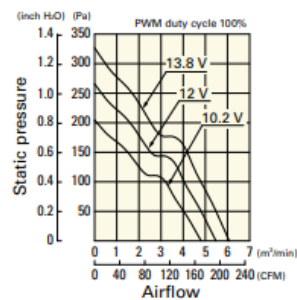
Airflow - Static Pressure Characteristics / PWM Duty - Speed Characteristics Example

9GV1212P1G01 With pulse sensor with PWM control function

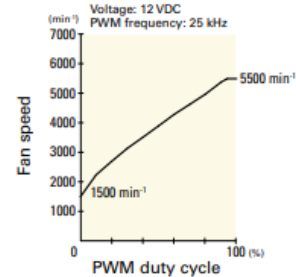
PWM duty cycle



Operating voltage range

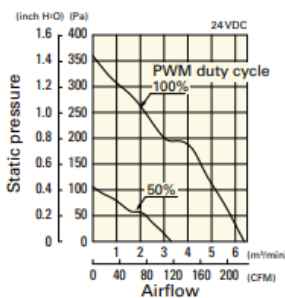


PWM duty - Speed characteristics example

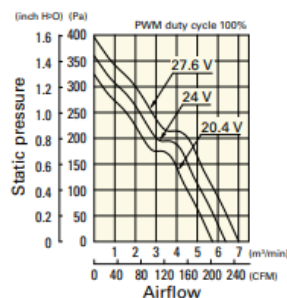


9GV1224P1J01 With pulse sensor with PWM control function

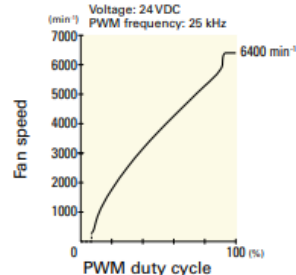
PWM duty cycle



Operating voltage range

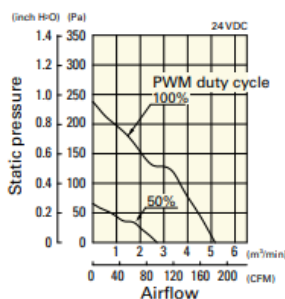


PWM duty - Speed characteristics example

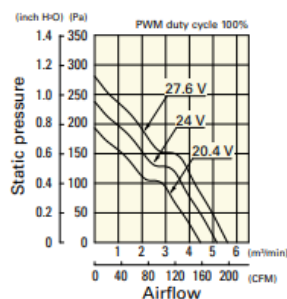


9GV1224P1H01 With pulse sensor with PWM control function

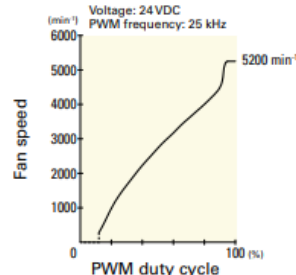
PWM duty cycle



Operating voltage range

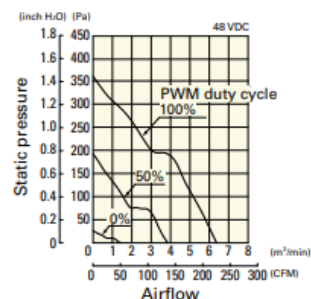


PWM duty - Speed characteristics example

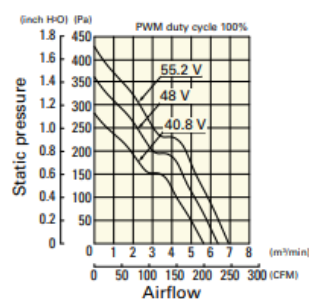


9GV1248P1J01 With pulse sensor with PWM control function

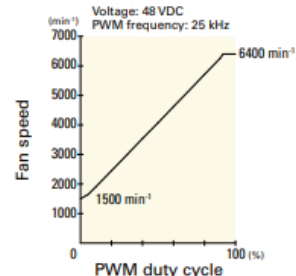
PWM duty cycle



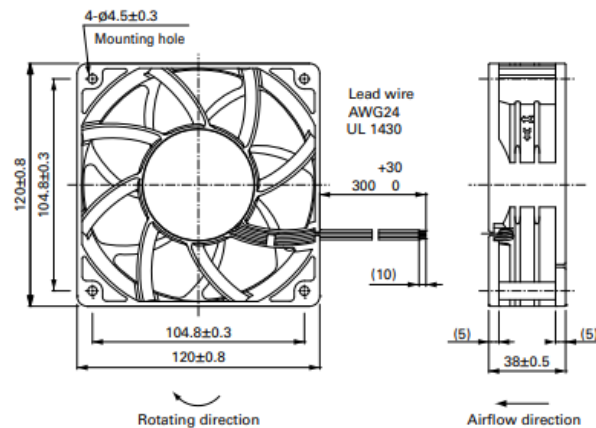
Operating voltage range



PWM duty - Speed characteristics example



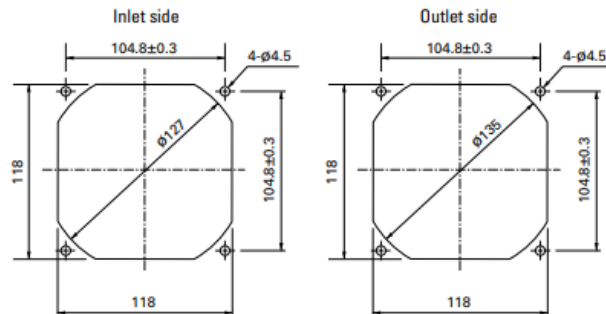
■ Dimensions (unit: mm) (With ribs)



DC

DC Fan 120 mm sq.

■ Reference Dimensions of Mounting Holes and Vent Opening (unit: mm)



■ Options

Finger guards

page: p. 514

Model no.: 109-019C, 109-019H, 109-019E, 109-019K

Resin finger guards

page: p. 520

Model no.: 109-1000G

Resin filter kits

page: p. 521

Model no.: 109-1000F13 (13PPI), 109-1000F20 (20PPI),
109-1000F30 (30PPI), 109-1000F40 (40PPI)

Appendix 4: Rules Enquiry Regarding Accumulator Clearance to Side Impact Structure

Rules Enquiry - NU Racing

F.11.2.3 All Non Crushable Items should have a minimum 25 mm clearance to the surface of the Accumulator Container.

Non Crushable Items include, but are not limited to motors, differentials, and the side or rear impact structure itself. Accumulator mounts do not require clearance

We are seeking clarification as to whether our accumulator is compliant with F.11.2.3, and in the case it isn't, whether the listed proposed changes will rectify it.

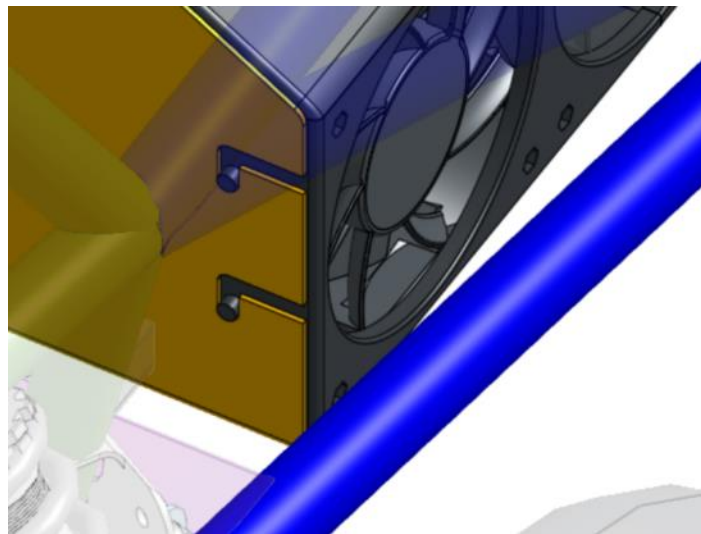


Figure 1 Chassis member within 25 mm of accumulator.

Our accumulator currently has a flange that supports fans against the side of the accumulator wall. While this flange is within 25 mm of the nearest chassis tube, shown in Figure 1, the external wall of the accumulator and cells are more than 25 mm from the chassis. This is more clearly illustrated in Figure 2.

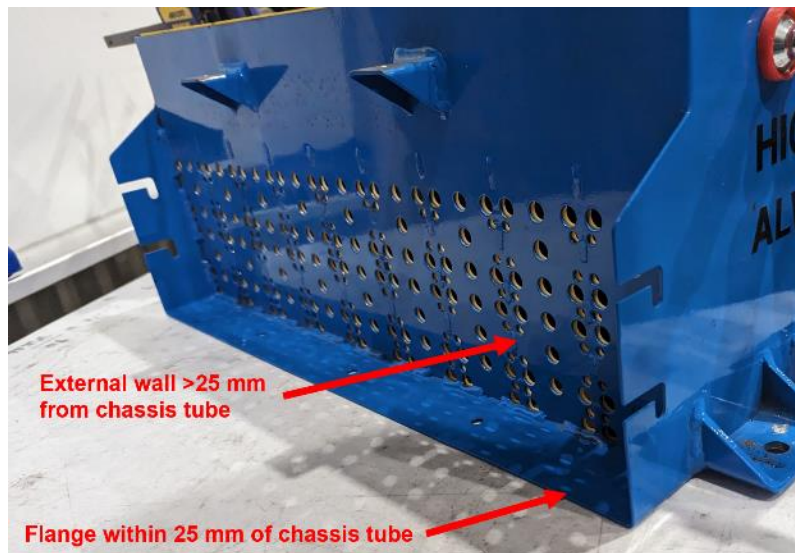


Figure 2 Accumulator container with flange.

We believe this merits the consideration alluded to in the Rules Enquiries Q&A shown below in Figure 3.

In cases where the 25 mm clearance to the exterior of the container wall is not met, we may take into account any additional interior clearance between the cells and the container wall in the problem area.

Figure 3 Exert from Rules Enquiries Q&A.

The fans and their manifold however are within 25 mm of the nearest chassis member as well. While the Rules Enquiries Q&A does state that fans are not considered crushable objects, it does also assume the body of fans to be metal. In our case, the fan is constructed entirely of plastic, except the motor and central hub. This means that all parts of the fan within 25 mm of the chassis member are plastic and therefore crushable.

If the motivation for these rules is to protect cells from chassis members deforming in the event of a collision, we believe our accumulator satisfies this motivation, with only crushable, and non-dangerous parts falling within the 25 mm region.

If, however it is deemed that this fails these rules, we propose one of the following changes in order of preference, with increasing modifications required in each.

1. Reduce the material in the flange.

Should the flange not be allowable, we propose removing material from the flange such that it becomes a crushable object. These cuts would be done as shown in Figure

4. These cuts would be made to the length of flange within 25 mm of the chassis member.

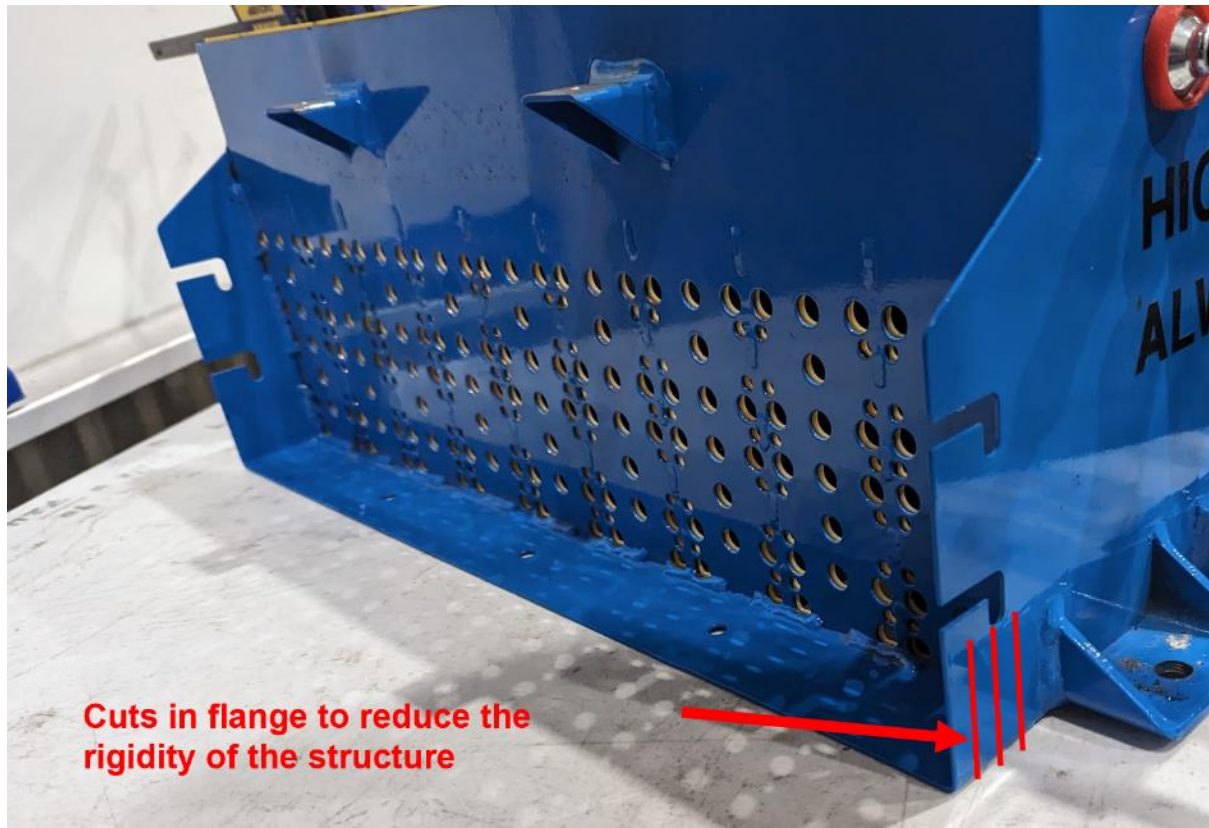


Figure 4 Cut locations for reducing rigidity of accumulator flange.

This would obviously only be applicable if the fan was considered crushable as reasoned earlier.

2. Removing the flange within 25 mm of chassis member.

If the first proposal is not considered rules compliant, the flange could be removed all together where it exists within 25 mm of the chassis member. This would leave just the fan manifold and fans within 25 mm of the chassis member, which as reasoned earlier, are crushable objects.

3. Remove flange and fan within 25 mm of chassis member.

Failing both the first and second proposals being compliant, the fan, manifold and flange within 25 mm of the chassis member will be removed. The manifold would be modified such that the fan is no longer within 25 mm all together of the chassis member.



However, as the manifold is plastic, it would likely remain within 25 mm of the chassis member, as it is a crushable object.

Appendix 5: FR4 Datasheet



ENGLISH

Datasheet**RS PRO Epoxy Glass FR-4**

Stock No: 2035184

**Technical Specifications:**

Colour	Green Black (carbon free)
Sheet Size	420 x 297mm
Maximum Temperature	130°C
Flammability Rating	UL94 V-0
Electrical Strength Condition A	800 V/mil

rspro.com

Features & Benefits:

- Engineered to provide NEMA grade FR-4 properties
- Woven glass fabric combined with epoxy resin system
- Certifiable to MIL-I-24768/27, Type GEE-F

General Physical Properties	Units	Value
Specific Gravity	-	1.85
Rockwell Hardness	M Scale	98
Moisture Absorption	%	0.1
Flexural Strength LW CW	Psi	65,000 52,000
Tensile Strength LW CW	Psi	40,000 32,000
Izod Impact Strength LW CW	Ft-lb/in Notched	7.9 7.3
Compressive Strength Flatwise	Psi	66,000
Bonding Strength	Lb	2,300
Shear Strength (Perpendicular)	Psi	21,500



rspro.com



Thermal & Electrical Properties	Units	Value
Temperature Index (electrical/mechanical)	°C	130
Coefficient of Thermal Expansion X-axis Y-axis	" / °C · 10 ⁻⁶	10.0 13.0
Flammability Rating UL94	Class	V-0
Breakdown Voltage Condition A	KV	66
Electric Strength Condition A	V/mil	800
Permittivity Condition D-24/23	-	4.8
Dissipation Factor Condition D-24/23	-	0.032
Arc Resistance D-495	Sec	130
Comparative Tracking Index D3638	-	300
T _g	°C	130


rspro.com


Appendix 6: Flame Retardant ABS Filament



TDS Rev 3.0

Technical Data Sheet: Firewire™ Flame Retardant ABS 3D Filament

Physical Properties	Standard	Unit	Typical Value
Density	ISO 1183	g/cc	1.18

Mechanical Properties	Standard	Unit	Typical Value
Tensile Strength, Break	ISO 527	MPa	45
Tensile Modulus	ISO 527	MPa	1990
Tensile Elongation, Break	ISO 527	%	4
Flexural Strength	ISO 178	MPa	80
Flexural Modulus	ISO 178	MPa	2250

Thermal Properties	Standard	Unit	Typical Value
Glass Transition Temperature (Tg)	DSC	°C	105
Deflection Temperature at 0.45 MPa (66psi)	ISO 75	°C	95

Fire Testing*	Standard	Unit	Typical Value
Flamability Rating (*Base Resin)	UL 94	-	V-0 @1.5mm

Printed Specimen Conditions
Printer: Open Source FDM/FFF
Nozzle: 0.4mm
Layer Height: 0.25mm
Infill: 100%, +/- 45°
Extrusion Temp: 235°C
Bed Temp: 110°C
Specimen Orientation: XY Flat

www.3dxttech.com

Disclaimer: The technical data contained on this data sheet is furnished without charge or obligation and accepted at the recipient's sole risk. This data should not be used to establish specifications limits or used alone as the basis of design. The data provided is not intended to substitute any testing that may be required to determine fitness for any specific use.

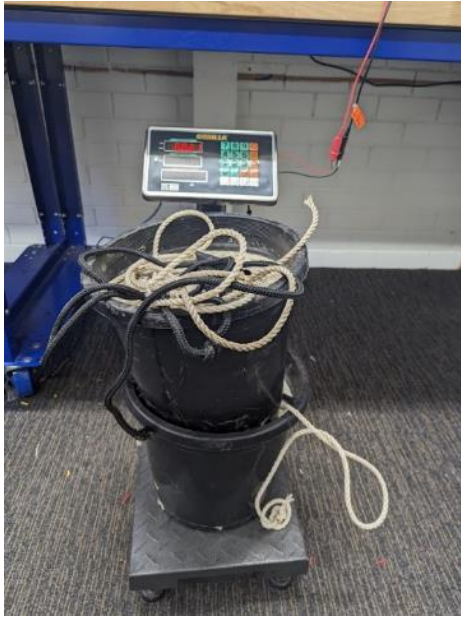
Appendix 7: Loading Test of Segment Housings

Fore and aft loading: [Error! Not a valid link.](#)

Weight required to simulate 40g loading:

$$\text{Weight} = 40 \times \text{Segment Weight} = 40 \times 2.95 \text{ kg} = 118 \text{ kg}$$

To reduce weighted objects required, only half the segment was loaded with half the weight.



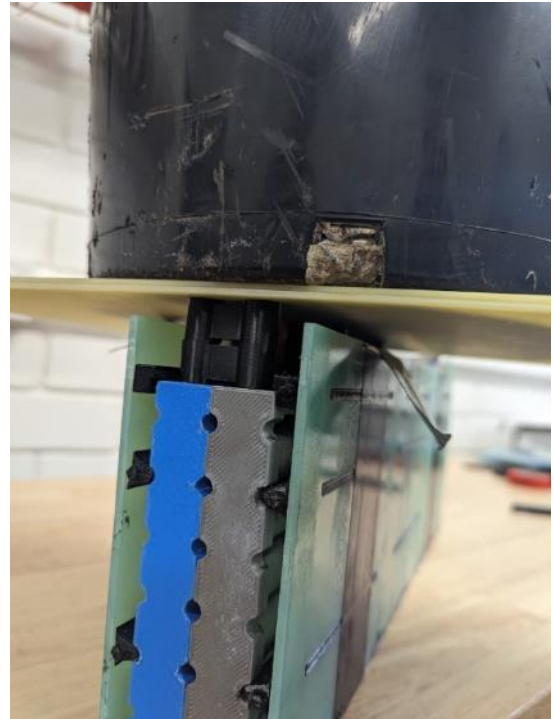
No signs of failure in housing.

Vertical Loading:

Required weight for simulating 20g vertical loading:

$$\text{Weight} = 20 \times \text{Segment Weight} = 20 \times 2.95 \text{ kg} = 59 \text{ kg}$$

Again, half the weight and only half the segment loaded.

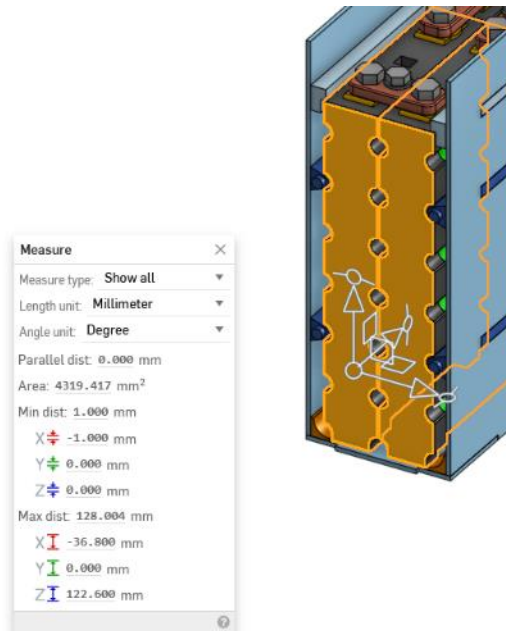


No signs of failure in the housing.

Lateral Loading:

$$\text{Stress} = \frac{\text{Segment weight} \times 40 \times 9.8}{\text{Segment Area}} = \frac{2.95\text{kg} \times 40 \times 9.8 \frac{\text{m}}{\text{s}^2}}{4913 \text{ mm}^2 \times 10^{-6}} = 235.375 \text{ kPa}$$

Segment area:



FR4 Data sheet:

General Physical Properties	Units	Value
Specific Gravity	-	1.85
Rockwell Hardness	M Scale	98
Moisture Absorption	%	0.1
Flexural Strength LW CW	Psi	65,000 52,000
Tensile Strength LW CW	Psi	40,000 32,000
Izod Impact Strength LW CW	Ft-lb/in Notched	7.9 7.3
Compressive Strength Flatwise	Psi	66,000
Bonding Strength	Lb	2,300
Shear Strength (Perpendicular)	Psi	21,500

Compressive stress of 235 kPa at 40g times segment weight is significantly lower than the 455 mPa (66,000 psi) rating of FR4.

Appendix 8: Flame Retardant Epoxy Datasheet

9200FR



Flame Retardant Structural Epoxy Adhesive

9200FR is a toughened, flame retardant, 2-part epoxy adhesive, designed to create long-lasting load-bearing joints. It creates tough vibration-resistant bonds and is especially useful for joining dissimilar materials that will experience thermal cycling stresses or harsh environmental conditions.

This smooth, non-sagging, thixotropic adhesive is excellent for use on vertical surfaces, for gap filling, and for potting electronics enclosures with gaps where a non-thixotropic encapsulant would flow through.



Features & Benefits

- Flame retardant—meets UL 94V-0
- Excellent bond strength with a wide variety of substrates
- Extreme resistance to vibration and temperature cycles
- Superior tensile, compressive and lap shear strength
- Excellent chemical resistance
- Excellent electrical insulating characteristics
- Low shrinkage
- RoHS 3 compliant

Available Packaging

Cat. No.	Packaging	Net Vol.	Net Wt.
9200FR-25ML	Dual syringe	25 mL	32 g
9200FR-50ML	Dual cartridge	45 mL	57.7 g

Contact Information

MG Chemicals, 1210 Corporate Drive
Burlington, Ontario, Canada L7L 5R6

Email: support@mgchemicals.com

Phone: North America: +(1)800-340-0772
International: +(1) 905-331-1396
Europe: +(44)1663 362888

Cured Properties

Breakdown Voltage	39 800 V
Dielectric Strength	497 V/mil
Resistivity	$1.1 \times 10^{13} \Omega \cdot \text{cm}$
Hardness	78 D
Tensile Strength	13 N/mm ²
Compressive Strength	46 N/mm ²
Lap Shear (stainless steel)	14 N/mm ²
(aluminum)	10 N/mm ²
Glass Transition Temperature (T _g)	59 °C
CTE Prior T _g	79 ppm/°C
CTE After T _g	126 ppm/°C
Thermal Conductivity @ 25 °C	0.4 W/(m·K)
Service Temperature Range	-40–150 °C

Usage Parameters

Working Time	30 min
Mix Ratio by Volume	1:1
Mix Ratio by Weight	1:0.92

Uncured Properties

Mixed Density	1.34 g/mL
Viscosity @ 25 °C	(A) 380 Pa·s
	(B) 370 Pa·s
Shelf Life	3 y

9200FR



Application Instructions

Read the product SDS and Application Guide for more detailed instructions before using this product (downloadable at www.mgchemicals.com).

Recommended Preparation

Clean the substrate with Isopropyl Alcohol, MG #824, so the surface is free of oils, dust, and other residues.

Syringe or Cartridge

1. Twist and remove the cap from the cartridge or syringe. Do not discard cap.
2. Dispense a small amount to ensure even flow of both parts.
3. (Optional) Attach a static mixer to the 9200FR-50ML.
 - a. Dispense and discard 3 to 5 mL of the product to ensure a homogeneous mixture.
 - b. After use, dispose of static mixer.
4. Without a static mixer, dispense material on a mixing surface or container, and thoroughly mix parts A and B together.
5. To stop the flow, pull back on the plunger.
6. Clean nozzle to prevent contamination and material buildup.
7. Replace the cap on the cartridge or syringe.

Dispensing Accessories

Consult the table below for accessory selection. See the Dispensing Accessories Application Guide for usage instructions. 8MT-50-FT should only be used with a pneumatic dispenser.

Cat. No.	Dispensing Gun	Static Mixer
9200FR-25ML	N/A	N/A
9200FR-50ML	8DG-50-1-1	8MT-50, 8MT-50FT

Cure Instructions

Allow to cure at room temperature for 48 hours, or cure the adhesive in an oven at one of these time/temperature options:

Temperature	40 °C	65 °C	80 °C	100 °C
Time	16 h	3 h	1.5 h	30 min

Storage and Handling

Store between -10 and 27 °C in a dry area, away from sunlight (see SDS). To maximize shelf life, recap product firmly when not in use.

Disclaimer

This information is believed to be accurate. It is intended for professional end-users who have the skills required to evaluate and use the data properly. M.G. Chemicals Ltd. does not guarantee the accuracy of the data and assumes no liability in connection with damages incurred while using it.

Appendix 9: Nomex Data Sheet



DuPont™ Nomex® 410

Technical Data Sheet

Nomex® 410 is a family of insulation papers that offer high inherent dielectric strength, mechanical toughness, flexibility and resilience. Nomex® 410, the original form of Nomex® paper, is widely used in a majority of electrical equipment applications. Available in 11 thicknesses ranging from 0.05 mm to 0.76 mm (2 mil to 30 mil), Nomex® 410 is used in almost every known electrical sheet insulation application.

Electrical Properties

The typical electrical property values for Nomex® 410 are shown in Table I. The AC Rapid Rise dielectric strength data in Table I represent voltage stress levels withstood for 10 to 20 seconds at a frequency of 60 Hz. These values differ from long-term strength potential. DuPont recommends that continuous stresses in transformers designed with Nomex® 410 not exceed 40 V/mil (1.6 kV/mm) to help minimize the risk of partial discharges. The full wave impulse dielectric

strength data shown in Table I are based on multiple sheets. These values are appropriate for applications that employ these materials in such configurations. Data based on single sheets of material are available upon request.

The geometry of the system has an effect on the actual impulse strength values of the material. The dielectric strength data are typical values and not recommended for design purposes. Design values can be supplied upon request.

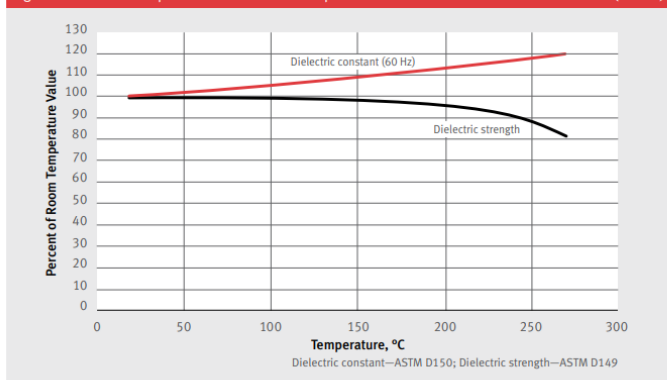
Table I. Typical Electrical Properties of DuPont™ Nomex® 410

Property	Nominal Thickness, mm (mil)											Test Method
	0.05 (2)	0.08 (3)	0.10 (4)	0.13 (5)	0.18 (7)	0.25 (10)	0.30 (12)	0.38 (15)	0.51 (20)	0.61 (24)	0.76 (30)	
Dielectric Strength AC Rapid rise, V/mil kV/mm	460 18	565 22	525 21	715 28	865 34	845 33	870 34	850 33	810 32	810 32	680 27	ASTM D149 ¹
Full Wave Impulse, V/mil kV/mm	1000 39	1000 39	900 36	1400 55	1400 55	1600 63	N/A N/A	1400 55	1400 55	N/A N/A	1250 49	ASTM D3426
Dielectric Constant at 60 Hz	1.6	1.6	1.8	2.4	2.7	2.7	2.9	3.2	3.4	3.7	3.7	ASTM D150
Dissipation Factor at 60 Hz (x 10 ⁻³)	4	5	6	6	6	6	7	7	7	7	7	ASTM D150

1. Using 50-mm (2-in.) electrodes, rapid rise; corresponds with IEC 243-1 subclause 9.1, except for electrodes set-up of 50 mm (2 in.).

Temperature has a minor effect on dielectric strength and dielectric constant, as shown in Figure 1.

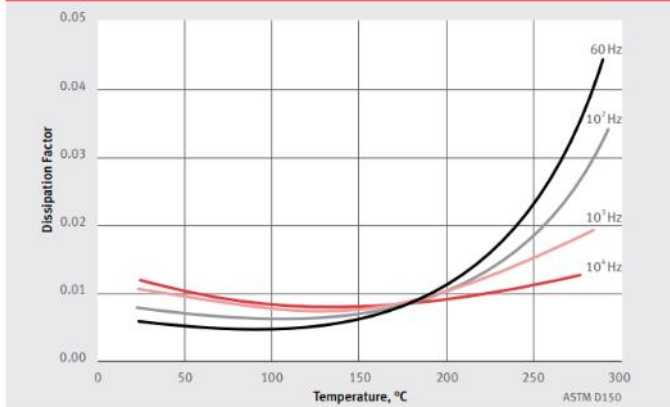
Figure 1. Effect of Temperature on Electrical Properties of DuPont™ Nomex® 410 — 0.25 mm (10 mil)





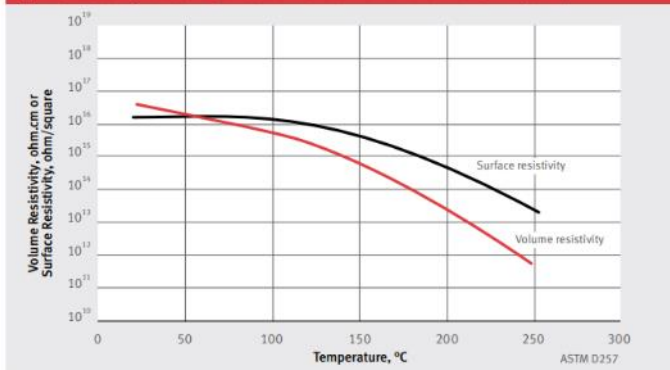
Variations in frequency up to 10^4 Hz have essentially no effect on the dielectric constant of Nomex[®] 410. The effects of temperature and frequency on dissipation factor of dry Nomex[®] 410 — 0.25 mm (10 mil) paper are shown in Figure 2. The 60 Hz dissipation factors of thinner papers are essentially the same as those for 0.25 mm (10 mil) at temperatures up to 200°C. At higher temperatures and frequencies, the thicker papers have somewhat higher dissipation factors than those shown for 0.25 mm (10 mil) paper.

Figure 2. Dissipation Factor versus Temperature and Frequency of DuPont[®] Nomex[®] 410 — 0.25 mm (10 mil)



Surface and volume resistivities of dry Nomex[®] 410 — 0.25 mm (10 mil) paper are shown in Figure 3 as functions of temperature. The corresponding values for other thicknesses of Nomex[®] 410 are very similar.

Figure 3. Resistivity versus Temperature of DuPont[®] Nomex[®] 410 — 0.25 mm (10 mil)



The relatively minor effects of moisture (humidity) on the electrical properties of Nomex[®] 410 — 0.25 mm (10 mil) are shown in Table II.

Table II. Humidity Effects on Electrical Properties of DuPont[®] Nomex[®] 410 — 0.25 mm (10 mil)

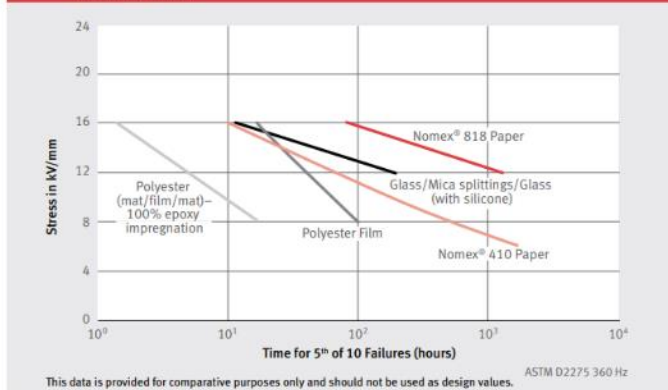
Property	Typical Value			Test Method
Relative Humidity, %	Oven Dry	50	96	
Dielectric Strength, V/mil kV/mm	850 33.5	845 33.2	780 30.7	ASTM D149 ¹
Dielectric Constant at 60 Hz at 1 kHz	2.5 2.3	2.7 2.6	3.2 3.1	ASTM D150
Dissipation Factor at 60 Hz ($\times 10^{-3}$) at 1 kHz ($\times 10^{-3}$)	6 13	6 14	11 25	ASTM D150
Volume Resistivity, ohm-cm	6×10^{16}	2×10^{16}	2×10^{14}	ASTM D257

1. Using 50-mm (2-in.) electrodes, rapid rise; corresponds with IEC 60243-1 subclause 9.1 except for electrode set-up of 50 mm (2 in.).



Like other organic insulating materials, Nomex® paper is gradually eroded under attack by partial discharges. Partial discharge intensity is a function of voltage stress, which, in turn, depends almost entirely on design parameters such as spacing between circuit elements, smooth vs. sharp contours, etc. Although corona does not occur during normal operation of properly designed electrical equipment, any device may be subject to occasional overvoltages that produce brief corona discharges; and it is important that the insulation not fail prematurely under these conditions. The voltage endurance (time to failure under corona attack) of Nomex® 410 is superior to other commonly used organic insulations and even compares favorably with some inorganic compositions, as shown in Figure 4. These data were obtained in all cases on single layers of 0.25 mm (10 mil) materials at room temperature, 50% relative humidity, and 360 Hz frequency. Times to failure at 50–60 Hz are approximately 6 to 7 times as long as indicated.

Figure 4. Voltage Endurance of Various Insulating Materials — Single-Layer DuPont® Nomex® 410 — 0.25 mm (10 mil)



Mechanical Properties

The typical mechanical property values for Nomex® 410 are shown in Table III.

Table III. Typical Mechanical Properties of DuPont® Nomex® 410

Property	Nominal Thickness, mm (mil)											Test Method
	0.05 (2)	0.08 (3)	0.10 (4)	0.13 (5)	0.18 (7)	0.25 (10)	0.30 (12)	0.38 (15)	0.51 (20)	0.61 (24)	0.076 (30)	
Typical Thickness, mm mil	0.06 2.2	0.08 3.1	0.11 4.2	0.13 5.2	0.18 7.2	0.26 10.2	0.31 12.2	0.39 15.3	0.52 20.4	0.61 24.2	0.78 30.6	ASTM D374 ¹
Basis Weight, g/m ²	41	64	88	115	174	249	310	395	549	692	839	ASTM D646
Density, g/cc	0.72	0.81	0.83	0.88	0.95	0.96	1.00	1.02	1.06	1.13	1.08	
Tensile Strength, N/cm MD XD	43 19	68 34	93 49	141 71	227 116	296 161	380 185	462 252	610 374	728 500	816 592	ASTM D828
Elongation, % MD XD	10 7	12 9	12 9	16 13	20 15	22 18	23 18	22 16	23 18	21 16	21 17	ASTM D828
Elmendorf Tear, N MD XD	0.8 1.5	1.2 2.4	1.9 4.4	2.3 4.8	3.7 7.2	5.6 10.6	7.1 13.7	9.0 16.7	14.3 24.8	N/A N/A	N/A N/A	TAPPI 414
Initial Tear Strength, N MD XD	11 6	16 9	24 14	31 17	48 27	69 42	88 55	110 71	158 114	191 153	233 193	ASTM D1004 ²
Shrinkage at 300°C, % MD XD	1.8 0.0	0.8 0.0	0.4 0.0	0.4 0.0	0.5 0.1	0.2 0.0	0.2 0.1	0.2 0.1	0.0 0.0	0.0 0.0	0.0 0.0	

MD = Machine Direction; XD = Cross Direction

1. Method D; using 17 N/cm².

2. Data presented for initial tear strength is listed in the direction of the sample per ASTM D1004. The tear is 90 degrees to sample direction; hence, for papers with a higher reported machine direction initial tear strength, the paper will be tougher to tear in the cross direction.



The effects of high temperatures on tensile strength and elongation are illustrated in Figure 5. Nomex® sheet structures also retain good mechanical properties at very low temperatures. At the boiling point of liquid nitrogen (minus 196°C or 77 K) the tensile strength of Nomex® 410 — 0.25 mm (10 mil) paper exceeds its room temperature value by 30% to 60% (depending on direction), while elongation to break is still greater than 3% (better than most inorganic materials at room temperature). This allows Nomex® 410 to work well in cryogenic applications.

The effects of moisture (humidity) on tensile strength and elongation are shown in Figure 6. Like elongation, the tear strength and toughness of Nomex® 410 are also improved at higher moisture contents.

The dimensions of bone-dry Nomex® 410 exposed to 95% relative humidity (RH) conditions will increase at most 1% in the machine direction and 2% in the cross direction (due to moisture absorption). This swelling is largely reversible when the paper is re-dried. The rate of change in dimensions will depend, of course, on paper thickness and configuration (e.g., individual sheets versus tightly wound rolls). Variations in environmental humidity will usually produce dimensional changes that will be less than 1%.

However, even small dimensional changes—especially if they are non-uniform—can cause or accentuate non-flatness (e.g., sag, puckers, etc.) in the sheet, which can cause problems in critical operations like laminating or creping. Therefore, Nomex® paper intended for these applications should be kept sealed in its protective polyethylene wrapper to maintain uniform moisture content until just before use. For more information on the effects of moisture on Nomex® insulation and how to appropriately protect the rolls, request our moisture brochure.

Figure 5. Effect of Temperature on Mechanical Properties of DuPont® Nomex® 410 — 0.25 mm (10 mil)

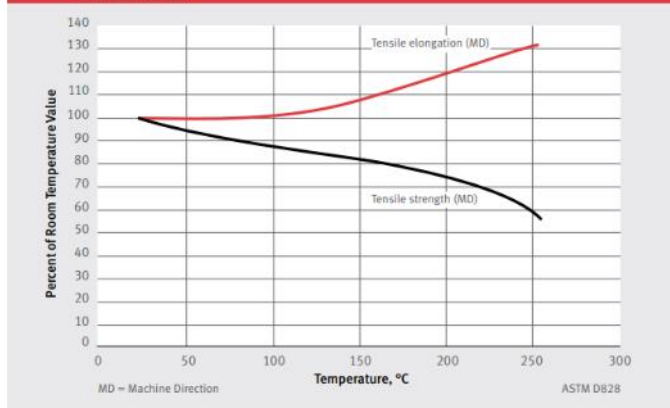
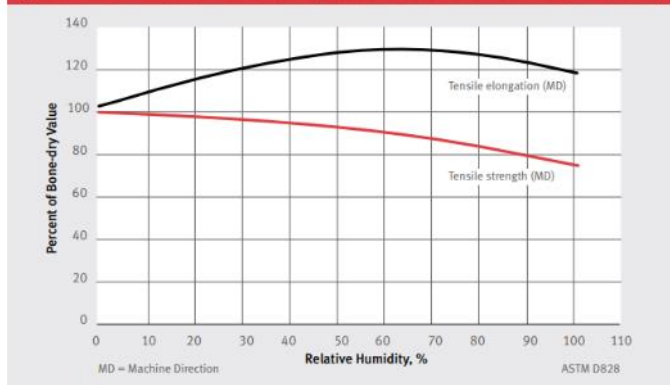


Figure 6. Effect of Moisture on Mechanical Properties of DuPont® Nomex® 410 — 0.25 mm (10 mil)





Thermal Properties

The effects of long-time exposure of Nomex® 410 — 0.25 mm (10 mil) to high temperature on important electrical and mechanical properties are shown in Figures 7 and 8. These Arrhenius plots of aging behavior are the basis for the recognition of Nomex® paper as a 220°C insulation by Underwriters Laboratories and have been utilized for almost 50 years in commercial applications. These curves can also be extrapolated to higher temperatures. Measurements show, for example, that Nomex® 410 will maintain 300 V/mil (12 kV/mm) dielectric strength for several hours at 400°C, which is the performance predicted by the Arrhenius plot.

Figure 7. Useful Life versus Temperature for DuPont® Nomex® 410 — 0.25 mm (10 mil)

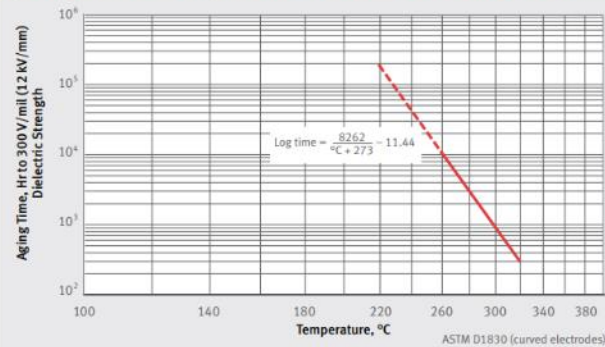
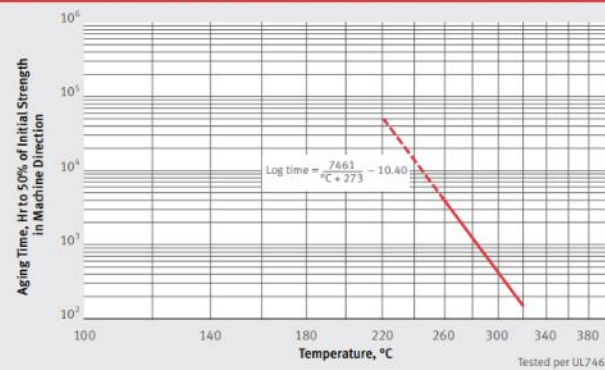


Figure 8. Useful Life versus Temperature for DuPont® Nomex® 410 — 0.25 mm (10 mil)





The thermal conductivity of Nomex® 410 — 0.25 mm (10 mil) paper is shown in Figure 9. These values are similar to those for cellulosic papers and, as with most materials, are primarily determined by specific gravity (density). Therefore, thinner grades of Nomex® 410 will have slightly lower conductivity and thicker grades will have higher conductivities, as is seen in Table IV. The total system construction may affect the overall thermal conductivity; therefore, care should be taken in applying individual sheet data to actual situations. For example, two sheet insulations with identical thermal conductivities may have quite different effects on heat transfer from a coil due to the differences in stiffness or winding tension, which affect the spacing between the insulation layers.

Chemical Stability

The compatibility of Nomex® papers and pressboards with virtually all classes of electrical varnishes and adhesives (polyimides, silicones, epoxies, polyesters, acrylics, phenolics, synthetic rubbers, etc.), as well as with other components of electrical equipment, is demonstrated by the many UL-recognized systems comprising Nomex®, including use in commercial applications for almost 50 years. Nomex® papers are also fully compatible (and in commercial use) with transformer fluids (mineral and silicone oils and other synthetics) and with lubricating oils and refrigerants used in hermetic systems. Common industrial solvents (alcohols, ketones, acetone, toluene, xylene) have a slight softening and swelling effect on Nomex® 410, similar to that of water. These effects are mainly reversible when the solvent is removed.

The Limiting Oxygen Index (LOI) of Nomex® 410 at room temperature ranges between 27% and 32% (depending on thickness and density); at 220°C, it ranges between 22% and 25%. Materials with LOI above 20.8% (ambient air) will not support combustion. Nomex® 410 must be heated between 240°C and 350°C (again depending on thickness) before its LOI declines below the flammability threshold. The LOI data for Nomex® 410 — 0.25 mm (10 mil) is shown in Figure 10.

Figure 9. Thermal Conductivity versus Temperature for DuPont® Nomex® 410 — 0.25 mm (10 mil)

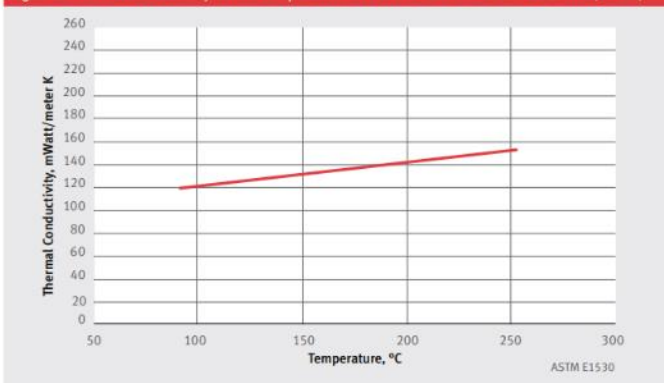


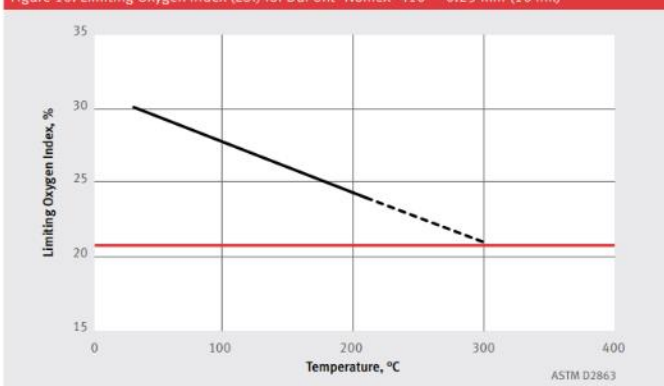
Table IV. Thermal Conductivity of DuPont® Nomex® 410

Nominal Thickness, mm mil	0.05 2	0.08 3	0.10 4	0.13 5	0.18 7	0.25 10	0.38 15	0.51 20	0.76 30
Density, g/cc	0.72	0.81	0.83	0.88	0.95	0.96	1.02	1.06	1.08
Thermal Conductivity, ¹ mWatt/meter K	103	114	117	123	143	139	149	157	175

¹. All data taken at 150°C

ASTM E1530

Figure 10. Limiting Oxygen Index (LOI) for DuPont® Nomex® 410 — 0.25 mm (10 mil)





Radiation Resistance

The effect of 6400 megarads (64 Mgy) of 2 MeV beta radiation on the mechanical and electrical properties of Nomex® 410 is shown in Table V. (By comparison, a laminate of polyester film

and polyester mat of the same thickness, 100% epoxy-impregnated, crumbled after 800 megarads, or 8 Mgy). Similar results were obtained on exposure to gamma radiation.

The outstanding radiation resistance of Nomex® paper has led to its use in critical control equipment for nuclear power installations.

Table V. Radiation Resistance of DuPont® Nomex® 410 — 0.25 mm (10 mil) to 2 MeV Electrons (Beta Rays)

Property	Dose, Mgy								Test Method
	0	1	2	4	8	16	32	64	
Tensile Strength, % of original									
MD	100	96	100	100	94	87	81	65	ASTM D828
XD	100	100	99	99	97	86	81	69	
Elongation, % of original									
MD	100	89	92	96	76	60	36	18	ASTM D828
XD	100	92	91	88	82	47	27	16	
Dielectric Strength, V/mil	860	860	840	840	840	860	890	790	ASTM D149 ¹
kV/mm	34	34	33	33	33	34	35	31	
Dielectric Constant									
at 60 Hz	3.1	3.0	3.0	3.0	3.0	3.1	2.3	2.5	ASTM D150
at 1 kHz	3.0	3.0	2.9	3.0	2.9	3.1	2.3	2.5	
at 10 kHz	2.9	2.9	2.9	2.9	2.8	3.0	2.2	2.4	
Dissipation Factor (x 10 ⁻³)									
at 60 Hz	8	14	10	12	9	14	7	10	ASTM D150
at 1 kHz	13	16	15	16	13	16	11	13	
at 10 kHz	18	21	20	20	19	20	15	17	

MD = Machine Direction; XD = Cross Direction
1. With a 6.4-mm (1/4-in.) diameter electrode.

UL Ratings

Table VI shows the UL ratings for Nomex® 410 papers. Descriptions of the numerical values

for each of the UL ratings are detailed in the UL website on Component Materials, which

can be accessed at iq.ul.com/ul/cert.aspx?ULID=230937

Table VI. UL Ratings for DuPont® Nomex 410

Nominal Thickness, mm	Nominal Thickness, mil	UL94 Flame Class	UL746A HWI Rating	UL746A HAI Rating	UL746B RTI Electrical	UL746B RTI Mechanical	UL746A HVTR Rating	UL746A CTI Rating
0.05	2	—	0	3	220	220	3	3
0.08	3	—	0	3	220	220	3	3
0.10	4	VTM-0	0	3	220	220	3	3
0.13	5	V-0	0	1	220	220	3	3
0.18	7	V-0	0	1	220	220	3	3
0.25	10	V-0	0	0	220	220	3	3
0.30	12	V-0	0	0	220	220	3	3
0.38	15	V-0	0	0	220	220	3	3
0.51	20	V-0	0	0	220	220	3	3
0.61	24	V-0	0	0	220	220	3	3
0.76	30	V-0	0	0	220	220	3	3